

The Central Nervous System

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CHAPTER CONTENTS

Cellular Pathology of the Central Nervous System, 1130

- Reactions of Neurons to Injury, 1130
- Reactions of Astrocytes to Injury, 1131
- Reactions of Microglia to Injury, 1131
- Reactions of Other Glial Cells to Injury, 1131

Cerebral Edema, Hydrocephalus, Raised Intracranial Pressure, and Herniation, 1132

- Cerebral Edema, 1132
- Hydrocephalus, 1132
- Raised Intracranial Pressure and Herniation, 1132

Malformations and Developmental Disorders, 1133

- Neural Tube Defects, 1133
- Forebrain Anomalies, 1134
- Posterior Fossa Anomalies, 1135
- Syringomyelia and Hydromyelia, 1135

Cerebrovascular Disease, 1136

- Hypoxia and Ischemia, 1136
 - Focal Cerebral Ischemia*, 1136
 - Global Cerebral Hypoxia/Ischemia*, 1138
- Intracranial Hemorrhage, 1139
 - Intraparenchymal Hemorrhage*, 1139
 - Subarachnoid Hemorrhage and Ruptured Saccular Aneurysm*, 1141
 - Vascular Malformations*, 1143
- Vascular Dementia, 1143

Trauma, 1143

- Skull Fractures, 1144
- Parenchymal Injuries, 1144
 - Concussion*, 1144
 - Contusions and Lacerations*, 1144
 - Diffuse Axonal Injury*, 1145
- Traumatic Vascular Injury, 1145
 - Epidural Hematoma*, 1145
 - Subdural Hematoma*, 1145
- Chronic Traumatic Encephalopathy, 1146
- Spinal Cord Injury, 1146

Perinatal Brain Injury, 1147

Infections, 1147

- Acute Meningitis, 1147
 - Acute Pyogenic (Bacterial) Meningitis*, 1147
 - Acute Aseptic (Viral) Meningitis*, 1149
- Acute Focal Suppurative Infections, 1149
 - Brain Abscess*, 1149
 - Subdural Empyema*, 1149
 - Extradural Abscess*, 1149
- Chronic Bacterial Meningoencephalitis, 1149
 - Tuberculosis*, 1149
 - Neurosyphilis*, 1150

Neuroborreliosis (Lyme Disease), 1150

Viral Meningoencephalitis, 1150

Arthropod-Borne Viral Encephalitis, 1150

Herpes Simplex Virus Type 1, 1151

Herpes Simplex Virus Type 2, 1151

Varicella-Zoster Virus, 1151

Cytomegalovirus, 1151

Poliomyelitis, 1152

Rabies, 1152

Subacute Sclerosing Panencephalitis, 1153

Human Immunodeficiency Virus, 1153

Progressive Multifocal Leukoencephalopathy, 1153

Fungal Meningoencephalitis, 1154

Other Infectious Diseases of the Nervous System, 1154

Demyelinating Diseases, 1156

- Multiple Sclerosis, 1156
- Neuromyelitis Optica, 1157
- Acute Disseminated Encephalomyelitis and Acute Necrotizing Hemorrhagic Encephalomyelitis, 1158
- Central Pontine Myelinolysis, 1158

Neurodegenerative Diseases, 1159

- Prion Diseases, 1159
 - Creutzfeldt-Jakob Disease*, 1159
 - Variant Creutzfeldt-Jakob Disease*, 1160
- Alzheimer Disease, 1161
- Frontotemporal Lobar Degenerations, 1164
 - FTLD-tau*, 1164
 - FTLD-TDP*, 1166
- Parkinson Disease, 1166
 - Dementia with Lewy Bodies*, 1168
- Atypical Parkinsonian Syndromes, 1169
 - Progressive Supranuclear Palsy*, 1169
 - Corticobasal Degeneration*, 1169
 - Multiple System Atrophy*, 1169
- Huntington Disease, 1169
- Spinocerebellar Degenerations, 1171
 - Spinocerebellar Ataxias*, 1171
 - Friedreich Ataxia*, 1171
 - Ataxia-Telangiectasia*, 1171
- Amyotrophic Lateral Sclerosis, 1172
- Other Motor Neuron Diseases, 1173
 - Spinal and Bulbar Muscular Atrophy (Kennedy Disease)*, 1173
 - Spinal Muscular Atrophy*, 1173
- Genetic Metabolic Diseases, 1173**
 - Neuronal Storage Diseases, 1174
 - Leukodystrophies, 1174
 - Mitochondrial Encephalomyopathies, 1174

Toxic and Acquired Metabolic Diseases, 1175

Vitamin Deficiencies, 1175

*Thiamine (Vitamin B₁) Deficiency, 1175**Vitamin B₁₂ Deficiency, 1175*

Neurologic Sequelae of Metabolic Disturbances, 1175

*Hypoglycemia, 1175**Hepatic Encephalopathy, 1176*

Toxic Disorders, 1176

*Carbon Monoxide, 1176**Ethanol, 1176**Radiation, 1176***Tumors, 1176**

Gliomas, 1177

*Diffuse Astrocytomas (Astrocytoma, IDH-mutant and**Glioblastoma, IDH-wildtype), 1177**Oligodendroglioma, IDH-mutant and 1p/19q-codeleted, 1178**Pilocytic Astrocytoma, 1180**Ependymoma, 1180*

Choroid Plexus Tumors, 1181

Neuronal and Glioneuronal Tumors, 1181

Embryonal Tumors, 1182

Medulloblastoma, 1182

Primary CNS Lymphoma, 1182

Meningiomas, 1182

Metastatic Tumors, 1184

Paraneoplastic Syndromes, 1184

Familial Tumor Syndromes, 1185

*Neurofibromatosis and Schwannomatosis, 1185**Tuberous Sclerosis Complex, 1185**Von Hippel-Lindau Disease, 1185*

The principal functional unit of the central nervous system (CNS) is the *neuron*. Neurons of different types and in different locations have distinct properties, including diverse functional roles, patterns of synaptic connections, neurotransmitters used, and metabolic requirements (which vary with electrical activity). A set of neurons, not necessarily clustered together in a region of the brain, may thus show *selective vulnerability* to an insult because the affected neurons share one or more of these properties. Because brain functions are anatomically compartmentalized, the pattern of clinical signs and symptoms that follow injury depend as much on the region of brain that is affected as on the pathologic process. Mature neurons are incapable of cell division, so destruction of even a small number of neurons essential for a specific function may produce a neurologic deficit.

In addition to neurons, the CNS contains other cells, such as *astrocytes* and *oligodendrocytes*, which make up the *glia*. Different cellular components of the CNS are affected by distinct unique neurologic disorders and respond to common insults (e.g., ischemia, infection) in a manner that is distinct from other tissues. We start our discussion of diseases of the CNS with an overview of the reactions of different brain cells to various insults.

CELLULAR PATHOLOGY OF THE CENTRAL NERVOUS SYSTEM

Neurons and glia of the CNS display a range of functional and morphologic changes following injury. Understanding these patterns can provide information about the mechanism of cellular injury and type of disease.

Reactions of Neurons to Injury

Neuronal injury may be an acute process, often following depletion of oxygen and/or glucose or brain trauma, or may develop over a period of years, as occurs in degenerative disorders of the brain. Neurons are highly metabolically active and require a continuous supply of oxygen and glucose to meet their metabolic needs. Moreover, since neurons are postmitotic cells that are incapable of proliferation, they must be maintained throughout life. Perhaps because of their long lifespan and high metabolic activity, neurons are unusually susceptible to the accumulation of misfolded proteins, which triggers the unfolded protein response (Chapter 2) and appears to be central to the pathogenesis of many degenerative disorders of the CNS.

MORPHOLOGY

Acute neuronal injury (“red neurons”) refers to changes seen following acute CNS hypoxia/ischemia, severe hypoglycemia, and other acute insults, and are the earliest morphologic markers of neuronal necrosis (see Fig. 28.15A). Red neurons are evident by 6 to 12 hours after an irreversible hypoxic/ischemic insult. Typical features include shrinkage of the cell body, pyknosis of the nucleus, disappearance of the nucleolus, loss of Nissl substance, and intense cytoplasmic eosinophilia.

Subacute and chronic neuronal injury (“degeneration”) refers to neuronal death that occurs over days to weeks (subacute) or as a result of a progressive disease of months to years duration (chronic), as is seen in slowly evolving neurodegenerative diseases such as amyotrophic lateral sclerosis (ALS) and Alzheimer disease. Initially, neurons often suffer a loss of synapses (sites of interneuronal communication), which may stem from aberrations in synaptic pruning (a process responsible for normal brain development and plasticity). Loss of synapses is followed by apoptotic cell death (often selectively involving functionally related groups of neurons) and reactive gliosis. At an early stage, the cell loss is difficult to appreciate; the associated reactive glial changes are often the best indicator of neuronal injury, although gliosis is not specific since it is seen in all forms of CNS injury.

Axonal reaction is a change observed in the cell body during regeneration of the axon; it is best seen in anterior horn cells of the spinal cord when motor axons are cut or seriously damaged. The increase in protein synthesis that occurs in response to the injury is reflected in enlargement and rounding up of the cell body, peripheral displacement of the nucleus, enlargement of the nucleolus, and dispersion of Nissl substance from the center to the periphery of the cell (central chromatolysis).

Neuronal damage may be associated with a wide range of subcellular alterations in the neuronal organelles and cytoskeleton. **Neuronal inclusions** may occur as a manifestation of aging and consist of intracytoplasmic accumulations of complex lipids (lipofuscin), proteins, or carbohydrates. Abnormal cytoplasmic accumulations of lipids and other substances also occur in certain inborn errors of metabolism, genetic disorders caused by mutations that lead to the loss of specific enzyme activities (Chapter 5). Viral infection can lead to the appearance of viral inclusions (e.g., in herpetic infection [Cowdry A or B bodies], rabies [Negri body], and cytomegalovirus infection).

Some degenerative diseases of the CNS are associated with neuronal intracytoplasmic inclusions, such as neurofibrillary tangles of Alzheimer disease and Lewy bodies of Parkinson disease; others cause abnormal vacuolization of the perikaryon and neuronal cell processes in the neuropil (Creutzfeldt-Jakob disease).

Wallerian degeneration refers to degeneration of axons distal to the site of acute axonal injury (see Chapter 27).

Reactions of Astrocytes to Injury

Gliosis is the most important histopathologic marker of CNS injury, regardless of etiology, and is characterized by hypertrophy and hyperplasia of astrocytes. The astrocyte derives its name from its star-shaped appearance. These cells have multipolar, extensively branched cytoplasmic processes that emanate from the cell body and express glial fibrillary acidic protein (GFAP), a cell type–specific intermediate filament. Astrocytes act as metabolic buffers and detoxifiers within the brain. Additionally, through the foot processes (which surround capillaries or extend to the subpial and subependymal zones), they contribute to barrier functions by controlling the flow of macromolecules between the blood, the cerebrospinal fluid (CSF), and the brain.

MORPHOLOGY

In gliosis, the nuclei of astrocytes (which are typically oval with evenly dispersed, pale chromatin) enlarge, become vesicular, and occasionally develop prominent nucleoli. The previously indistinct cytoplasm becomes bright pink due to increased expression of GFAP, and the cells develop numerous stout, ramifying processes; these cells are called **reactive** or **gemistocytic astrocytes** (Fig. 28.1). Despite this uniform morphology, there is no prototypical reactive astrocyte; instead, they adopt multiple context-dependent states that show a spectrum of different molecular and functional changes. The diversity and complexity of reactive astrocytes, along with their role in repair and disease, are areas of ongoing research.

Acute astrocytic injury, as occurs in hypoxia, hypoglycemia, and toxic injuries, is manifested by cellular swelling, as in other cells (Chapter 2). The **Alzheimer type II astrocyte** (unrelated to the disease but first described by the same neuroscientist) is a gray matter cell with a large (two to three times normal) nucleus, pale-staining central chromatin, an intranuclear glycogen droplet, and a prominent nuclear membrane and nucleolus. Alzheimer type II astrocytes are mainly seen in individuals with hyperammonemia due to chronic liver disease or hereditary metabolic disorders of the urea cycle.

Other types of astrocyte injury lead to the formation of cytoplasmic inclusion bodies. **Rosenthal fibers** are thick, elongated, brightly eosinophilic, irregular structures that occur within astrocytic processes and contain two heat-shock proteins (α B-crystallin and HSP27) as well as ubiquitin. Rosenthal fibers are typically found in regions of long-standing gliosis; they are also characteristic of one type of glial tumor, pilocytic astrocytoma (see Fig. 28.50C). In Alexander disease, a leukodystrophy associated with mutations in the gene encoding GFAP, abundant Rosenthal fibers are found in periventricular, perivascular, and subpial locations. More commonly seen are **corpora amylacea** (also called polyglucosan bodies). These are round, faintly basophilic, periodic acid–Schiff (PAS)–positive, concentrically lamellated structures of 5 to 50 μ m in diameter that are located wherever there are astrocytic end processes, especially in the subpial and perivascular zones. They consist primarily of glycosaminoglycan polymers, as well as heat-shock proteins and ubiquitin. Corpora amylacea occur in increasing numbers with advancing age and are thought to represent a degenerative change in the astrocyte. **Lafora bodies**, which are seen in the cytoplasm of neurons (as well as hepatocytes, myocytes, and other cells) in a variant of myoclonic epilepsy, have a similar structure and biochemical composition.

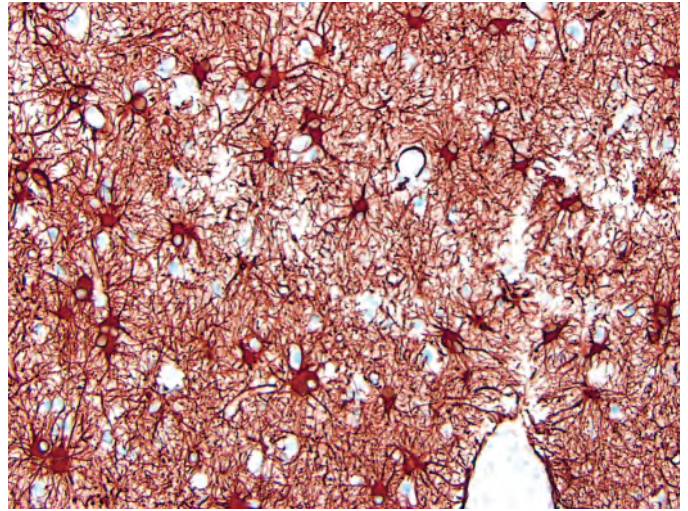


Fig. 28.1 Reactive astrocytes. Immunohistochemical staining for GFAP (glial fibrillary acidic protein) highlights the stellate astrocytic processes (brown) from which the cell's name is derived.

through a phagocytic process that may involve the complement system; aberrant reactivation of this developmental process has recently been implicated in several different brain diseases, including schizophrenia, encephalitis, Alzheimer disease, and frontotemporal dementia. Microglia respond to injury by (1) proliferating; (2) developing elongated nuclei; (3) forming aggregates around small foci of tissue necrosis (*microglial nodules*); or (4) congregating around cell bodies of dying neurons (*neuronophagia*). In addition to resident microglia, blood-derived macrophages may also be present in inflammatory foci.

Reactions of Other Glial Cells to Injury

Oligodendrocytes are cells that wrap their cytoplasmic processes around axons and form myelin. Each oligodendrocyte myelinates numerous internodes on multiple axons, in contrast to Schwann cells in peripheral nerves, which only myelinate the internode of a single axon. Injury or apoptosis of oligodendrocytes is a feature of acquired demyelinating disorders and leukodystrophies. Oligodendroglial nuclei harbor viral inclusions in *progressive multifocal leukoencephalopathy* (PML). Glial cytoplasmic inclusions, primarily composed of α -synuclein, are found in oligodendrocytes in *multiple system atrophy* (MSA).

Ependymal cells, the ciliated columnar epithelial cells lining the ventricles, do not show specific reaction patterns. When there is inflammation or marked dilation of the ventricular system, disruption of the ependymal lining is paired with proliferation of subependymal astrocytes to produce small irregularities on the ventricular surfaces (*ependymal granulations*). Certain infectious agents, particularly CMV, may produce extensive ependymal injury, with viral inclusions in ependymal cells. However, neither oligodendrocytes nor ependymal cells mediate significant responses to most forms of injury in the CNS.

KEY CONCEPTS

Cellular Pathology of the Central Nervous System

- Each cellular component of the nervous system responds to injury in a distinctive way.
- Neuronal injury commonly results in cell death, either by apoptosis or necrosis. Loss of neurons or synapses is often difficult to detect morphologically, but is a major contributor to neurologic dysfunction.

Reactions of Microglia to Injury

Microglia are phagocytic cells, derived early in embryonic development from the yolk sac or fetal liver, that serve as the resident macrophages of the CNS and share many surface markers with bone marrow–derived peripheral monocytes/macrophages. At rest, microglia have highly branched, amoeboid processes. During development, microglia prune unused neuronal synaptic connections, most likely

- Astrocytes respond to injury through apparent hypertrophy (increased cytoplasm due to accumulation of the intermediate filament protein GFAP) and hyperplasia; this response can be beneficial or harmful, depending on the type of injurious stimulus and the exact nature of astrocytic response.
- Microglia, the resident macrophages of the CNS, proliferate and accumulate in response to injury. Aberrant activation of microglia- and complement-mediated synaptic pruning plays an important role in the pathogenesis of many different neurologic disorders.

CEREBRAL EDEMA, HYDROCEPHALUS, RAISED INTRACRANIAL PRESSURE, AND HERNIATION

The brain is encased and protected by the rigid skull and dural reflections, and the spinal cord by the bony spinal canal. The pressure within the cranial cavity may rise in any one of three commonly observed clinical settings: generalized brain edema, increased CSF volume, and focally expanding mass lesions. Depending on the degree and speed of the pressure increase and the nature of the underlying lesion, the consequences range from subtle neurologic deficits to death.

Cerebral Edema

Cerebral edema (more precisely, brain parenchymal edema) is the result of increased fluid leakage from blood vessels and injury to various cells of the CNS. There are two main pathways of edema formation in the brain.

- *Vasogenic edema* is an increase in extracellular fluid caused by blood-brain barrier disruption and increased vascular permeability, allowing fluid to shift from the intravascular compartment to the intercellular spaces of the brain. The paucity of lymphatics in the brain slows the resorption of excess extracellular fluid. Vasogenic edema may be either localized (e.g., adjacent to inflammation or a neoplasm) or generalized, as may occur following a global ischemic injury.
- *Cytotoxic edema* is an increase in intracellular fluid secondary to neuronal, glial, or endothelial cell membrane injury, as might be encountered with a generalized hypoxic/ischemic insult or with a metabolic derangement that prevents maintenance of the normal membrane ionic gradients.

In practice, conditions associated with generalized edema often have elements of both vasogenic and cytotoxic edema. In generalized edema, the gyri are flattened, the intervening sulci are narrowed, and the ventricular cavities are compressed (Fig. 28.2). As the brain expands, herniation may occur.

Hydrocephalus

Hydrocephalus is the accumulation of excessive CSF within the ventricular system (Fig. 28.3). The choroid plexus within the ventricular system produces CSF, which normally circulates through the ventricular system and enters the cisterna magna at the base of the brainstem through the foramina of Luschka and Magendie. Subarachnoid CSF bathes the superior cerebral convexities and is absorbed by the arachnoid granulations. Most cases of hydrocephalus are a consequence of impaired flow and resorption of CSF; overproduction is a rare cause that can accompany tumors of the choroid plexus. Regardless of cause, the increased volume of CSF expands the ventricles and can elevate the intracranial pressure.

When hydrocephalus develops in infancy before closure of the cranial sutures, the head enlarges. By contrast, after the sutures close, hydrocephalus is associated with expansion of the ventricles and increased



Fig. 28.2 Cerebral edema. The surfaces of the gyri are flattened because of compression of the expanding brain by the dura mater and inner surface of the skull. Such changes are associated with a dangerous increase in intracranial pressure.

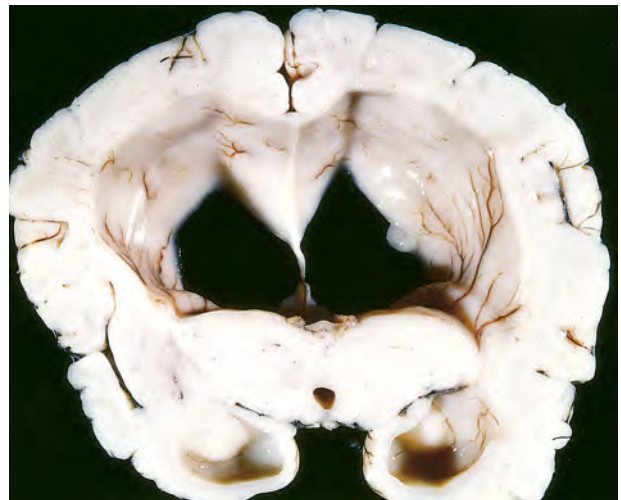


Fig. 28.3 Hydrocephalus. Dilated lateral ventricles seen in a coronal section through the mid-thalamus.

intracranial pressure without a change in head circumference. If the ventricular system is only focally obstructed, due to a mass in the third ventricle or to stenosis of the aqueduct, it is called *noncommunicating (obstructive) hydrocephalus*. In *communicating hydrocephalus*, the ventricular system remains in continuity with the subarachnoid space and there is enlargement of the entire ventricular system; causes include overproduction of CSF from a choroid plexus tumor and arachnoid fibrosis following meningitis. The term *hydrocephalus ex vacuo* refers to a compensatory increase in ventricular volume secondary to a loss of brain parenchyma.

Raised Intracranial Pressure and Herniation

Herniation is the displacement of brain tissue past rigid dural folds (the falx and tentorium) or through openings in the skull because of increased intracranial pressure. As the volume of the brain increases, CSF is displaced, leading to increasing pressure within the cranial cavity; herniation occurs when this pressure exceeds the brain's limited capacity to accommodate the increased intracranial pressure. Brain herniation is mostly caused by mass effects, either diffuse (generalized brain edema) or focal (tumors, abscesses, or hemorrhages). Elevated

intracranial pressure may also compress the vasculature and reduce perfusion of the brain, causing ischemic injury and further exacerbating cerebral edema.

MORPHOLOGY

The brain may herniate through different openings, and if the expansion is sufficiently severe, herniation may occur simultaneously in several locations (Fig. 28.4):

- **Subfalcine (cingulate) herniation** occurs when unilateral or asymmetric expansion of a cerebral hemisphere displaces the cingulate gyrus under the falx. This may lead to compression of the anterior cerebral artery and its branches, resulting in secondary infarcts.
- **Transtentorial (uncal, mesial temporal) herniation** occurs when the medial aspect of the temporal lobe is compressed against the free margin of the tentorium. With increasing displacement of the temporal lobe, the third cranial nerve is compromised, resulting in pupillary dilation and impaired ocular movements on the side of the lesion (ipsilateral). The ipsilateral posterior cerebral artery may also be compressed, resulting in an infarct of its territory (which includes the primary visual cortex); this manifests as contralateral homonymous hemianopia. When the extent of herniation is large enough, the contralateral cerebral peduncle may be compressed, resulting in false localizing hemiparesis ipsilateral to the side of the herniation (Kernohan notch phenomenon). Progression of transtentorial herniation is often accompanied by secondary hemorrhagic lesions in the midbrain and pons, termed **Duret hemorrhages** (Fig. 28.5). These linear or flame-shaped lesions usually occur in the midline and paramedian regions and are believed to be due to distortion or tearing of penetrating veins and arteries supplying the upper brainstem.
- **Tonsillar herniation** refers to displacement of the cerebellar tonsils through the foramen magnum. This pattern of herniation is life-threatening because it causes brainstem compression and compromises vital respiratory and cardiac centers in the medulla.

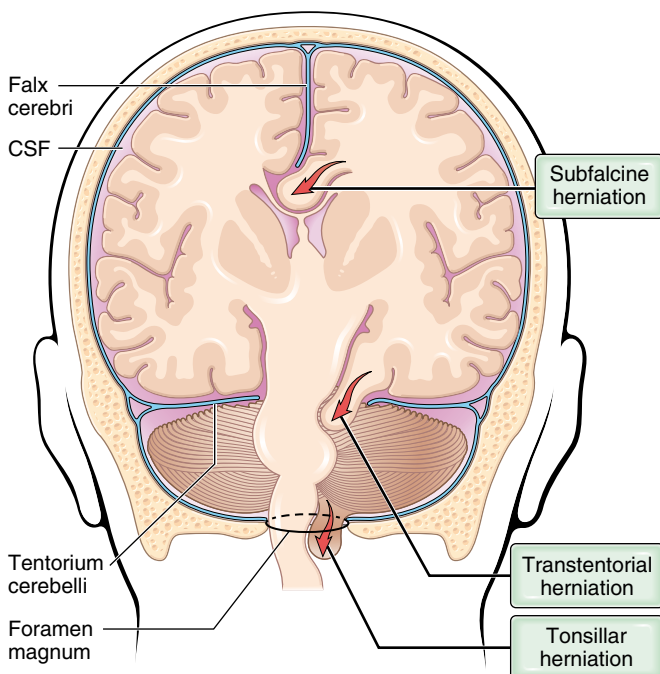


Fig. 28.4 Major sites of brain herniation: subfalcine, transtentorial, and tonsillar. CSF, Cerebrospinal fluid.

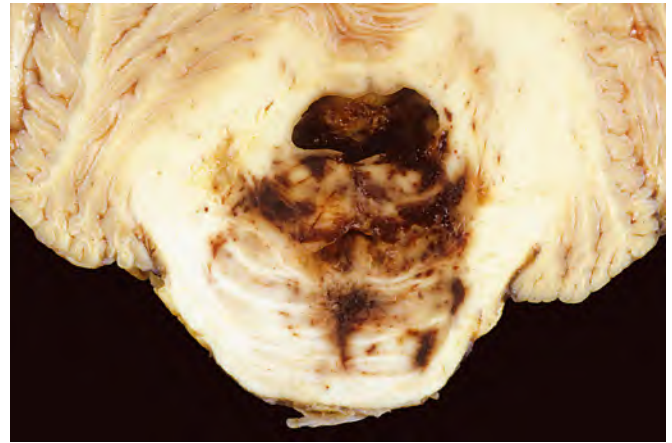


Fig. 28.5 Duret hemorrhage. As mass effect displaces the brain downward, there is disruption of the vessels that enter the pons along the midline, leading to hemorrhage.

KEY CONCEPTS

Cerebral Edema, Hydrocephalus, Raised Intracranial Pressure, and Herniation

- Cerebral edema is the accumulation of excess fluid within the brain parenchyma. Hydrocephalus is an increase in CSF volume within all, or a part, of the ventricular system.
- Increase in the intracranial volume (due to increased CSF volume, parenchymal edema, hemorrhage, or tumor) raises intracranial pressure.
- Increased intracranial pressure may result in cerebral herniation, decreased perfusion, and secondary infarction of the affected areas.

MALFORMATIONS AND DEVELOPMENTAL DISORDERS

Although the pathogenesis and etiology of many CNS malformations remain unknown, both genetic and environmental influences appear to be involved. Genomic sequencing has begun to uncover a range of genetic variants that are associated with certain malformations. The causal relationship between these genetic alterations and the pathogenesis of the malformations is the subject of active research. Besides genetic factors, many toxic compounds and infectious agents also have teratogenic effects and may cause brain malformations.

Neural Tube Defects

Neural tube defects are midline malformations that involve some combination of neural tissue, meninges, and overlying bone or soft tissue; collectively, they are the most common CNS malformations. Two distinct pathogenic mechanisms are contributory: (1) failure of neural tube closure, in which secondary mesenchymal tissue defects stem from aberrant skeletal modeling around the malformed tube (e.g., anencephaly and myelomeningocele); and (2) primary bony defects that are caused by abnormal axial mesoderm development and lead to secondary CNS abnormalities (e.g., encephalocele, meningocele, and spina bifida).

MORPHOLOGY

- **Anencephaly** is a malformation of the anterior end of the neural tube that leads to absence of most of the brain and calvarium. Forebrain development is disrupted at approximately 28 days of gestation, and all that remains in

its place is the **area cerebrovasculosa**, a flattened remnant of disorganized brain tissue with admixed ependyma, choroid plexus, and meningotheelial cells. The posterior fossa structures may be spared, depending on the extent of the skull deficit; descending tracts associated with disrupted structures are, as expected, absent.

- **Myelomeningocele** (or meningocele) refers to extension of spinal cord tissue through a defect in the vertebral column; the term **meningocele** applies when there is only a meningeal extrusion. Myelomeningoceles occur most commonly in the lumbosacral region (Fig. 28.6). Affected individuals have motor and sensory deficits in the lower extremities as well as disturbances of bowel and bladder control. Often, there is superimposed infection of the spinal cord due to defective barrier function of the thin, overlying skin.
- **Encephalocele** refers to an extrusion of malformed brain tissue through a midline defect in the cranium. It most often occurs in the occiput, although nasofrontal variants involving the orbit, ethmoid, or cribriform plate ("nasal glial heterotopia," sometimes misleadingly referred to as a "nasal glioma") can be seen.
- **Spinal dysraphism** or **spina bifida** (the most common neural tube defect) ranges from an asymptomatic bony defect (spina bifida occulta) to a severe malformation with a flattened, disorganized segment of spinal cord, associated with an overlying meningeal outpouching.

Clinical Features

The frequency of neural tube defects varies significantly among populations and geographically. The overall recurrence rate for a neural tube defect in subsequent pregnancies is estimated at 4% to 5%. Folate deficiency during the first several weeks of gestation is a well-established risk factor; differences in rates of neural tube defects among populations can be attributed in part to polymorphisms in enzymes involved in folic acid metabolism. Folate supplementation can lower the risk of neural tube defects, but because neural tube closure is normally complete by day 28 of embryonic development (before most pregnancies are recognized), it must be given to females throughout their reproductive years to be fully effective. Precisely how folate deficiency increases the risk is uncertain; effects on DNA methylation (an important epigenetic mechanism of gene regulation) are suspected.

Forebrain Anomalies

Abnormalities in the generation and migration of neurons result in malformations of the forebrain that may be focal or involve large segments. The pool of proliferating precursor cells in the developing brain lies in the germinal matrix adjacent to the ventricular system. The total number of neurons is determined by the fraction of proliferating cells that undergo transition into migrating cells with each cell cycle. The migration of neurons from the germinal matrix to the cerebral cortex follows two paths: radial migration, for progenitor cells destined to become excitatory neurons; and tangential migration, for those that will become inhibitory interneurons.

Several different anatomic malformation patterns have been defined. Recently, it has become clear that many of these patterns are caused by mutations in genes that are required for proper cerebral development. Changes may be seen in the complexity of the brain surface (either too few or too many gyri present), the organization of the brain into normal lobes, the structure of the cerebral cortex, or the distribution of neurons within the brain.

- The **volume of brain** may be abnormally large (*megalencephaly*) or abnormally small (*microencephaly*). Microencephaly, by far the more common of the two, is typically accompanied by a small head circumference; it is associated with many different conditions, including chromosome abnormalities, fetal alcohol syndrome, and viral



Fig. 28.6 Myelomeningocele. Both meninges and spinal cord parenchyma are included in the cyst-like structure visible just above the buttocks.

infection acquired in utero (e.g., with human immunodeficiency virus 1 [HIV-1] or Zika virus). It is postulated that the underlying anomaly is a reduction in the number of neurons that reach the neo-cortex, which leads to a simplification of gyral folding, a mechanism supported by experimental results in mouse models.

- **Lissencephaly** is a malformation characterized by reduction in the number of gyri, which in the extreme case may show no gyral pattern (*agyria*). Two patterns are observed, a classic smooth-surface form (type I), and a rough- or cobblestone-surface form (type II). In general, type I forms are associated with mutations that disrupt mechanisms involved in cell migration, such as mutations in the cytoskeletal "motor" proteins that drive migration of neuroblasts. In contrast, type II lissencephaly is most commonly associated with genetic alterations that disrupt the "stop signal" for migration. This signal depends on a set of specifically glycosylated proteins, and mutations in the enzymes that place sugars onto these proteins are the most common causes of this form of lissencephaly.
- **Polymicrogyria** is characterized by numerous small, irregularly formed cerebral convolutions with shallow sulci. The cerebral cortex is composed of four or fewer layers (instead of the normal six layers), with fusion of the molecular layers between gyri. Polymicrogyria can be induced by localized tissue injury toward the end of neuronal migration.
- **Neuronal heterotopias** are a group of neuron migration disorders defined by collections of subcortical neurons in inappropriate locations along the pathway of migration. As might be expected, one location is along the ventricular surface—as though the cells never left their place of birth. Periventricular nodular heterotopias can be caused by mutations in the gene encoding filamin A, an actin-binding protein that plays a role in cytoskeletal organization and cell motility. This gene is on the X chromosome, and the

mutant allele is lethal for individuals with a single X chromosome; in individuals with two X chromosomes, the process of X inactivation separates neurons into those with a normal allele (in the correct location) and those with the mutant allele (in the heterotopia). A microtubule-associated protein, doublecortin (DCX), is also encoded by a gene on the X chromosome; mutations in this gene result in lissencephaly in individuals with a single X chromosome, while individuals with two X chromosomes develop subcortical bandlike heterotopias that impart the impression of a “double cortex.”

- **Holoprosencephaly** is a spectrum of malformations characterized by incomplete separation of the cerebral hemispheres across the midline. Severe forms manifest midline facial abnormalities, including cyclopia; less severe variants (*arrhinencephaly*) show absence of the olfactory cranial nerves and related structures. Intrauterine diagnosis of severe forms by ultrasonography or fetal MRI is now possible. Holoprosencephaly is associated with trisomy 13 as well as other genetic syndromes. Mutations in genes that encode components of the sonic hedgehog signaling pathway may also produce holoprosencephaly.
- **Agenesis of the corpus callosum**, a relatively common malformation, is the absence of the white matter bundles that carry cortical projections from one hemisphere to the other (Fig. 28.7). Radiologic imaging studies show misshapen lateral ventricles (“batwing” deformity). Agenesis of the corpus callosum is sometimes associated with intellectual disability but may also be found in healthy individuals. It may be sporadic or familial and can be present in isolation or in association with a range of other malformations.

Posterior Fossa Anomalies

A distinct set of malformations primarily affects the brainstem and the cerebellum, which often show dramatic changes in size and shape. These may be accompanied by morphologic changes in other regions of the brain.

MORPHOLOGY

- **Chiari type I malformation** consists of low-lying cerebellar tonsils that extend down into the vertebral canal. It may be a silent abnormality or may become symptomatic because of impaired CSF flow, medullary compression, or associated hydromyelia/syringomyelia (see later); if present, these symptoms can usually be corrected by neurosurgical intervention.
- **Chiari type II malformation** (also known as **Arnold-Chiari malformation**) is more severe and produces a small posterior fossa, a misshapen midline cerebellum with downward extension of vermis through the foramen magnum (Fig. 28.8), and, almost invariably, hydrocephalus and a lumbar myelomeningocele. Other associated changes may include caudal displacement of the medulla, malformation of the tectum, aqueductal stenosis, cerebral heterotopias, and hydromyelia/syringomyelia.
- **Dandy-Walker malformation** is characterized by an enlarged posterior fossa. The cerebellar vermis is absent or present only in rudimentary form in its anterior portion; in its place is a large midline cyst that is lined by ependyma and is contiguous with leptomeninges on its outer surface. This cyst represents the expanded, roofless fourth ventricle in the absence of a normally formed vermis. Dysplasias of brainstem nuclei are commonly found in association with Dandy-Walker malformation.
- **Joubert syndrome** and related disorders share hypoplasia of the cerebellar vermis with rotation and apparent elongation of the superior cerebellar peduncles and an altered brainstem shape; together these changes give rise to the “molar tooth sign” on imaging. This group of malformations has been found to be caused by diverse mutations affecting genes that encode components of the primary (nonmotile) cilium.

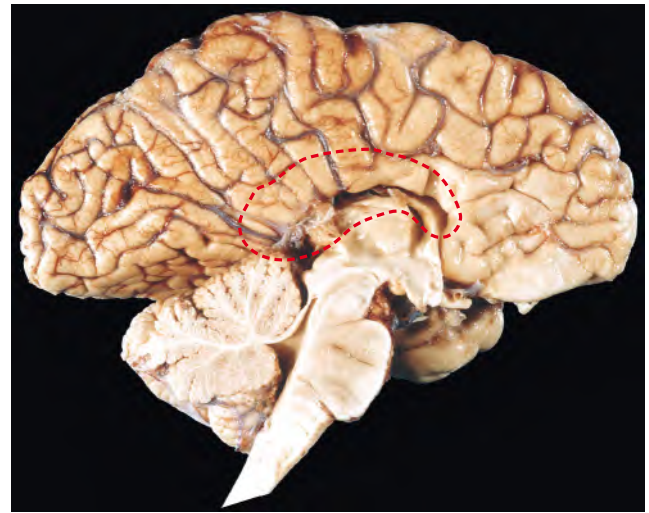


Fig. 28.7 Agenesis of the corpus callosum. The midsagittal view of the left hemisphere shows the lack of a corpus callosum and cingulate gyrus (area noted by dashed line).



Fig. 28.8 Chiari type II malformation. Midsagittal section showing small posterior fossa contents, downward displacement of the cerebellar vermis, and deformity of the medulla (arrows indicate the approximate level of the foramen magnum).

Syringomyelia and Hydromyelia

These disorders are characterized by expansion of the ependyma-lined central canal of the cord (*hydromyelia*) or by the formation of a fluid-filled cleft-like cavity in the inner portion of the cord (*syringomyelia*, *syrinx*) that may extend into the brainstem (*syringobulbia*).

Syringomyelia may be associated with a Chiari malformation; it may also occur in association with intraspinal tumors or following traumatic injury. In general, the histologic appearance is similar in all these conditions, with destruction of the adjacent gray and white matter, surrounded by dense reactive gliosis. The disease generally becomes manifest in the second or third decade of life. The distinctive symptoms and signs of a syrinx are the isolated loss of pain and temperature sensation in the upper extremities because of disruption of the crossing anterior spinal commissural fibers of the spinal cord.

KEY CONCEPTS

Malformations and Developmental Disorders

- Malformations may be associated with single-gene mutations, larger-scale genetic alterations, or exogenous factors.
- Overall, the earlier in development a malformation occurs, the more severe the morphologic and functional phenotype.
- Neural tube defects are caused by the failure of the tube to close or by skeletal bony defects that lead to secondary tube abnormalities; they range from incidental findings to severe malformations.
- Cortical development depends on proper orchestration of progenitor cell proliferation in the germinal matrix and migration of progenitors upward into the developing cortex. Disruption of these processes can alter the size, shape, and organization of the brain.
- Malformations involving the posterior fossa are typically distinct from those that affect the cerebral hemispheres.

CEREBROVASCULAR DISEASE

Cerebrovascular disease—injury to the brain because of altered blood flow—can be grouped into ischemic and hemorrhagic etiologies, with tissue infarction the ultimate consequence of both. *Stroke* is the clinical designation applied to these conditions and is defined as neurologic signs and symptoms that can be explained by a vascular mechanism, have an acute onset, and persist beyond 24 hours. (If symptoms resolve within 24 hours, the event is termed *transient ischemic attack*.) Cerebrovascular disease is the third leading cause of death (after heart disease and cancer) in the United States, and the most prevalent cause of morbidity and mortality from neurologic disease. Stroke stems from two major mechanisms:

- *Hypoxia and/or ischemia* resulting from impairment of oxygenation and blood supply of CNS tissue. This can be either a global or focal process, with the clinical manifestations determined by the region of brain affected. In the brain, embolism is a more common cause of vascular occlusion than thrombosis.
- *Hemorrhage* resulting from rupture of CNS vessels. Common etiologies include hypertension, vascular anomalies (aneurysms and malformations), and brain trauma (discussed later).

Hypoxia and Ischemia

Although the brain accounts for only 1% to 2% of body weight, it receives approximately 15% of the resting cardiac output and accounts for 20% of the body's oxygen consumption. Cerebral blood flow remains relatively constant over a wide range of blood pressure and intracranial pressure because of autoregulation of vascular resistance. The brain is strictly dependent on aerobic metabolism to meet its constant energy demands. It may be deprived of oxygen by either hypoxemia (low blood oxygen content) or ischemia (inadequate blood flow). Inadequate blood flow may result from a reduction in perfusion pressure (as in hypotension), small- or large-vessel occlusion, or both.

When blood flow to a portion of the brain is reduced, the survival of the tissue at risk depends on the presence of collateral circulation, the duration of ischemia, and the magnitude and rapidity of the reduction of flow. These factors determine, in turn, the anatomic site and size of the lesion and, consequently, the clinical deficit.

The general biochemical changes in cells resulting from ischemia are discussed in Chapter 2. In addition to processes shared with ischemia in other parts of the body, ischemia in the CNS can result in inappropriate release of excitatory amino acid neurotransmitters such as glutamate, which can damage neurons by allowing excessive influx of calcium ions through N-methyl-D-aspartate (NMDA)-type glutamate receptors; this phenomenon is termed *excitotoxicity*. In the region of

transition between necrotic tissue and the normal brain, there is an area of “at-risk” brain, referred to as the *penumbra*; this region can be rescued from cell death in many animal models with a variety of anti-apoptotic interventions, suggesting that ischemic neurons die by apoptosis as well as necrosis.

Focal Cerebral Ischemia

Focal cerebral ischemia follows reduction or cessation of blood flow to a localized area of the brain due to partial or complete arterial occlusion. When the ischemia is sustained, infarction follows in the territory of the compromised vessel. The size, location, and shape of the infarct and the extent of tissue damage that results are influenced by the duration of the ischemia and the adequacy of collateral flow. The major source of collateral flow is the circle of Willis (supplemented by external carotid-ophthalmic artery collaterals). Inconstant collateral leptomeningeal vessels from the surface of the brain may also supply the distal branches of the anterior, middle, and posterior cerebral arteries through cortical-leptomeningeal anastomoses; in contrast, there is little if any collateral flow for the deep penetrating vessels of the thalamus, basal ganglia, and deep white matter.

Occlusive vascular disease of severity sufficient to lead to cerebral infarction may be due to embolization from a distant source, in situ thrombosis, or various vasculitides; the pathology of these conditions is also discussed in Chapters 4 and 11.

- *Embolism* to the brain occurs from a variety of sources. Cardiac mural thrombi are among the most common culprits; myocardial infarct, valvular disease, and atrial fibrillation are important predisposing factors. Next in frequency are thromboemboli originating from arteries, most often atheromatous plaques in the carotids. Other sources of emboli include paradoxical thromboemboli, particularly in children with congenital heart disease; thromboemboli associated with cardiac surgery; and emboli of other types (tumor, fat, or air). The territory supplied by the middle cerebral artery—the direct extension of the internal carotid artery—is most frequently affected by embolic infarction; the incidence is about equal in the two hemispheres. Emboli tend to lodge where blood vessels branch or in areas of preexisting luminal stenosis. “Shower embolization,” as in fat embolism, may occur after fractures; affected individuals manifest generalized cerebral dysfunction with disturbances of higher cortical function and consciousness, often without localizing signs. Widespread hemorrhagic lesions involving the white matter are characteristic of embolization of bone marrow after trauma.
- *Thrombotic occlusion of the cerebral arteries* is most commonly caused by acute change of vulnerable atherosclerotic plaques, as in coronary artery disease (Chapter 12). The most common sites are the carotid bifurcation, the origin of the middle cerebral artery, and either end of the basilar artery. Thrombi cause progressive narrowing of the lumen, may be accompanied by anterograde extension, and may progress to fragmentation and distal embolization. Atherosclerotic cerebrovascular disease is frequently associated with systemic diseases such as hypertension and diabetes.
- *Hypertension* affects the deep penetrating arteries and arterioles that supply the basal ganglia and hemispheric white matter as well as the brainstem; these cerebral vessels develop *arteriolosclerosis*, also known as small vessel disease, because it affects arteries 40 to 900 μm in diameter (described in more detail later). If this disease process progresses to thrombosis and complete vessel occlusion, the end-result is development of small cavitary infarcts known as *lacunes* or *lacunar infarcts*. Depending on their location in the CNS, lacunar infarcts can be clinically silent or cause severe neurologic impairment. Affected vessels may also be associated with widening of the perivascular spaces without tissue infarction (*état criblé*).

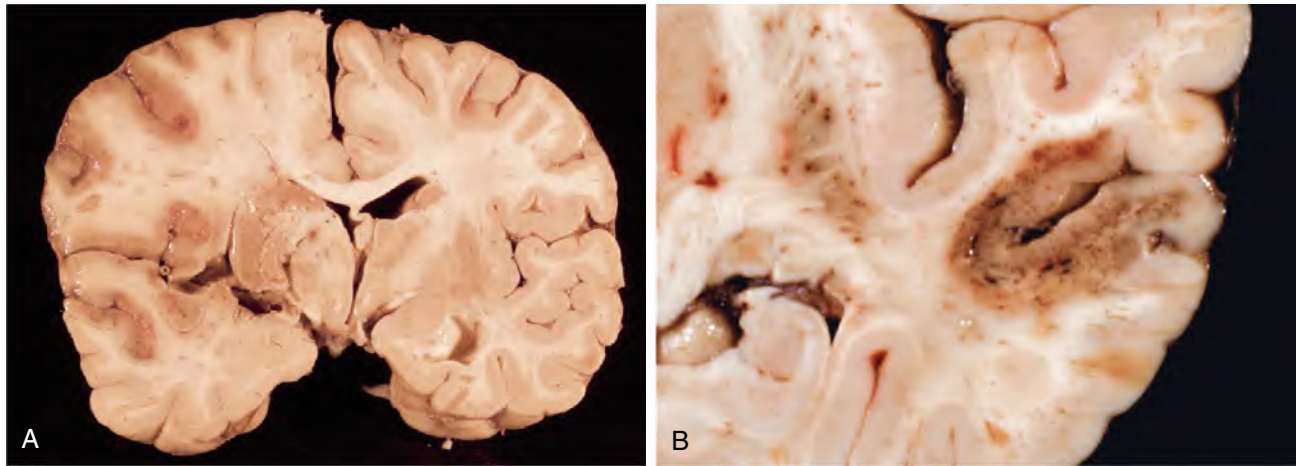


Fig. 28.9 (A) An ischemic infarct involves the territory of the middle cerebral artery, including the striatum, on the left side of this brain. (B) A hemorrhagic infarct with punctate hemorrhages, consistent with ischemia-reperfusion injury, is present in the temporal lobe.

- *Inflammatory processes* that involve blood vessels may also lead to luminal narrowing, thrombosis, and cerebral infarcts. Although infectious vasculitis of small and large vessels occurs with syphilis and tuberculosis, it is now more common in the setting of immunosuppression and opportunistic infection (e.g., aspergillosis). *Polyarteritis nodosa* and other noninfectious vasculitides may involve cerebral vessels and cause single or multiple infarcts throughout the brain. *Primary angiitis of the CNS* can also develop in the absence of systemic vasculitis.
- *Other conditions* that may cause thrombosis (and intracranial hemorrhage) include hypercoagulable states, dissecting aneurysms of extracranial arteries in the neck that supply the brain, and substance use (amphetamines, heroin, cocaine).

Brain infarcts are subdivided into two broad groups based on the presence of secondary hemorrhage. Because the brain has end-organ circulation with limited collateral supply, occlusive brain infarcts generally start as nonhemorrhagic (pale/anemic; Fig. 28.9A); clinically, these nonhemorrhagic infarcts are called ischemic, a confusing term because every infarct, not just this type, is caused by tissue ischemia. Secondary hemorrhage can occur from ischemia-reperfusion injury following spontaneous or therapeutic dissolution or fragmentation of the intravascular occlusive material. This process (termed *secondary hemorrhagic transformation* and leading to a hemorrhagic infarct) develops if the ischemic event lasts long enough to damage small blood vessels in the affected area; the resulting reperfusion hemorrhages are largely punctate in nature but may be multiple or even confluent (Fig. 28.9B). The clinical management of patients with nonhemorrhagic and hemorrhagic infarcts differs greatly, although the underlying causes are the same (for instance, thrombolytic therapy is contraindicated in a patient with brain hemorrhage of any etiology).

MORPHOLOGY

Both the gross and microscopic appearance of a **nonhemorrhagic infarct** changes over time. Grossly, there is little change in appearance during the first 6 hours of irreversible injury. By 48 hours, however, the tissue becomes pale, soft, and swollen, and the gray-white matter junction becomes indistinct. From 2 to 10 days, the brain becomes gelatinous and friable, and the previously ill-defined boundary between normal and infarcted tissue becomes more distinct as edema resolves in the viable adjacent tissue. From 10 days to 3 weeks, the tissue liquefies, eventually leaving a fluid-filled cavity that continues to expand until all the dead tissue has been removed (Fig. 28.10).

Microscopically, the tissue reaction evolves along the following sequence:

- **Acute infarct** (Fig. 28.11A). After the first 6 to 12 hours, neurons in the affected area show eosinophilic neuronal necrosis (increased eosinophilia of the cytoplasm followed by nuclear pyknosis and karyorrhexis (“dead red neurons”); both cytotoxic and vasogenic edema are present. There is loss of the usual tinctorial characteristics of white- and gray-matter structures. Endothelial and glial cells, mainly astrocytes, swell, and myelinated fibers begin to disintegrate. Up to 48 hours, neutrophilic emigration progressively increases (but is never as prominent as in myocardial infarction) and then falls off.
- **Subacute (evolving) infarct** (Fig. 28.11B). Phagocytic cells, derived from circulating monocytes and activated microglia, are evident at 48 to 72 hours and become the predominant cell type in the ensuing 2 to 3 weeks. The macrophages become stuffed with the products of myelin breakdown or blood and may persist in the lesion for months to years. Reactive astrocytes and newly formed vessels can be seen at the periphery of the lesion as early as 1 week after the insult. As the process of liquefaction and phagocytosis proceeds, astrocytes at the edges of the lesion progressively enlarge, divide, and develop a prominent network of cytoplasmic extensions.
- **Healed (remote) infarct** (Fig. 28.11C). After several months, the astrocytic response recedes, leaving behind a cystic cavity with dense meshwork of glial fibers admixed with new capillaries at the rim (“glial scar”). In the cerebral cortex, the cavity is separated from the meninges and subarachnoid space by a gliotic layer of tissue, which is derived from the molecular layer of the cortex. The pia and arachnoid are not affected. Infarcts undergo these reactive and reparative changes from the edges inward; thus, different areas of a lesion may appear chronologically divergent, revealing the natural progression of the response.

The features and temporal evolution of **hemorrhagic infarcts** parallel ischemic infarcts, with the addition of blood extravasation and resorption. In individuals receiving anticoagulant treatment, hemorrhagic infarcts may be associated with extensive intracerebral hematomas. Venous infarcts are often hemorrhagic and may occur after thrombotic occlusion of the superior sagittal sinus or other sinuses, or after occlusion of the deep cerebral veins. Neoplasms, localized infections, and other conditions leading to a hypercoagulable state increase the risk for venous thrombosis.

Lacunar infarcts (Fig. 28.12) are small occlusive infarcts that can be single or multiple and involve the putamen, globus pallidus, thalamus, internal capsule, deep white matter, caudate nucleus, and pons, in descending order of frequency. Grossly, they are lake-like spaces, arbitrarily defined as less than 15 mm in largest dimension; on microscopic examination, there is tissue loss surrounded by gliosis.

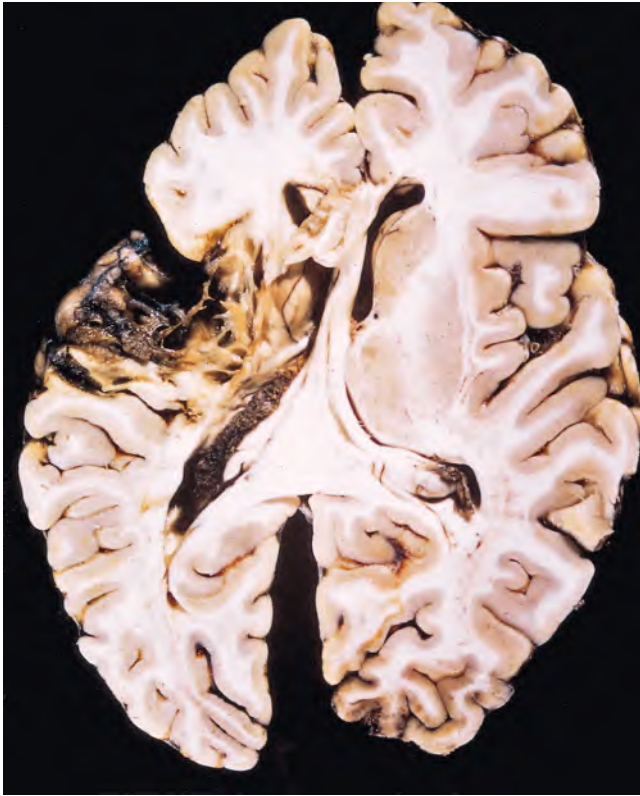


Fig. 28.10 Old cystic infarct showing cavitation from loss of brain parenchyma.

Clinical Features

Deficits produced by infarction are determined by the anatomic distribution of the damage rather than the underlying cause. Neurologic symptoms referable to the area of injury often develop rapidly, over minutes, and may continue to evolve over hours. Some recovery of

function is possible with reversal of injury in the ischemic penumbra and resolution of associated local edema. In general, there is slow improvement during a period of months. Because strokes are frequently associated with cardiovascular disease, many of the genetic and lifestyle risk factors are shared. Early diagnosis is of critical importance, as rapid treatment of nonhemorrhagic strokes with thrombolytic agents can limit or entirely prevent the development of permanent neurologic deficits.

Global Cerebral Hypoxia/Ischemia

Global cerebral hypoxia or ischemia occurs when there is a generalized reduction of cerebral perfusion (as in cardiac arrest, shock, and severe hypotension) or decreased oxygen-carrying capacity of the blood (e.g., in carbon monoxide poisoning). The clinical outcome varies with the severity and length of the insult. In mild cases, there may only be a transient postischemic confusional state, followed by complete recovery. Nevertheless, irreversible CNS damage may occur in individuals who experience more severe global hypoxic and/or ischemic insults (*diffuse hypoxic/ischemic encephalopathy*). Among CNS cells, there is a hierarchy of sensitivity to hypoxia/ischemia: neurons are the most sensitive, but glial cells (oligodendrocytes and astrocytes) are also vulnerable. The most sensitive neurons in the brain are pyramidal neurons in the hippocampus (especially area CA1, also referred to as Sommer sector), cerebellar Purkinje cells, and pyramidal neurons in the cerebral cortex (especially layers III and V); the molecular mechanisms underlying this selective vulnerability are not understood. With severe global cerebral hypoxia/ischemia, widespread neuronal death occurs, irrespective of regional vulnerability; patients who survive this injury often develop unresponsive wakefulness syndrome (previously known as persistent vegetative state). Other patients meet the current clinical criteria for “brain death,” including evidence of irreversible diffuse cortical injury (isoelectric, or “flat,” electroencephalogram), brainstem damage (such as absent reflexes and respiratory drive), and absent cerebral perfusion. When individuals with this severe form of injury are maintained on mechanical ventilation, the reperfused brain gradually undergoes widespread liquefaction,

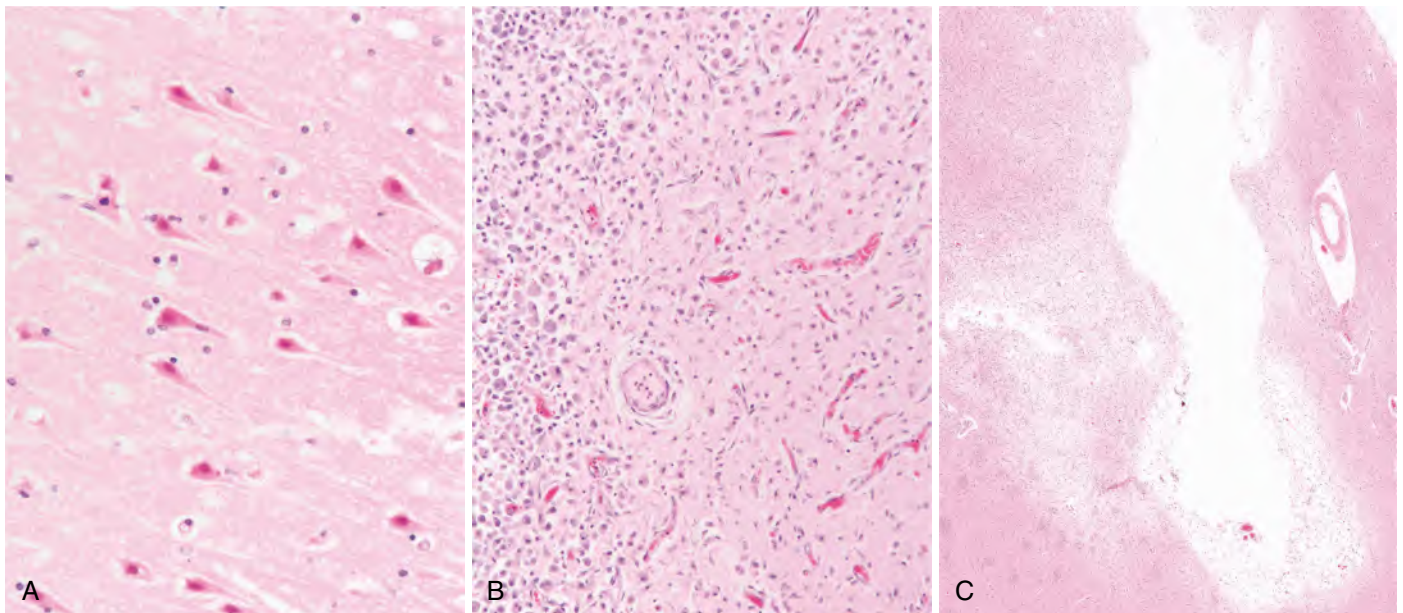


Fig. 28.11 Cerebral infarcts. (A) Acute ischemic injury causes diffuse eosinophilia of neurons, which are beginning to shrink. (B) After about 10 days, the lesion is characterized by the presence of foamy macrophages (best seen on the *left*) and adjacent reactive astroglia with neovascularization (*right*). (C) Remote small infarct is seen as an area of tissue loss surrounded by residual astroglia.



Fig. 28.12 Lacunar infarcts in the caudate nucleus and putamen (arrows).

producing so-called “respirator brain,” a misleading and medicolegally objectionable term, since the respirator does not cause the pathology.

Border zone (“watershed”) infarcts occur in the regions of the brain or spinal cord that lie at the most distal reaches of the arterial blood supply (i.e., the border zones between arterial territories). In the cerebral hemispheres, the border zone between the anterior and the middle cerebral artery distributions is at greatest risk. Damage to this region, located a few centimeters lateral to the interhemispheric fissure, results in a cortical wedge-shaped infarct that usually shows secondary hemorrhagic transformation (Fig. 28.13) and is often bilateral. Border zone infarcts usually develop after severe hypotensive episodes and are most commonly seen in patients resuscitated after cardiac arrest.

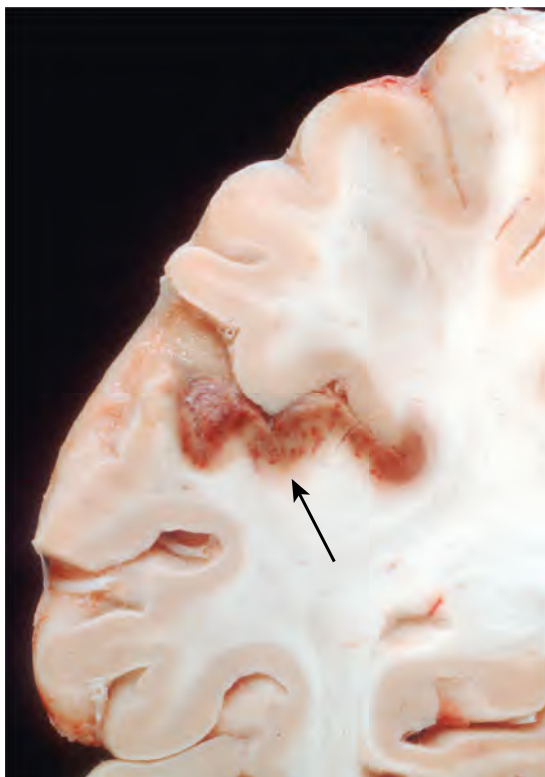


Fig. 28.13 Classic watershed infarct with secondary hemorrhagic transformation (arrow); boundary between anterior and middle cerebral artery circulations.

MORPHOLOGY

In the setting of global ischemia, the brain becomes edematous and swollen, producing widening of the gyri and narrowing of the sulci. The cut surface shows poor demarcation between gray and white matter. The microscopic features of irreversible ischemic injury evolve over time and mimic the changes seen in infarcts. **Early changes**, occurring 6 to 12 hours after the insult, are seen in neurons (dead red neurons described earlier); similar acute changes occur somewhat later in astrocytes and oligodendroglia. **Subacute changes**, occurring at 24 hours to 2 weeks, include tissue necrosis, influx of macrophages, vascular proliferation, and reactive gliosis. **Repair**, robust after approximately 2 weeks, is characterized by removal of necrotic tissue, cystic degeneration, loss of normal CNS architecture, and gliosis. In the cerebral neocortex, neuronal loss and gliosis are uneven, with preservation of some layers and destruction of others, producing a pattern of injury termed **laminar necrosis**.

Intracranial Hemorrhage

Hemorrhages may occur at any site within the cranium, outside the brain or within it (intraparenchymal) (Table 28.1). Hemorrhages in the epidural or subdural space are typically associated with trauma and will be discussed later. Hemorrhages within the brain parenchyma and in the subarachnoid space are more often a manifestation of underlying cerebrovascular disease and are discussed in this section.

Intraparenchymal Hemorrhage

Rupture of a small intraparenchymal vessel can result in a primary hemorrhage within the brain, often associated with sudden onset of neurologic symptoms (stroke); this should not be confused with the secondary hemorrhagic transformation of an occlusive infarct (described earlier). Spontaneous (nontraumatic) intraparenchymal hemorrhages occur most commonly in middle to late adult life, with a peak incidence at about 60 years of age. Hemorrhages in the basal ganglia and thalamus are commonly designated *ganglionic hemorrhages*, whereas those that occur more superficially within the cerebral hemispheres are called *lobar hemorrhages*; the two major causes of these patterns of hemorrhage are hypertension and cerebral amyloid angiopathy, respectively. In addition, other local and systemic factors may cause or contribute to nontraumatic hemorrhage, including systemic coagulation disorders, neoplasms, vasculitis, aneurysms, and vascular malformations.

Hypertension is the risk factor most commonly associated with deep brain parenchymal hemorrhages, accounting for more than 50% of clinically significant hemorrhages and for roughly 15% of deaths among individuals with chronic hypertension. *Hypertensive intraparenchymal hemorrhage* may originate in the putamen (50% to 60% of cases), thalamus, cerebellar hemispheres, pons, and other regions of the brain (Fig. 28.14A). Hypertension leads to several different vessel wall abnormalities, including accelerated atherosclerosis in larger arteries, hyaline arteriosclerosis in smaller arteries, and (in severe cases) proliferative changes and frank necrosis of arterioles. Arteriolar walls affected by hyaline change (Fig. 28.14B) are thickened but rigid and more vulnerable to rupture than normal vessels; these changes are most prominent in the basal ganglia and the subcortical white matter. As described earlier, if small arteries affected by hyaline arteriosclerosis are occluded, the result is lacunar infarction.

Cerebral amyloid angiopathy (CAA) is the risk factor most often associated with lobar hemorrhages (Fig. 28.14C). In CAA, amyloidogenic peptides, usually the same ones found in Alzheimer disease (A β ; see later), are deposited in the walls of medium- and small-caliber meningeal, cortical, and cerebellar vessels; involved vessels are rigid, and as a result fail to collapse during tissue processing and sectioning. Although similar to hyaline arteriosclerosis on

Table 28.1 Patterns of Hemorrhage in the Central Nervous System

Compartment	Common Etiology	Additional Features
Epidural space	Trauma	Usually associated with skull fracture (in adults); rapidly evolving neurologic symptoms (often after a short lucid period); requires rapid surgical intervention
Subdural space	Trauma	May follow minor trauma; slowly evolving neurologic symptoms, often with a delay from the time of injury
Subarachnoid space	Trauma	Typically associated with underlying parenchymal injury
	Vascular abnormality (arteriovenous malformation or aneurysm)	Sudden onset of severe headache, often with rapid neurologic deterioration; secondary injury may emerge and is associated with vasospasm
Intraparenchymal space	Trauma (contusions)	Selective involvement of the crests of gyri, where the brain is in contact with the inner surface of the skull (frontal and temporal tips, orbitofrontal surface)
	Ischemia (hemorrhagic conversion of an ischemic infarct)	Punctate hemorrhages in an area of previously ischemic brain parenchyma, usually following the cortical ribbon
	Cerebral amyloid angiopathy	“Lobar” hemorrhage, involving subcortical white matter, often with extension into the subarachnoid space
	Hypertension	“Ganglionic” hemorrhage centered in the putamen, thalamus, cerebellum, or brainstem; may extend into the ventricular system
	Tumors (primary or metastatic)	Associated with high-grade gliomas or certain metastases (melanoma, choriocarcinoma, renal cell carcinoma)

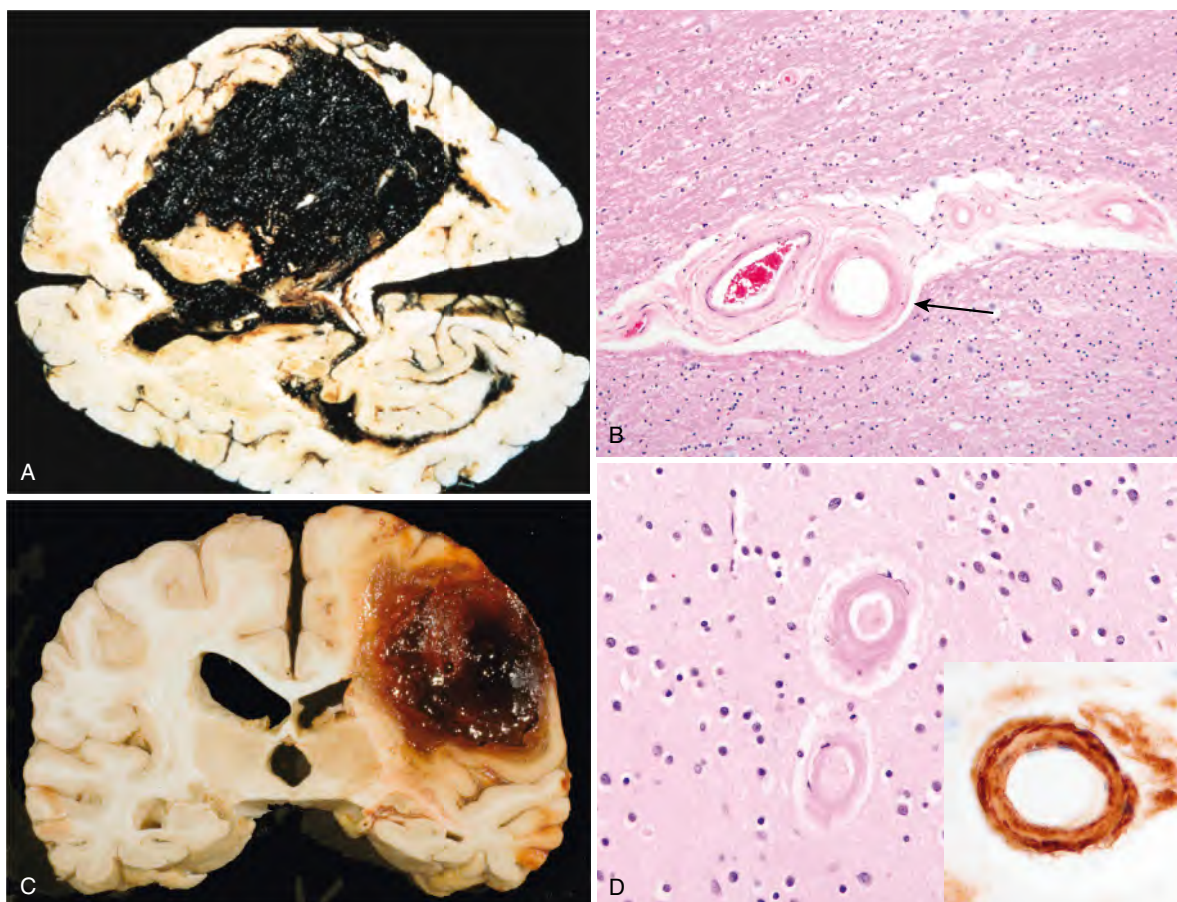


Fig. 28.14 Intraparenchymal cerebral hemorrhage. (A) Massive hypertensive ganglionic hemorrhage rupturing into a lateral ventricle. (B) Hyaline arteriosclerosis (fibrosis and thickening of the arteriolar walls) in the basal ganglia and subcortical white matter of patients with long-standing hypertension, a risk factor for hypertensive hemorrhages and lacunar infarcts. (C) Large lobar hemorrhage due to cerebral amyloid angiopathy focally dissects into the subarachnoid space. (D) Amyloid deposition in cortical arterioles in cerebral amyloid angiopathy; *inset*, immunohistochemical staining for A β peptide highlights the material deposited in the vessel wall. (Courtesy Dr. Dimitri Agamanolis, <https://neuropathology-web.org>.)

routine hematoxylin and eosin (H&E) stain, the hyaline material in CAA consists of β amyloid rather than collagen (Fig. 28.14D) and is primarily seen in the leptomeningeal and cortical (rather than basal ganglia and white matter) vessels. Amyloid deposition can weaken the vessel wall and lead to hemorrhage; as a result, many individuals with CAA have evidence of numerous small hemorrhages within the brain (“microbleeds”) that can be visualized by various imaging methods. As with Alzheimer disease (discussed later), there is a relationship between polymorphisms in the gene that encodes apolipoprotein E (ApoE) and risk of disease; specifically, the presence of either an $\epsilon 2$ or $\epsilon 4$ allele increases the risk of bleeding. Autosomal dominant forms of CAA are associated with certain mutations in the *APP* gene, which encodes the precursor for the $A\beta$ peptides that are prone to deposit as amyloid.

Other forms of hereditary small-vessel diseases of the CNS have been identified. *Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy* (CADASIL) is an autosomal dominant disorder caused by mutations in the *NOTCH3* gene that lead to misfolding of the extracellular domain of the NOTCH3 receptor. The disease is characterized by recurrent small vessel strokes (usually infarcts, less often hemorrhages) and dementia. Imaging studies show that the first detectable changes are in white matter, usually presenting around 35 years of age and then progressing further over time. Other forms of the heritable small vessel disease include a disorder associated with mutations in the gene encoding collagen COL4A1, a component of the vascular basement membrane. In addition to brain hemorrhage, this mutation causes a number of conditions including cataract, hematuria, and epilepsy.

MORPHOLOGY

Acute primary intraparenchymal hemorrhages are characterized by a central core of clotted blood that compresses the adjacent parenchyma; this compression leads to secondary infarction of the affected brain tissue, with anoxic neuronal and glial changes as well as edema. Eventually the edema resolves, hemosiderin- and lipid-laden macrophages appear, and proliferation of reactive astrocytes is seen at the periphery of the lesion; the cellular events then follow the same time course that is observed after cerebral infarction. Old hemorrhages show areas of parenchymal cavitory destruction with a rim of brownish discoloration.

Clinical Features

Intracerebral hemorrhage can be devastating if it involves a large part of the brain or extends into the ventricular system. When hemorrhage affects smaller regions, it is either silent or evolves like an infarct; over weeks or months, there is a gradual removal of the hematoma, sometimes with considerable clinical improvement. Again, the location of the hemorrhage determines the clinical manifestations.

Subarachnoid Hemorrhage and Ruptured Saccular Aneurysm

The most frequent cause of spontaneous subarachnoid hemorrhage is rupture of a saccular (“berry”) aneurysm in a cerebral artery. Non-traumatic subarachnoid hemorrhage may also result from rupture of a primary intracerebral hemorrhage into the ventricular system and subarachnoid space, vascular malformation, hematologic disturbances, and tumors. Trauma is another major cause of subarachnoid hemorrhage, usually in the context of other serious injuries.

Saccular aneurysm is the most common type of intracranial aneurysm; other aneurysm types include atherosclerotic (fusiform; mostly of the basilar artery), mycotic, traumatic, and dissecting. These latter three, like saccular aneurysms, are most often found in the anterior

circulation, but typically cause cerebral infarction rather than subarachnoid hemorrhage.

Saccular aneurysms are found in about 2% of the population according to community-based radiologic studies. About 90% of saccular aneurysms are found near major arterial branch points in the anterior circulation (Fig. 28.15); multiple aneurysms exist in 20% to 30% of cases.

Pathogenesis

Although the etiology of saccular aneurysms remains obscure, the structural abnormality of the involved vessels (absence of smooth muscle and internal elastic lamina) suggests they are developmental anomalies. The majority occur sporadically, but genetic factors may be important as there is an increased incidence in first-degree relatives of those affected. There is also an increased incidence in individuals with certain Mendelian disorders (e.g., autosomal dominant polycystic kidney disease, Ehlers-Danlos syndrome type IV, neurofibromatosis type 1 [NF1], and Marfan syndrome), fibromuscular dysplasia of extracranial arteries, and coarctation of the aorta. Other predisposing factors include smoking and hypertension (estimated to be present in about half of affected individuals). Endothelial dysfunction resulting from hemodynamic stress near the branch points of vessel walls may play an important role in aneurysm formation. Although the aneurysms are sometimes referred to as “congenital,” they are not present at birth but develop over time because of an underlying defect in the media of the vessel wall.

MORPHOLOGY

An unruptured saccular aneurysm is a thin-walled outpouching, usually at an arterial branch point along the circle of Willis or a major vessel just beyond. Saccular aneurysms measure from a few millimeters to 2 or 3 cm in diameter and have a bright red, shiny surface, and a thin, translucent wall (Fig. 28.16). Atheromatous plaques, calcification, or thrombi may be found in the wall or lumen of the aneurysm. Sometimes there is evidence of prior hemorrhage, in the form of orange-brown discoloration of the adjacent brain and meninges. The neck of the aneurysm may be wide or narrow. Rupture usually occurs at the apex of the sac and leads to extravasation of blood into the subarachnoid space, brain parenchyma, or both. The arterial wall adjacent to the neck of the aneurysm often shows some intimal thickening and attenuation of the media. Smooth muscle and internal elastic lamina do not extend into the neck and are absent from the aneurysm sac itself, which is made up of thickened hyalinized intima and a covering of adventitia.

Clinical Features

Rupture of an aneurysm leading to subarachnoid hemorrhage is most frequent in the fifth decade and is slightly more frequent in females. Overall, aneurysms rupture at a rate of 1.3% per year, but the risk is higher for larger aneurysms; for example, aneurysms greater than 10 mm in diameter have a roughly 50% risk of bleeding per year. Rupture may occur at any time, but in about one-third of cases it is associated with acute increases in intracranial pressure, such as with straining at stool or sexual orgasm. Blood under arterial pressure is forced into the subarachnoid space, and affected individuals are stricken with a sudden, excruciating headache (“the worst headache I’ve ever had”) and rapidly lose consciousness. Between 25% and 50% of patients die with the first rupture, but patients who survive often improve and recover consciousness in minutes. Repeat bleeding is common in survivors and unpredictable in timing. With each episode of bleeding, the prognosis is worse.

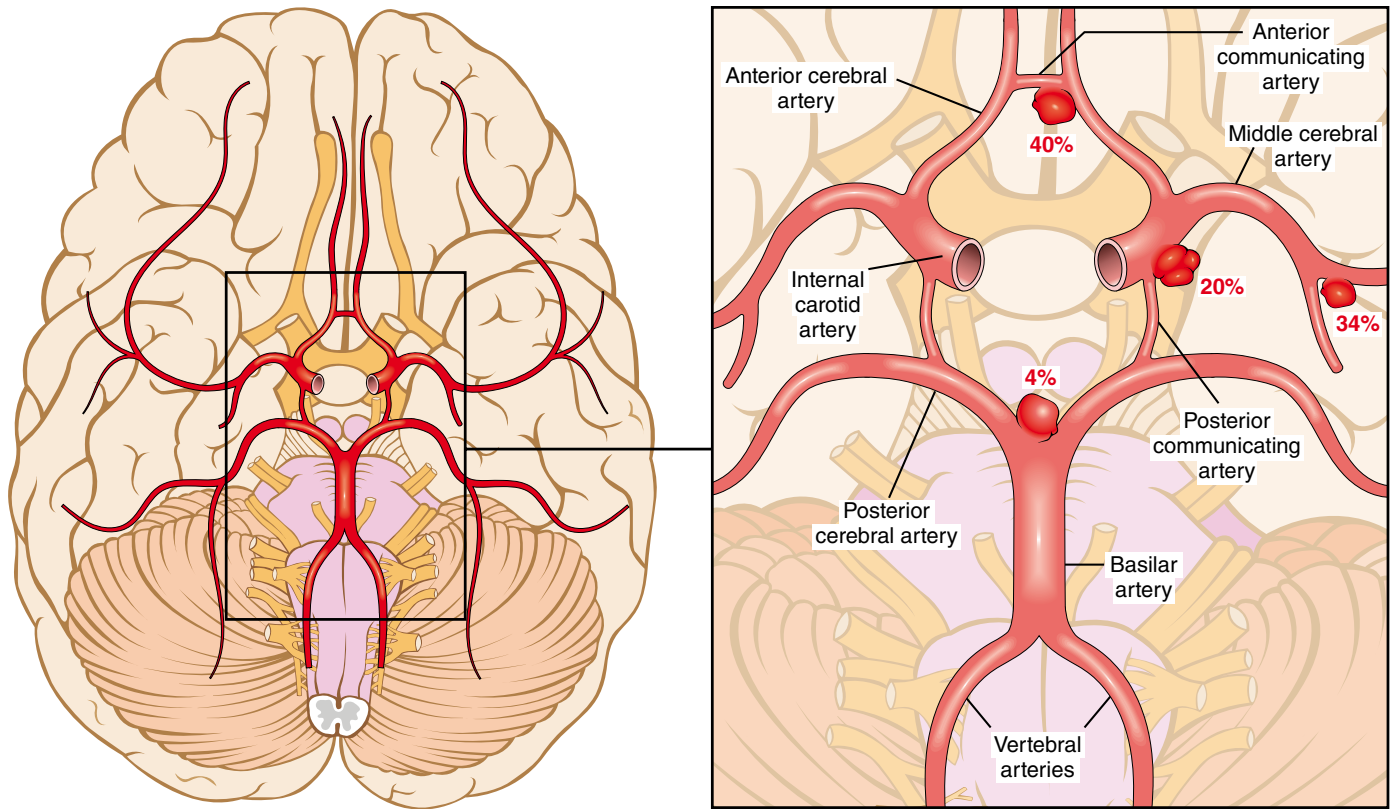


Fig. 28.15 Common sites of saccular (berry) aneurysms in the circle of Willis.

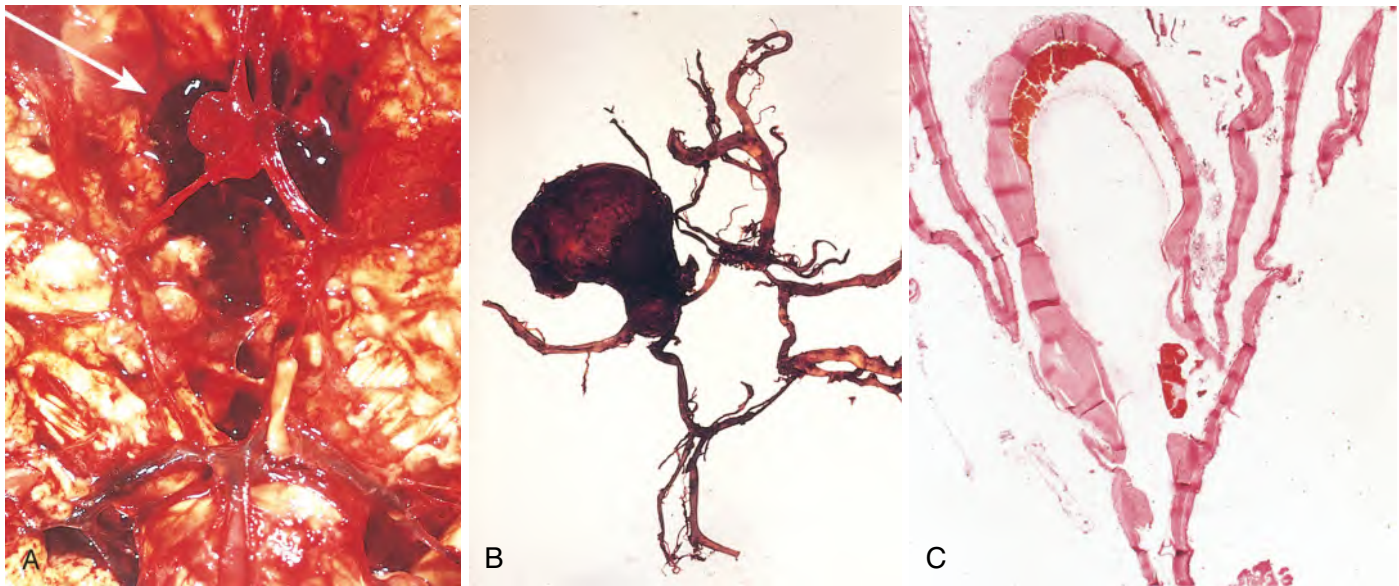


Fig. 28.16 Saccular (berry) aneurysm. (A) View of the base of the brain, dissected to show the circle of Willis with an aneurysm of the anterior cerebral artery (*arrow*). (B) Dissected circle of Willis to show large aneurysm. (C) Section through a saccular aneurysm showing the hyalinized fibrous vessel wall (hematoxylin and eosin).

The clinical consequences of blood in the subarachnoid space can be separated into acute events (occurring within hours to days after the hemorrhage) and late sequelae associated with the healing process. In the first few days after a subarachnoid hemorrhage, regardless of the etiology, there is an increased risk of additional ischemic injury from vasospasm affecting vessels bathed in the extravasated blood.

This problem is of greatest significance in cases of basal subarachnoid hemorrhage, in which vasospasm can involve major vessels of the circle of Willis. Various mediators have been proposed to have a role in this process, including endothelins, nitric oxide, and arachidonic acid metabolites. In the healing phase of subarachnoid hemorrhage, meningeal fibrosis and scarring occur, sometimes leading to

obstruction of CSF flow as well as interruption of the normal pathways of CSF resorption.

Vascular Malformations

Vascular malformations of the brain are classified into four principal groups: arteriovenous malformations, cavernous malformations, capillary telangiectasias, and venous angiomas. Of these, the first two are the types associated with risk of hemorrhage and development of neurologic symptoms. Although lacking morphologic features that typify neoplasms, arteriovenous malformations are frequently associated with activating somatic mutations in the *KRAS* oncogene within the endothelial cells that line the malformed vessels, suggesting that dysregulated RAS signaling has a role in their pathogenesis.

As with arteriovenous malformations, the pathogenesis of cavernous malformations involves a cancer-like accumulation of gene mutations in endothelial cells: rapidly growing, clinically significant cavernous malformations typically show not only loss-of-function mutations in genes that encode the components of a heterotrimeric cerebral cavernous malformation (CCM) protein complex (which limits vessel growth) but also gain-of-function mutations in the *PIK3CA* gene, which promotes vascular malformation.

MORPHOLOGY

Arteriovenous malformations are tangled networks of wormlike vascular channels with high blood flow due to prominent, pulsatile arteriovenous shunting that may involve vessels in the subarachnoid space, the brain, or both (Fig. 28.17). They are composed of greatly enlarged blood vessels separated by gliotic tissue, often with evidence of prior hemorrhage. Some vessels can be recognized as arteries with duplication and fragmentation of the internal elastic lamina, while others show marked thickening or partial replacement of the media by hyalinized connective tissue.

Cavernous malformations consist of distended, loosely organized vascular channels arranged back-to-back with collagenized walls of variable thickness; there is usually, but not invariably, no brain parenchyma between vessels in this type of malformation. They occur most often in the cerebellum, pons, and subcortical regions, in decreasing order of frequency, and are “low-flow” channels that do not participate in arteriovenous shunting. Foci of old hemorrhage, infarction, and calcification frequently surround the abnormal vessels.

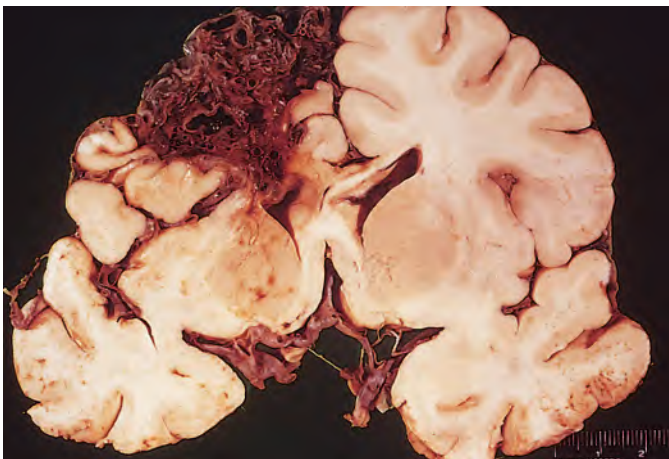


Fig. 28.17 Large arteriovenous malformation in the left cerebral hemisphere.

Clinical Features

Arteriovenous malformations are the most common clinically significant vascular malformations. The lesion often presents between 10 and 30 years of age as a seizure disorder, an intracerebral hemorrhage, or a subarachnoid hemorrhage. The most common site is the territory of the middle cerebral artery, particularly its posterior branches. Large arteriovenous malformations occurring in the newborn period can lead to congestive heart failure because of shunting of blood, especially if the malformation involves the vein of Galen. (In a vein of Galen malformation, misshapen arteries in the brain connect directly with veins instead of connecting with capillaries, which helps slow blood flow. This causes a rush of high-pressure blood into the veins.) With cavernous malformations, familial forms (which are caused by autosomal dominant mutations in genes that encode the subunits of the CCM protein complex) are relatively common; these patients generally have multiple lesions.

Vascular Dementia

Individuals who over the course of many months and years experience multiple, bilateral, gray matter and white matter infarcts may develop a distinctive syndrome characterized by dementia, gait abnormalities, and pseudobulbar signs, often with superimposed focal neurologic deficits. This syndrome, referred to as *vascular dementia*, is caused by multifocal vascular disease of several types, including (1) cerebral atherosclerosis, (2) vessel thrombosis or embolization from carotid vessels or from the heart, and (3) cerebral arteriolosclerosis from chronic hypertension. When the pattern of injury preferentially involves large areas of the subcortical white matter, the disorder is referred to as *Binswanger disease* (*subcortical arteriosclerotic encephalopathy*); this distribution of vascular white-matter injury must be distinguished clinically and radiologically from other diseases that affect the hemispheric white matter. In addition, many individuals with neurodegenerative diseases resulting in cognitive impairment or dementia also have evidence of cerebrovascular disease, which increases the risk of neurologic impairment for a given level of severity of degenerative lesions.

KEY CONCEPTS

Cerebrovascular Diseases

- Stroke is the clinical term for acute-onset neurologic deficits that last longer than 24 hours and that are a consequence of vessel occlusion or vessel rupture.
- Cerebral infarction follows loss of blood supply and can be focal, widespread, or restricted to regions with the least robust vascular supply (boundary zones).
- Large artery infarcts are most commonly embolic; with subsequent dissolution of the embolus and reperfusion, a nonhemorrhagic infarct can become hemorrhagic.
- Primary intraparenchymal hemorrhages lead to infarction of the adjacent brain parenchyma and are typically either ganglionic (most commonly due to hypertension) or lobar (most commonly due to cerebral amyloid angiopathy).
- Hyaline arteriolosclerosis (small vessel disease), which is caused by a long-standing hypertension, can lead to either lumen occlusion (resulting in a lacunar infarct) or wall rupture (resulting in intraparenchymal hemorrhage).
- Spontaneous subarachnoid hemorrhage is usually caused by a structural vascular abnormality, such as an aneurysm or arteriovenous malformation.

TRAUMA

The consequences of CNS trauma are determined by the location of the injury and the limited capacity of the brain to repair damage and recover function. Injury of several cubic centimeters of brain parenchyma

may be clinically silent (e.g., in the frontal lobe), severely disabling (e.g., in the spinal cord), or fatal (e.g., in the brainstem).

The physical forces associated with head injury may result in skull fractures, parenchymal injury, and vascular injury; all three can coexist. The magnitude and distribution of a traumatic brain lesion depends on the shape of the object causing the trauma, the force of impact, and whether the head is in motion at the time of injury. A blow to the head may be penetrating or blunt, and it may cause either an open or a closed injury.

Skull Fractures

A fracture in which bone is displaced into the cranial cavity by a distance greater than the thickness of the bone is called a *displaced skull fracture*. The thickness of the cranial bones varies; therefore, their resistance to fracture differs greatly. Also, the relative incidence of fractures among skull bones is related to the pattern of falls. For example, when an individual falls while awake (such as might occur when stepping off a ladder), the site of impact is often the occipital portion of the skull; in contrast, a fall that follows loss of consciousness (as might follow a syncopal attack) can result in either frontal or occipital impact. Symptoms referable to the lower cranial nerves or the cervicomedullary region, and the presence of orbital or mastoid hematomas distant from the point of impact, raise the suspicion of a basal skull fracture, which typically follows impact to the occiput or sides of the head. CSF leakage from the nose or ear and infection (meningitis) may follow.

Parenchymal Injuries

The structural consequences and clinical manifestations of parenchymal injury depend on the nature and severity of the causative head trauma and range from mild (concussion) to severe (contusions and lacerations).

Concussion

Concussion is a clinical syndrome of altered consciousness secondary to head injury, typically brought about by a change in the momentum of the head (e.g., following impact with a rigid object). The characteristic clinical picture includes the sudden onset of transient neurologic dysfunction, including loss of consciousness, temporary respiratory arrest, and loss of reflexes. Although neurologic recovery is usually complete, amnesia for the event often persists. The pathogenesis of the disruption of neurologic function is unknown, but likely involves dysregulation of the reticular activating system in the brainstem. Postconcussive neuropsychiatric syndromes, typically associated with repetitive injuries, are well recognized, and there is increasing evidence that significant cognitive impairment can emerge along with distinct pathologic findings termed chronic traumatic encephalopathy (discussed later).

Contusions and Lacerations

Contusions and *lacerations* are direct parenchymal injuries caused by transmission of kinetic energy to the brain. A contusion is analogous to the familiar bruise caused by blunt trauma, and a laceration is an injury caused by penetration of an object and tearing of tissue. As with any other organ, a blow to the surface of the brain, transmitted through the skull, leads to rapid tissue displacement, disruption of vascular channels, and subsequent hemorrhage, tissue injury, and edema. Hemorrhage can extend into the subarachnoid space from these lesions. The crests of gyri are most susceptible because this is where the direct force is greatest. The most common locations for contusions correspond to the most frequent sites of direct impact and to regions of the brain that overlie a rough and irregular inner skull surface, such as the frontal lobes along the orbital ridges and the temporal lobes. Contusions are less frequent over the

occipital lobes, brainstem, and cerebellum, but may be seen when there is an adjacent skull fracture (fracture contusions).

A person who sustains a blow to the head may develop a contusion at the point of contact (a *coup* injury) or on the brain surface diametrically opposite to it (a *contrecoup* injury). Their macroscopic and microscopic appearances are indistinguishable, and the distinction between them is based on identification of the point of impact. In general, if the head is immobile at the time of trauma, only a coup injury is found. If the head is mobile, both coup and contrecoup lesions may be found, though the latter predominate and are thought to develop when the brain strikes the opposite inner surface of the skull after sudden deceleration.

Sudden impacts that result in violent posterior or lateral hyperextension of the neck (as occurs when a pedestrian is struck from the rear by a vehicle) may avulse the pons from the medulla or the medulla from the cervical cord, causing instant death.

MORPHOLOGY

When seen on cross-section, contusions are wedge shaped, with the broad base lying along the surface at the point of impact (Fig. 28.18A). The appearance of contusions is similar regardless of the source of the trauma. In the earliest stages, there is edema and hemorrhage, which is often pericapillary. During the next few hours, the extravasation of blood extends throughout the

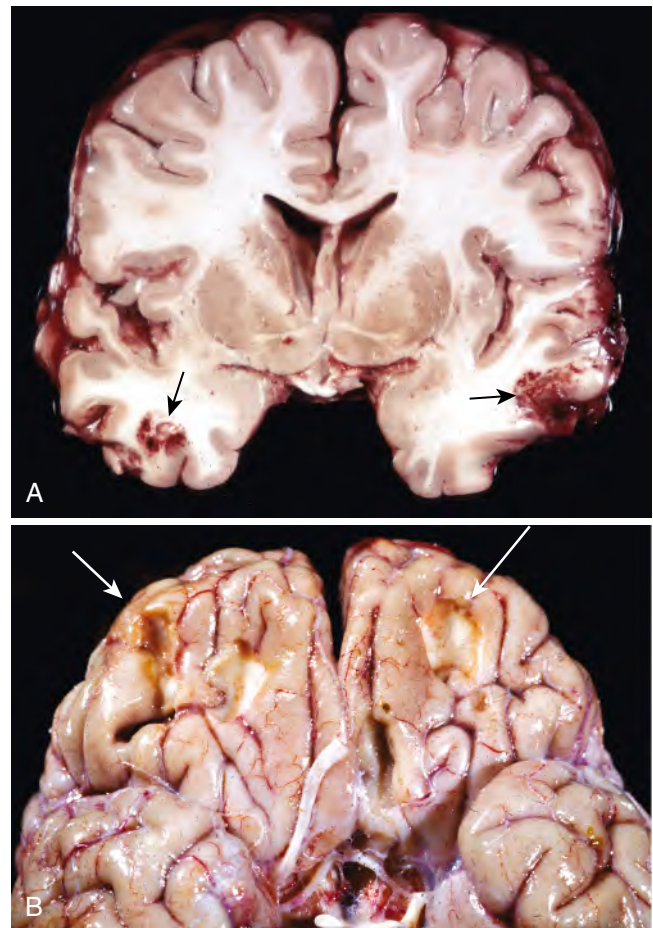


Fig. 28.18 Contusions. (A) Acute contusions are present in both temporal lobes, with areas of hemorrhage and tissue disruption (arrows). (B) Remote contusions (arrows) are present on the inferior frontal surface of this brain and have a yellow color (plaque jaune) that reflects hemosiderin accumulation.

involved tissue, across the width of the cerebral cortex, and into the white matter and subarachnoid space. Morphologic evidence of neuronal injury (pyknosis of the nucleus, eosinophilia of the cytoplasm, and disintegration of the cell) takes 6 to 12 hours to appear, although functional deficits generally occur earlier. Axonal swellings develop along the full axon length of the damaged neurons. The inflammatory response to the injured tissue follows its usual course, with the appearance of sparse neutrophils followed by abundant macrophages. Old traumatic lesions on the surface of the brain have a characteristic gross appearance. They are depressed, retracted, yellowish brown patches involving the crests of gyri, most commonly those that are located at the sites of contrecoup injuries (inferior frontal cortex, temporal and occipital poles). These lesions, called **plaque jaune** (Fig. 28.18B), can become epileptic foci. More extensive hemorrhagic regions of brain trauma give rise to larger cavitated lesions, which resemble remote infarcts. In old contusions, gliosis and residual hemosiderin-laden macrophages predominate.

Diffuse Axonal Injury

Traumatic brain injuries may also damage the deep white matter regions, cerebral peduncles, superior colliculi, and deep reticular formation in the brainstem. The microscopic findings include axonal swelling, indicative of *diffuse axonal injury*, and focal hemorrhagic lesions. As many as 50% of individuals who develop coma shortly after trauma, even without cerebral contusions, are believed to have diffuse axonal injury. Axons are injured directly by mechanical forces, with subsequent alterations in axoplasmic flow. Such injuries may occur from marked changes in angular acceleration (e.g., from blast injuries), even in the absence of physical impacts involving the skull.

MORPHOLOGY

Diffuse axonal injury is characterized by widespread, often asymmetric axonal swellings that appear within hours of the injury and may persist for much longer. The swelling is best demonstrated with silver impregnation techniques or with immunoperoxidase stains for axonally transported proteins, such as amyloid β precursor protein and α -synuclein. Later, increased numbers of microglia are seen in damaged areas of the cerebral cortex, and subsequently there is degeneration of the involved fiber tracts.

Traumatic Vascular Injury

Vascular injury is a frequent component of CNS trauma. It results from disruption of the vessel wall and leads to hemorrhage in different anatomic sites (see Table 28.1). Depending on the position of the ruptured vessel, hemorrhage may occur in the epidural, subdural, subarachnoid, and intraparenchymal compartments, sometimes in combination (Fig. 28.19). Epidural and subdural hemorrhages rarely occur outside of the setting of trauma; in some conditions, such as coagulopathy or significant cerebral atrophy, subdural hemorrhages can follow even minor trauma. Subarachnoid hemorrhage almost always accompanies parenchymal trauma but can also develop spontaneously secondary to vascular anomalies (discussed earlier).

Epidural Hematoma

The dura is fused with the periosteum and is supplied by several dural arteries. These arteries, most notably the middle meningeal artery, are vulnerable to traumatic injury. In adults, this most often occurs with temporal skull fractures in which the fracture crosses the course of the vessel. In children, in whom the skull is deformable, a temporary

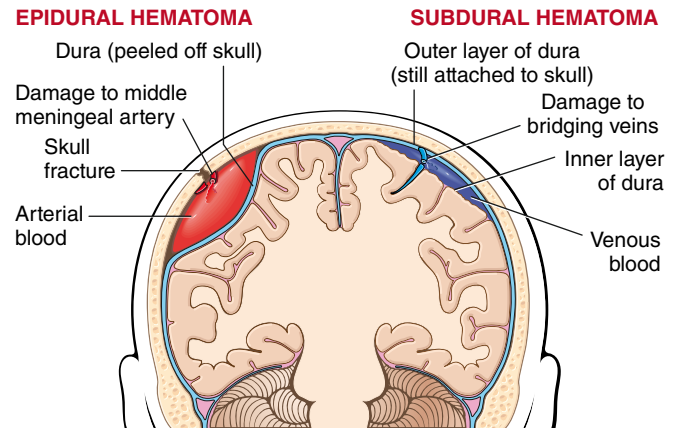


Fig. 28.19 Traumatic brain hemorrhage. Epidural hematoma (*left*) in which rupture of a meningeal artery, usually associated with a skull fracture, leads to accumulation of arterial blood between the dura and the skull. In a subdural hematoma (*right*), damage to bridging veins between the brain and the superior sagittal sinus leads to the accumulation of blood between the dura and the arachnoid.

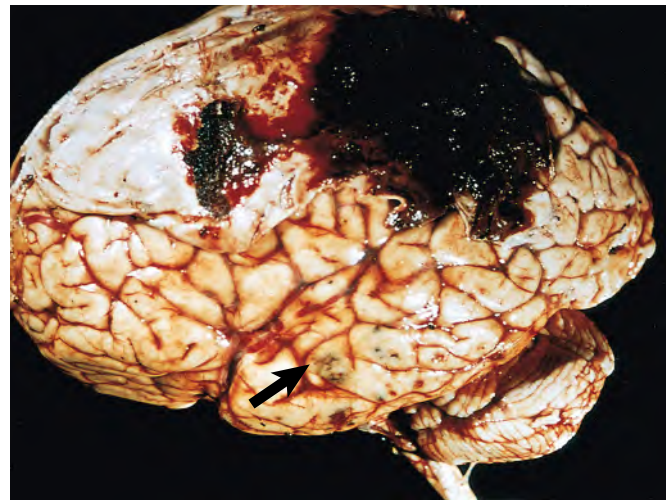


Fig. 28.20 Epidural hematoma covering a portion of the dura. Also present are multiple small contusions in the temporal lobe (arrow). (Courtesy Dr. Raymond D. Adams, Massachusetts General Hospital, Boston.)

displacement of the skull bones leading to laceration of a vessel can occur in the absence of a skull fracture.

Once a vessel has been torn, the extravasation of blood under arterial pressure can cause the dura to separate from the periosteum, creating a space (Fig. 28.20). The expanding hematoma compresses the underlying brain. When blood accumulates slowly, patients may experience a lucid period before the onset of neurologic signs. A symptomatic epidural hematoma is a neurosurgical emergency; without prompt diagnosis and drainage, fatal brain herniation may occur within a few hours.

Subdural Hematoma

The dura is composed of two layers—an external collagenous layer and an inner more cellular layer containing fibroblasts. Bridging veins travel from the convexities of the cerebral hemispheres through the subarachnoid space and dura to empty into the dural sinuses. The brain is suspended in CSF, but the venous sinuses are fixed relative to

the dura; as a result, traumatic displacement of the brain can tear the veins at the point where they penetrate the dura. The extravasated blood dissects through both layers of the dura, producing a subdural hematoma. In older individuals with brain atrophy, the bridging veins are stretched, hence the increasing incidence of subdural hematoma with aging. Infants are also very susceptible to subdural hematomas because their bridging veins are thin walled.

MORPHOLOGY

Grossly, **acute subdural hematomas** appear as a collection of freshly clotted blood along the brain surface, without extension into the depths of sulci (Fig. 28.21). The underlying brain is flattened, and the subarachnoid space is often clear. Usually, venous bleeding is self-limited, and the resulting hematoma is broken down and organized over time; this most often occurs in the following sequence:

- Lysis of the clot (about 1 week)
- Growth of fibroblasts from the dural surface into the hematoma (2 weeks)
- Early development of hyalinized connective tissue (1 to 3 months)

Typically, the organized hematoma is firmly attached to the inner surface of the dura by ingrowing fibrous tissue and is free of the underlying arachnoid, which does not contribute to healing. The lesion can eventually retract as the granulation tissue matures until only a thin layer of reactive connective tissue remains ("subdural membranes"). In other cases, however, multiple recurrent episodes of bleeding can occur (**chronic subdural hematoma**), presumably from the thin-walled vessels of the granulation tissue.

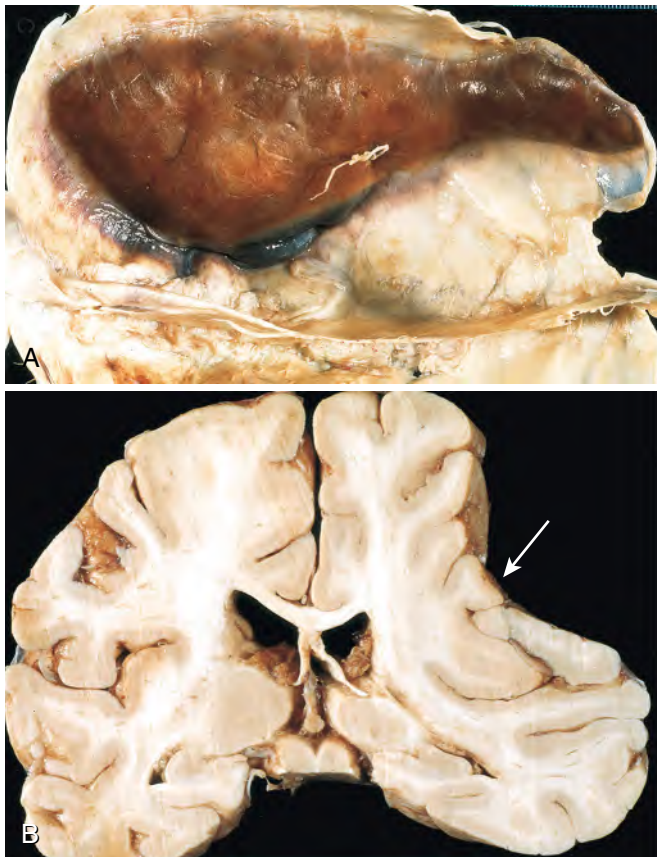


Fig. 28.21 Subdural hematoma. (A) Large organizing subdural hematoma attached to the dura. (B) Coronal section of the brain showing atrophy of the hemisphere compressed by the subdural hematoma shown in A (arrow).

Clinical Features

Symptomatic subdural hematomas typically manifest within 48 hours of injury. They are most common over the lateral aspects of the cerebral hemispheres and are bilateral in about 10% of cases. Neurologic signs are attributable to the pressure exerted on the adjacent brain. There may be focal signs, but often the clinical manifestations are nonlocalizing and include headache and confusion. Slowly progressive neurologic deterioration is typical, but acute decompensation may also occur. The treatment of subdural hematomas is to remove the blood and associated organizing tissue. The risk of repeat bleeding is greatest in the first few months after the initial hemorrhage.

Chronic Traumatic Encephalopathy

Chronic traumatic encephalopathy (CTE, previously referred to as "dementia pugilistica" or "boxer's dementia") is a dementing illness that develops after repeated head trauma; it has gained increasing notice in the context of litigation involving issues of compensation for professional athletes and individuals exposed to repetitive trauma in the civilian work force or during military service.

The presentation of CTE is variable, but includes behavioral, cognitive, mood, and motor symptoms. Affected brains are atrophic, with enlarged ventricles, and show accumulation of tau-containing neurofibrillary tangles (but not β amyloid plaques) in a characteristic pattern involving gyral depths and perivascular regions in the frontal and temporal lobe cortices; these are the areas that experience greatest physical stress and deformation during impact injury. White matter rarefaction and hippocampal sclerosis, with inclusions containing the protein TDP-43 (discussed later) in hippocampal neurons, can also develop. Although concussions are thought not to have obvious structural consequences, it is clear that repeated concussive events are antecedents to CTE. It is not fully established which factors determine whether encephalopathy will ultimately develop, but the length of time an individual participates in "risky" activities (which likely represents a proxy for the number and frequency of traumatic events) correlates with clinical severity and the CTE stage found at autopsy.

Other important sequelae of brain trauma include *posttraumatic hydrocephalus* (due to obstruction of CSF resorption by hemorrhage into the subarachnoid space), posttraumatic epilepsy, and increased infection risk.

Spinal Cord Injury

The spinal cord is highly vulnerable to trauma; most injuries that damage the cord are associated with the transient or permanent displacement of the vertebral column. The level of cord injury determines the extent of the neurologic manifestations: lesions involving the thoracic vertebrae or below can lead to paraplegia; cervical lesions result in quadriplegia; and those above C4 can, in addition, lead to respiratory compromise from paralysis of the diaphragm. Damage at the region of impact to descending and ascending white matter tracts isolates the distal spinal cord from the rest of the brain. This interruption, paired with the localized gray matter damage at the level of the impact, is the principal cause of neurologic deficits.

MORPHOLOGY

The histologic features of traumatic injury of the spinal cord resemble those found at other sites in the CNS. At the level of injury, the acute phase consists of hemorrhage, necrosis, and axonal swelling in the surrounding white matter. The lesion tapers above and below the level of injury. In time, central areas of neuronal destruction become cystic and gliotic; cord sections above and below the lesion show secondary ascending and descending Wallerian degeneration, respectively, involving the long white matter tracts.

KEY CONCEPTS

Trauma

- Physical injury to the brain occurs when the inside of the skull comes into forceful contact with the brain. In blunt trauma, if the head is mobile there may be brain injury at the original point of contact (coup injury) and on the opposite side of the brain (contrecoup injury).
- Parenchymal injuries take the form of contusions, with hemorrhage extending into the subarachnoid space.
- Rapid displacement of the head and brain can tear axons (diffuse axonal injury), often immediately causing severe, irreversible neurologic deficits.
- Traumatic tearings of blood vessels—mainly middle meningeal artery and bridging veins—lead to epidural or subdural hematoma, respectively.
- Repetitive head injury can lead to chronic traumatic encephalopathy, a tauopathy that results in neurobehavioral, cognitive, and less frequently motor symptoms.

PERINATAL BRAIN INJURY

Brain injury occurring in the perinatal period is an important cause of childhood-onset neurologic disability. Injuries that occur early in gestation may destroy brain tissue without eliciting the reactive changes observed in adult brains and, therefore, may be difficult to distinguish from malformations. Different patterns of injury may occur.

- The term *cerebral palsy* refers to a nonprogressive neurologic motor deficit characterized by spasticity, dystonia, ataxia/athetosis, and paresis, attributable to brain injury occurring during the prenatal and perinatal periods. Signs and symptoms may not be apparent at birth and only declare themselves later, as development proceeds. Postmortem examinations of children with cerebral palsy have shown a wide range of neuropathologic findings, including destructive lesions traced to remote events that may have caused hemorrhage and infarction.
- In premature infants, brain injury and hypoxic-ischemic events are associated with an increased risk of *germinal matrix hemorrhage*, often near the junction between the developing thalamus and caudate nucleus. Hemorrhages may remain small and localized or extend into the ventricular system and subarachnoid space, sometimes leading to hydrocephalus and death in severe cases.
- Infarcts may occur in the supratentorial periventricular white matter (*periventricular leukomalacia*), especially in premature infants. They take the form of chalky yellow plaques that consist of discrete white matter necrosis, often with dystrophic calcification. Ultimately, the infarcted areas develop into large cystic lesions (Fig. 28.22); when damage is extensive and involves both gray and white matter, the condition is termed *multicystic encephalopathy*.
- In *perinatal ischemic lesions of the cerebral cortex*, the depths of sulci bear the brunt of injury and result in mushroom-shaped gyri with thinned-out, gliotic stalks (*ulegyria*). The basal ganglia and thalamus may also sustain ischemic injury, with patchy neuronal loss and reactive gliosis. Later, aberrant and irregular myelinization gives rise to a marble-like appearance of the deep nuclei (*status marmoratus*). Because the lesions are in the caudate, putamen, and thalamus, movement disorders such as choreoathetosis are common clinical sequelae.

KEY CONCEPTS

Perinatal Brain Injury

- Timing of injury is critical, with earlier events resulting in greater damage and deficits.
- Cerebral palsy is the term used for nonprogressive deficits associated with injury during the prenatal and perinatal periods.
- Although they can be difficult to distinguish from developmental malformations, perinatal brain injuries are typically caused by prenatal or perinatal ischemia.

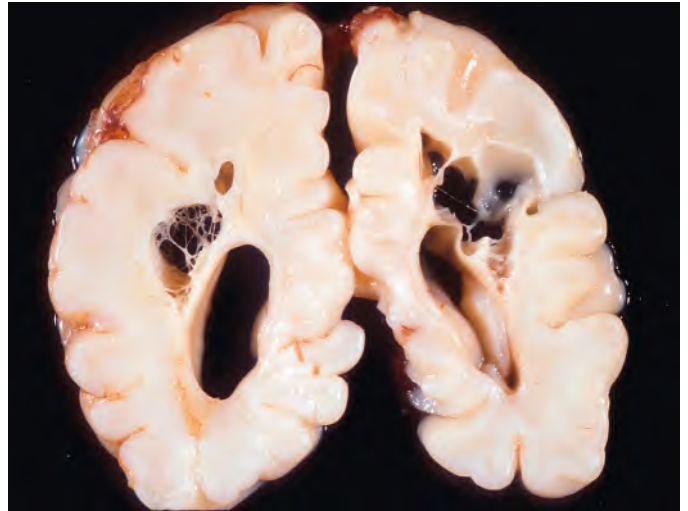


Fig. 28.22 Chronic stage of periventricular leukomalacia. Large cystic spaces in the periventricular white matter (seen in both hemispheres of this brain) are the long-term sequelae of a severe prenatal or perinatal ischemic injury.

INFECTIONS

Infection may damage the nervous system directly through injury of neurons or glia by the infectious agent, or indirectly through microbial toxins, the destructive effects of the inflammatory response, or via immune-mediated mechanisms. There are four principal routes by which microbes enter the nervous system:

- Hematogenous spread* is the most common; infectious agents ordinarily gain access through the arterial circulation, but retrograde venous spread can occur via anastomoses with veins of the face.
- Direct implantation* of microorganisms is most often due to trauma but can sometimes stem from congenital malformations (e.g., meningocele) that provide ready access for microorganisms.
- Local extension* can originate from infected adjacent structures, such as air sinuses, teeth, skull, or vertebrae.
- Viruses also may be *transported along the peripheral nervous system*, as occurs with rabies and herpes zoster viruses.

General aspects of the pathology of infectious agents are discussed in Chapter 8; distinctive forms of CNS infections are described herein (Table 28.2).

Acute Meningitis

Meningitis is an inflammatory process involving the leptomeninges and CSF within the subarachnoid space, usually caused by an infection. *Meningoencephalitis* refers to inflammation of the meninges and brain parenchyma (most commonly viral). Although infections are the most common causes of meningitis and meningoencephalitis, this reaction may also occur in response to an irritant introduced into the subarachnoid space (*chemical meningitis*) or in the context of a systemic autoimmune disease. Based on the etiology and clinical evolution of the illness, infectious meningitis is broadly classified as *acute pyogenic* (usually bacterial), *aseptic* (usually acute or subacute viral), and *chronic* (usually tuberculous, spirochetal, or cryptococcal). Each type is accompanied by characteristic changes in the CSF.

Acute Pyogenic (Bacterial) Meningitis

Distinctive microorganisms cause acute pyogenic meningitis in various age groups: *Escherichia coli* and the group B streptococci in neonates; *Streptococcus pneumoniae* and *Listeria monocytogenes* in the

Table 28.2 Common Central Nervous System Infections

Type of Infection	Clinical Disease	Common Causative Organisms
Bacterial Infections		
Meningitis	Acute pyogenic meningitis	<i>Escherichia coli</i> or group B streptococci (infants)
		<i>Neisseria meningitidis</i> (young adults)
		<i>Streptococcus pneumoniae</i> or <i>Listeria monocytogenes</i> (older adults)
	Chronic meningitis	<i>Mycobacterium tuberculosis</i>
Localized infections	Abscess	Streptococci and staphylococci
	Empyema	Polymicrobial (staphylococci, anaerobic gram-negative)
Viral Infections		
Meningitis	Acute aseptic meningitis	Enteroviruses
		Influenza species
		Lymphocytic choriomeningitis virus
Encephalitis	Encephalitic syndromes	Herpes simplex (HSV-1, HSV-2)
		Cytomegalovirus
		Human immunodeficiency virus
		JC polyomavirus (progressive multifocal leukoencephalopathy)
	Arthropod-borne encephalitis	West Nile virus
		Eastern equine encephalitis virus
		Western equine encephalitis virus
		St. Louis encephalitis virus
		La Crosse encephalitis virus
		Venezuelan equine encephalitis virus
		Japanese encephalitis virus
		Tick-borne encephalitis virus
		Brainstem and spinal cord syndromes
Acute flaccid myelitis/ poliomyelitis	Poliovirus	
	West Nile virus	
	Enterovirus D68	
Spirochetes and Fungi		
Meningoencephalitic syndromes	Neurosyphilis	<i>Treponema pallidum</i>
	Lyme disease (neuroborreliosis)	<i>Borrelia burgdorferi</i>
	Fungal meningitis	<i>Cryptococcus neoformans</i> <i>Candida albicans</i>
Protozoa and Metazoa		
Encephalitis	Amoebic encephalitis	<i>Balamuthia</i> and <i>Acanthamoeba</i> species
		<i>Naegleria</i> species
Localized infections	Toxoplasmosis	<i>Toxoplasma gondii</i>
	Cysticercosis	<i>Taenia solium</i>

elderly; and *Neisseria meningitidis* in adolescents and young adults, with clusters of cases raising public health concerns. The introduction of immunization against *Haemophilus influenzae* has markedly reduced the incidence of this infection in high-resource countries, particularly among infants (who used to be at high risk).

Affected individuals typically show systemic signs of infection superimposed on symptoms related to meningeal irritation and neurologic impairment, including headache, photophobia, irritability, clouding of consciousness, and neck stiffness. A lumbar puncture yields cloudy or frankly purulent CSF with numerous neutrophils, increased CSF pressure, increased protein concentration, and markedly reduced glucose content. Untreated pyogenic meningitis can be fatal, while effective treatment with antibiotics sharply diminishes mortality. One feared complication is the *Waterhouse-Friderichsen syndrome*, which results from meningitis-associated septicemia and hemorrhagic infarction of the adrenal glands (Chapter 24). It occurs most often with meningococcal and pneumococcal meningitis. In the immunosuppressed individual, purulent meningitis may be caused by several other infectious agents, such as *Klebsiella* or anaerobic organisms, and may have an atypical clinical course and uncharacteristic CSF findings, rendering timely diagnosis more difficult.

MORPHOLOGY

In **acute meningitis**, an exudate is evident within the leptomeninges over the surface of the brain (Fig. 28.23). The meningeal vessels are engorged and stand out prominently. The anatomic distribution of the exudate varies; in *H. influenzae* meningitis, for example, it is usually basal, whereas in pneumococcal meningitis it is often densest over the cerebral convexities near the sagittal sinus. From the areas of greatest accumulation, tracts of pus follow along blood vessels on the surface of the brain. When the meningitis is fulminant, the inflammation may extend to the ventricles, producing ventriculitis. The ventricles may also be the portal for CSF involvement by blood-borne infections because the choroid plexus lacks a blood-brain barrier.

On microscopic examination, neutrophils fill the subarachnoid space in severely affected areas and are found predominantly around the leptomeningeal blood vessels in less severe cases. Particularly in untreated meningitis, Gram stain reveals variable numbers of bacteria. Although pia is rarely breached in bacterial meningitis, in fulminant cases the inflammatory cells (mostly neutrophils) infiltrate the walls of the leptomeningeal veins and may extend focally into the brain parenchyma (cerebritis); secondary vasculitis and venous thrombosis may lead to hemorrhagic cerebral infarction.

Leptomeningeal fibrosis may follow pyogenic meningitis and cause hydrocephalus. Particularly in pneumococcal meningitis, large quantities of the organism-derived capsular polysaccharide produce a gelatinous exudate that promotes arachnoid fibrosis, a condition referred to as **chronic adhesive arachnoiditis**.

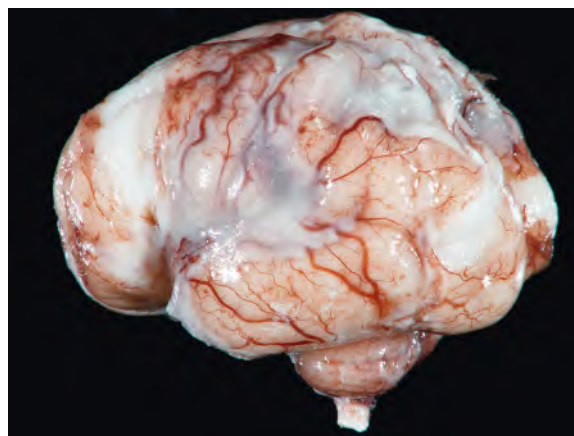


Fig. 28.23 Pyogenic meningitis. A thick layer of suppurative exudate covers the brain surface and thickens the leptomeninges.

Acute Aseptic (Viral) Meningitis

Aseptic meningitis is a clinical term for a disorder with manifestations of meningitis (meningeal irritation, fever, and alterations of consciousness of relatively acute onset) but no organisms identifiable by bacterial culture. The disease is generally of viral etiology, but may be bacterial, rickettsial, or autoimmune in origin. The clinical course is less fulminant than that of pyogenic meningitis, and the CSF findings also differ; in aseptic meningitis, there is a lymphocytic pleocytosis, the protein elevation is only moderate, and the glucose content is nearly always normal. The viral aseptic meningitides are usually self-limited and are treated symptomatically. Remarkably, the etiologic agent is identified in only a minority of cases; however, this may change through use of more sensitive and specific detection techniques (such as next-generation sequencing), which are under development. When pathogens are identified, enteroviruses are the most common etiology, accounting for 80% of cases. The spectrum of pathogens varies seasonally and geographically. An aseptic meningitis-like picture may also develop following rupture of an epidermoid cyst into the subarachnoid space or the introduction of a chemical irritant (chemical meningitis). The CSF is sterile in these cases, and there is neutrophilic pleocytosis and an increased protein concentration, but the sugar content is usually normal.

Acute Focal Suppurative Infections

Focal suppurative infections are typically caused by pyogenic bacteria or fungi and can arise in several different compartments including the brain parenchyma, subdural space, and extradural space.

Brain Abscess

A brain abscess is a localized focus of necrosis with accompanying acute inflammation, usually caused by a bacterial infection, although fungal abscess is encountered on occasion. Brain abscesses may arise by direct implantation of organisms, local extension from adjacent foci (mastoiditis, paranasal sinusitis), or hematogenous spread (usually from a primary site in the heart, lungs, or bones, or secondary to bacteremia from dental procedures). Predisposing conditions include acute bacterial endocarditis, which may give rise to multiple brain abscesses; congenital heart disease with right-to-left shunting and loss of pulmonary filtration of organisms; chronic pulmonary infection, as in bronchiectasis; and systemic disease with immunosuppression. Streptococci and staphylococci are the most common causative organisms identified in patients who are not immunosuppressed.

MORPHOLOGY

Abscesses are discrete lesions with central pus and liquefactive necrosis surrounded by brain swelling (Fig. 28.24). At the outer margin of the necrotic lesion, there is exuberant neovascularization. The newly formed vessels are abnormally permeable, accounting for marked vasogenic edema in the adjacent brain tissue. In well-established lesions, a collagenous capsule is produced by fibroblasts derived from the walls of blood vessels. Outside the fibrous capsule is a zone of reactive gliosis containing numerous gemistocytic astrocytes.

Clinical Features

Cerebral abscesses are destructive lesions, and patients often present with progressive focal neurologic deficits; signs and symptoms related to increased intracranial pressure may also develop. On MRI, they are rim-enhancing masses similar to glioblastomas and metastatic cancers; however, the central cavity of an abscess typically shows

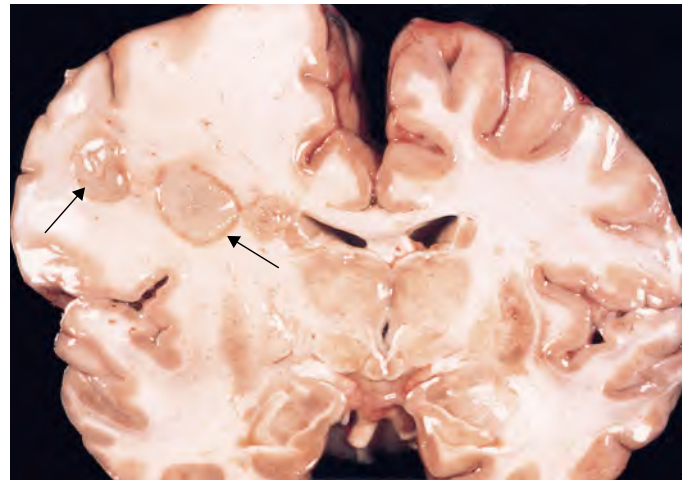


Fig. 28.24 Cerebral abscesses (arrows).

restricted diffusion, in contrast to most tumors. Typically, the CSF has a high white cell count and increased protein concentration, but the glucose content is normal. The source of infection may be apparent or may be traced to a small distant focus that is not symptomatic. The increased intracranial pressure can lead to fatal herniation; other complications include abscess rupture with ventriculitis or meningitis, and venous sinus thrombosis. With surgery and antibiotic treatment, the otherwise high mortality rate can be reduced to less than 10%.

Subdural Empyema

Bacterial, and rarely fungal, infections of the skull bones or air sinuses can spread to the subdural space, producing a subdural empyema. Although the underlying arachnoid and subarachnoid spaces are usually not affected, a large subdural empyema with mass effect and/or thrombophlebitis of the bridging veins can cause venous occlusion and infarction of the brain. In addition to symptoms referable to the source of the infection, most patients are febrile and have headache and neck stiffness. The CSF profile is similar to that seen in brain abscesses because both are parameningeal infectious processes. If untreated, focal neurologic signs, lethargy, and coma may develop. With prompt diagnosis and treatment, including surgical drainage, resolution and full recovery is possible, the only residuum being a thickened dura.

Extradural Abscess

Extradural abscess, commonly associated with osteomyelitis, often arises from an adjacent focus of infection, such as sinusitis, or following a surgical procedure. When the process occurs in the spinal epidural space, it may cause spinal cord compression and constitute a neurosurgical emergency.

Chronic Bacterial Meningoencephalitis

Chronic bacterial infection of the meninges and the brain may be caused by *Mycobacterium tuberculosis*, *Treponema pallidum*, and *Borrelia* species.

Tuberculosis

Tuberculosis of the CNS may be part of known active disease or appear in isolation following seeding from silent lesions in other organs, usually the lungs. It may involve the meninges or the brain.

MORPHOLOGY

The most common pattern of tuberculous involvement of the CNS is a diffuse **meningoencephalitis**. The subarachnoid space contains a gelatinous or fibrinous exudate that characteristically involves the base of the brain, effacing the cisterns and encasing cranial nerves. There may be discrete, white areas of inflammation scattered over the leptomeninges. On microscopic examination, involved areas contain a mixed inflammatory infiltrate with lymphocytes, plasma cells, and macrophages. Florid cases show well-formed granulomas with caseous necrosis and giant cells. Arteries running through the subarachnoid space may show obliterative endarteritis and marked intimal thickening. Organisms may be seen with acid-fast stains but are often rare and require a painstaking search by the pathologist. The infectious process may spread to the choroid plexus and ependymal surface, traveling through the CSF. In long-standing cases, a dense, fibrous adhesive arachnoiditis may develop, most conspicuous around the base of the brain. Hydrocephalus may result.

CNS involvement may also take the form of one or more well-circumscribed intraparenchymal masses (**tuberculomas**), which may be associated with meningitis. A tuberculoma may be as large as several centimeters in diameter, causing significant mass effect. These granulomas usually have central caseous necrosis; calcification may occur in inactive lesions.

Clinical Features

Patients with tuberculous meningitis usually have headache, malaise, mental confusion, and vomiting. The CSF typically shows a pleocytosis made up of mononuclear cells or a mixture of neutrophils and mononuclear cells, an elevated protein concentration (often strikingly so), and a moderately reduced or normal glucose. The most serious complications of chronic tuberculous meningitis are arachnoid fibrosis (leading to hydrocephalus) and obliterative endarteritis (marked intimal thickening leading to arterial occlusion and brain infarction). When the process involves the spinal cord subarachnoid space, nerve roots may also be affected. Tuberculomas produce symptoms typical of space-occupying brain lesions and must be distinguished from CNS tumors.

CNS tuberculosis in patients with acquired immunodeficiency syndrome (AIDS) is pathologically similar, but there is usually less host reaction than in immunocompetent individuals.

Neurosyphilis

Neurosyphilis is a manifestation of the tertiary stage of syphilis. It occurs in about 10% of individuals with untreated infection but is rarely observed in the modern era of effective antibiotic therapy. The major patterns of CNS involvement are meningovascular neurosyphilis, parietic neurosyphilis, and tabes dorsalis. Affected individuals often show an incomplete or mixed picture, most commonly the combination of tabes dorsalis and parietic disease (taboparesis). Because of impaired cell-mediated immunity, individuals infected with HIV are at increased risk for neurosyphilis, particularly acute syphilitic meningitis or meningovascular disease. The rate of disease progression and severity are also accelerated in those with HIV.

MORPHOLOGY

Neurosyphilis presents in several distinct forms.

- **Meningovascular neurosyphilis** is chronic meningitis involving the base of the brain and more variably the cerebral convexities and spinal leptomeninges. In addition, there may be an associated obliterative endarteritis (Heubner arteritis) accompanied by a distinctive perivascular inflammatory reaction rich in plasma cells and lymphocytes. Cerebral gummas (plasma

cell-rich mass lesions) may also occur in the meninges and extend into the parenchyma.

- **Paretic neurosyphilis** is chronic meningoencephalitis caused by invasion of the brain by *T. pallidum* that manifests as insidious but progressive cognitive impairment associated with mood alterations (including delusions of grandeur) that terminate in severe dementia (**"general paresis of the insane," paralytic dementia**). Parenchymal damage of the cerebral cortex is particularly common in the frontal lobe, but also occurs in other areas of the neocortex. The lesions are characterized by loss of neurons, proliferation of microglia, gliosis, and iron deposits; the latter are demonstrable with the Prussian blue stain perivascularly and in the neuropil and are presumably the sequelae of small bleeds stemming from microvascular damage. The spirochetes can, at times, be demonstrated in tissue sections.
- **Tabes dorsalis** is the result of damage to the sensory axons in the dorsal roots. This causes impaired joint position sense and ataxia (locomotor ataxia); loss of pain sensation, leading to skin and joint damage (Charcot joints); other sensory disturbances, particularly the characteristic "lightning pains"; and absence of deep tendon reflexes. On microscopic examination, there is loss of both axons and myelin in the dorsal roots, with corresponding pallor and atrophy in the dorsal columns of the spinal cord. Organisms are not demonstrable in the cord lesions.

Neuroborreliosis (Lyme Disease)

Lyme disease is caused by the spirochete *Borrelia burgdorferi*, which is transmitted by various species of *Ixodes* tick (Chapter 8). Involvement of the nervous system is referred to as **neuroborreliosis**. Neurologic symptoms are highly variable and include aseptic meningitis, facial nerve palsies and other polyneuropathies, as well as encephalopathy. The rare cases that have come to autopsy have shown a focal proliferation of microglial cells in the brain as well as scattered extracellular organisms.

Viral Meningoencephalitis

Viral encephalitis is a parenchymal infection of the brain almost invariably associated with meningeal inflammation (meningoencephalitis) and sometimes with simultaneous involvement of the spinal cord (encephalomyelitis).

Some viruses have a propensity to infect the nervous system. Such neural tropism takes several forms: some infect specific cell types (e.g., oligodendrocytes), while others preferentially involve particular areas of the brain (e.g., medial temporal lobes or the limbic system). Latency is an important phase of several viral infections of the CNS (e.g., herpes zoster and progressive multifocal leukoencephalopathy). Systemic viral infections in the absence of direct evidence of viral penetration into the CNS may be followed by an immune-mediated disease, such as perivenous demyelination. Viral infection of the fetus may cause congenital malformations, as occurs with rubella and Zika viruses. A slowly progressive degenerative disease syndrome may develop many years after a viral illness; an example is postencephalitic parkinsonism after the 1918 viral influenza pandemic.

Arthropod-Borne Viral Encephalitis

Arboviruses are an important cause of epidemic encephalitis, especially in tropical regions of the world, and are capable of causing serious morbidity and high mortality. In the Western hemisphere, the most important arboviruses causing encephalitis are Eastern and Western equine, West Nile, Venezuelan, St. Louis, and La Crosse; elsewhere in the world, pathogenic arboviruses include Japanese B (East Asia), Murray Valley (Australia and New Guinea), and tick-borne (Russia and Eastern Europe).

All these viruses have animal hosts and insect vectors. Clinically, affected individuals develop generalized neurologic deficits, such as seizures, confusion, delirium, and stupor or coma, as well as focal signs, such as reflex asymmetry and ocular palsies. Involvement of the spinal cord in West Nile encephalitis can lead to a polio-like syndrome with paralysis. In general, the CSF is usually colorless, with slightly elevated pressure, an elevated protein level, and a normal glucose reading. Initially the CSF exhibits a neutrophilic pleocytosis, but this rapidly converts to a lymphocytosis.

MORPHOLOGY

The encephalitides caused by various arboviruses produce similar histopathologic changes that vary only in severity and extent. Characteristically, there is a meningoencephalitis marked by the perivascular accumulation of lymphocytes (Fig. 28.25A). Multiple foci of necrosis of gray and white matter are found; in particular, there is evidence of single-cell neuronal necrosis with phagocytosis of the debris (**neuronophagia**). Microglial cells form small aggregates, called **microglial nodules** (Fig. 28.25B). In severe cases, there may be a necrotizing vasculitis with associated focal hemorrhages. Although some viruses declare their presence by formation of intracellular inclusions, the causative virus is most often identified in tissue samples by a combination of ultrastructural, immunohistochemical, and molecular methods.

Herpes Simplex Virus Type 1

Herpes simplex virus type 1 (HSV-1) encephalitis occurs most commonly in children and young adults; only about 10% of affected individuals have a history of prior herpetic infection. The typical presenting symptoms are alterations in mood, memory, and behavior. Antiviral agents now provide effective treatment in many cases, with a significant reduction in the mortality rate. In some individuals, HSV-1 encephalitis follows a subacute course with clinical manifestations (weakness, lethargy, ataxia, seizures) that evolve during a more protracted period (4 to 6 weeks).

MORPHOLOGY

This encephalitis starts in and most severely involves the inferior and medial regions of the temporal lobes and the orbital gyri of the frontal lobes (Fig. 28.26A). The infection is necrotizing and often hemorrhagic in the most

severely affected regions. Perivascular inflammatory infiltrates are usually present, and Cowdry type A intranuclear viral inclusions may be found in both neurons and glia (Fig. 28.26B). In individuals with slowly evolving HSV-1 encephalitis, there is more diffuse involvement of the brain.

Herpes Simplex Virus Type 2

Herpes simplex virus type 2 (HSV-2) can infect the nervous system. In adults it causes meningitis, but as many as 50% of neonates born by vaginal delivery to females with active primary HSV genital infections are infected during passage through the birth canal and develop severe encephalitis. In individuals with active HIV infection, HSV-2 may cause an acute hemorrhagic and necrotizing encephalitis.

Varicella-Zoster Virus

Primary infection with varicella-zoster virus causes one of the childhood exanthems (*chickenpox*), without evidence of neurologic involvement. Following the cutaneous infection, the virus enters a latent phase within sensory neurons of the dorsal root or trigeminal ganglia. Reactivation of infection in adults (*shingles* or *herpes zoster*) usually manifests as a painful, vesicular skin eruption confined to one or several dermatomes. Herpes zoster is typically self-limited, but there may be a postherpetic neuralgia syndrome (particularly in people over 60 years of age) characterized by persistent pain, sometimes induced by stimuli that are otherwise not painful. Vaccination is now available to prevent these complications of varicella-zoster virus reactivation.

Cytomegalovirus

Cytomegalovirus (CMV) infection of the nervous system occurs in fetuses and immunosuppressed individuals. The outcome of infection in utero is periventricular necrosis that produces severe brain injury, followed later by microcephaly and periventricular calcification. CMV is a common opportunistic viral pathogen in individuals with AIDS, with CNS involvement also occurring in this setting.

MORPHOLOGY

In the immunosuppressed individual, CMV most commonly causes subacute encephalitis, which may be associated with CMV inclusion-bearing cells (see Fig. 8.15). The infection tends to localize in the subependymal regions

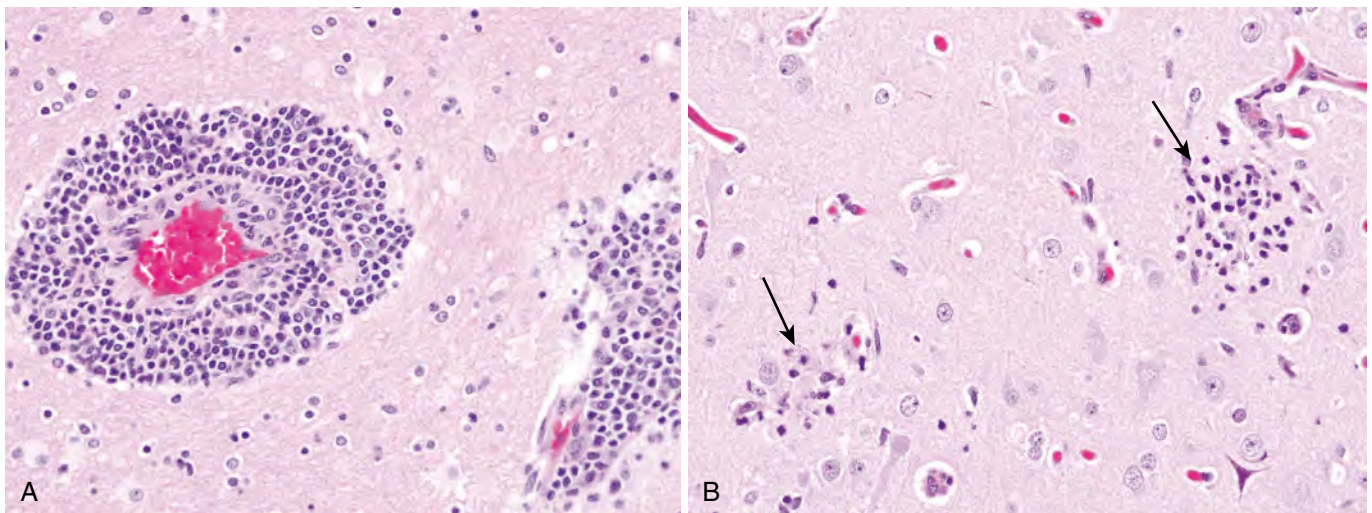


Fig. 28.25 Viral encephalitis. Characteristic findings include (A) perivascular cuffs of lymphocytes and (B) microglial nodules (arrows).

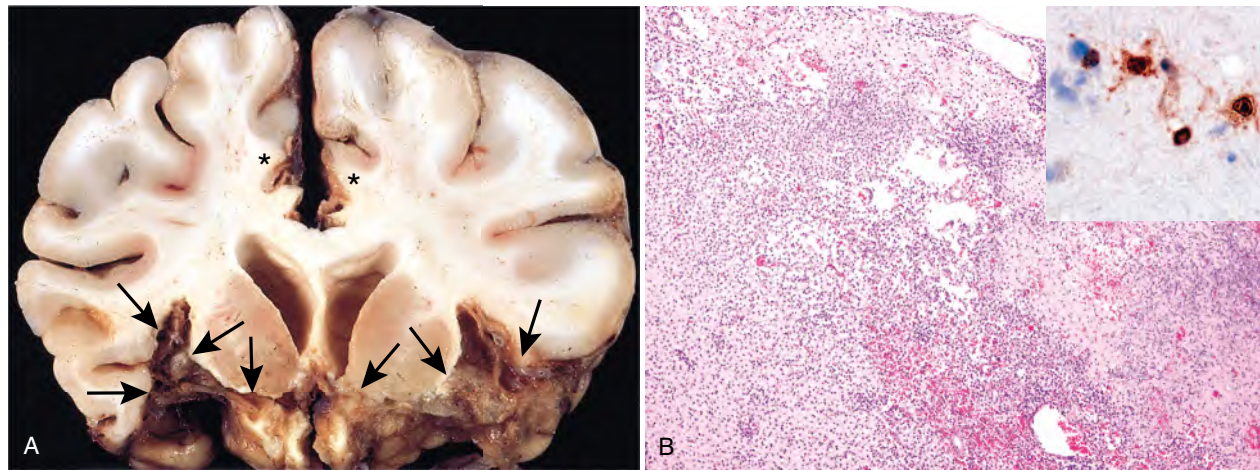


Fig. 28.26 Herpes encephalitis. (A) Extensive destruction of inferior frontal and anterior temporal lobes (arrows) and the cingulate gyri (asterisks). (B) Necrotizing inflammatory process characterizes acute herpes encephalitis; nuclear viral inclusions can be highlighted by immunostaining (inset). (A, Courtesy Dr. T.W. Smith, University of Massachusetts Medical School, Worcester.)

of the brain, where it results in a severe hemorrhagic necrotizing ventriculo-encephalitis and choroid plexitis. The virus can also attack the lower spinal cord and roots, producing a painful radiculoneuritis. Any cell in the CNS (neurons, glia, ependyma, or endothelium) may be infected. Prominent enlarged cells with intranuclear and intracytoplasmic inclusions can be readily identified by conventional light microscopy. CMV infection is confirmed by immunohistochemistry.

Poliomyelitis

Although paralytic poliomyelitis has been eradicated by vaccination in many parts of the world, there are still a few countries where it remains a serious problem. In addition, a cluster of cases of acute flaccid myelitis (a polio-like syndrome) has recently occurred in the United States and seems to have been caused by a novel strain of enterovirus D68 (which usually causes only a mild respiratory infection). In nonimmunized individuals, poliovirus infection causes a subclinical or mild gastroenteritis, similar to that caused by other members of the picornavirus group of enteroviruses. In a small fraction of the vulnerable population, however, the virus secondarily invades the nervous system.

MORPHOLOGY

Acute cases show perivascular mononuclear cell cuffs and neuronophagia of the anterior horn motor neurons of the spinal cord. The inflammatory reaction is usually confined to the anterior horns but may extend into the posterior horns, and the damage is occasionally severe enough to produce cavitation. Poliovirus RNA has been detected in anterior horn cell motor neurons; the cranial motor nuclei are sometimes involved as well. Postmortem examination in long-term survivors of symptomatic poliomyelitis shows loss of neurons and gliosis in the affected anterior horns of the spinal cord, atrophy of the anterior (motor) spinal roots, and neurogenic atrophy of denervated skeletal muscles.

Clinical Features

CNS infection manifests initially with meningeal irritation and a CSF picture consistent with aseptic meningitis; the disease may progress no further or may advance to involve the spinal cord. When the disease affects the motor neurons of the spinal cord, it produces a flaccid paralysis associated with muscle wasting and hyporeflexia in the corresponding

region of the body—the permanent neurologic residue of poliomyelitis. Because of the destruction of motor neurons, paresis or paralysis follows; when the diaphragm and intercostal muscles are affected, severe respiratory compromise and even death may occur. A myocarditis sometimes complicates the acute infection. *Postpolio syndrome* can develop in patients 25 to 35 years after resolution of the initial illness. It is characterized by progressive weakness associated with decreased muscle mass and pain and has been attributed to age-dependent loss of remaining motor neurons, without any convincing evidence of viral reactivation.

Rabies

Rabies is severe encephalitis transmitted to humans by the bite of a rabid animal, usually a dog or various wild mammals that are natural reservoirs. Exposure to certain species of bats, even without a known bite, can also lead to rabies.

MORPHOLOGY

External examination of the brain shows intense edema and vascular congestion. Microscopically, there is widespread neuronal degeneration and an inflammatory reaction that is most severe in the brainstem; the basal ganglia, spinal cord, and dorsal root ganglia may also be involved. **Negri bodies**, the pathognomonic microscopic finding, are cytoplasmic, round to oval, eosinophilic inclusions found in pyramidal neurons of the hippocampus and Purkinje cells of the cerebellum, sites usually devoid of inflammation (Fig. 28.27). Rabies virus can be detected within Negri bodies by ultrastructural and immunohistochemical methods.

Clinical Features

Because the virus enters the CNS by ascending along the peripheral nerves from the wound site, the incubation period (usually between 1 and 3 months) depends on the distance between the wound and the brain. The disease begins with nonspecific symptoms such as malaise, headache, and fever, but the conjunction of these symptoms with local paresthesias around the wound is diagnostic. As the infection advances, the affected individual exhibits extraordinary CNS excitability; the slightest touch is painful and produces violent motor responses or even convulsions. Contracture of the pharyngeal musculature on swallowing produces foaming at the mouth, which may create an aversion to swallowing even water (hydrophobia). There are signs of meningeal

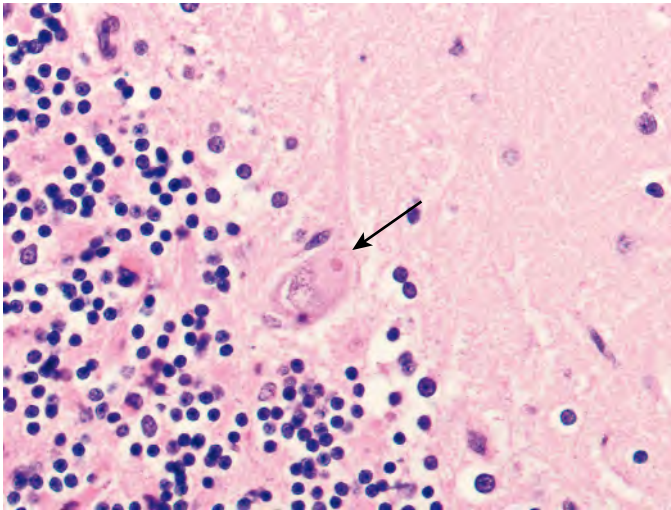


Fig. 28.27 Rabies encephalitis. The diagnostic histologic finding is the eosinophilic Negri body, as seen here in a Purkinje cell (arrow).

irritation and, as the disease progresses, flaccid paralysis. Alternating periods of mania and stupor progress to coma and eventually death ensues from respiratory failure.

Subacute Sclerosing Panencephalitis

Subacute sclerosing panencephalitis (SSPE) is a rare complication of persistent measles virus infection that usually develops years, or even decades, after the initial infection. It results from chronic production of viral proteins in neurons and glial cells, where the virus is concealed from the immune system. Patients present with dementia and abnormal movements and rigidity that may progress to paralysis and loss of consciousness. The serum and CSF contain antibodies specific for the measles virus. There is no treatment, but vaccination prevents the disease.

Human Immunodeficiency Virus

In the period before the availability of effective antiretroviral therapy, neuropathologic changes were demonstrated at postmortem examination in as many as 80% to 90% of cases of AIDS. These changes stem from direct effects of the virus on the nervous system, opportunistic infections, and primary CNS lymphoma, most commonly an EBV-positive B-cell tumor. There has been a decrease in the frequency of these secondary effects of HIV infection thanks to the efficacy of multidrug antiretroviral therapy.

HIV aseptic meningitis occurs within 1 to 2 weeks of seroconversion in about 10% of patients; antibodies to HIV can be demonstrated, and the virus can be isolated from the CSF. The neuropathologic studies of the early, acute phases of HIV invasion of the CNS have shown mild lymphocytic meningitis, perivascular inflammation, and some myelin loss. Among CNS cell types, only microglia express both the CD4 coreceptor and the chemokine receptors (CCR5 or CXCR4) that are required in combination for efficient infection by HIV. When symptomatic individuals come to autopsy during the chronic phase of HIV infection, HIV encephalitis is commonly found.

An *immune reconstitution inflammatory syndrome* (IRIS) can develop in patients with AIDS following effective treatment. The syndrome is recognized as a paradoxical deterioration after starting therapy and consists of an exuberant “reconstituted” inflammatory response while on antiretroviral therapy (Chapter 6). In the CNS of patients with opportunistic infections, IRIS has caused paradoxical exacerbation of symptoms; neuropathologic studies confirm intense inflammation with an influx of CD8+ lymphocytes.

MORPHOLOGY

HIV encephalitis is a chronic inflammatory reaction associated with widely distributed **microglial nodules**, often containing macrophage-derived **multinucleated giant cells** (Fig. 28.28); foci of tissue necrosis and reactive gliosis are sometimes seen together with these lesions. Some of the microglial nodules are found near small blood vessels, which show abnormally prominent endothelial cells and perivascular foamy or pigment-laden macrophages. These changes are especially prominent in the subcortical white matter, diencephalon, and brainstem. In some cases, there is also a disorder of white matter characterized by multifocal or diffuse areas of myelin pallor, axonal swelling, and gliosis. HIV can be detected in CD4+ microglia and mononuclear or multinucleated macrophages.

Clinical Features

Cognitive changes, some mild and others severe enough to be termed *HIV-associated dementia*, appear to have persisted in the era of effective anti-HIV treatment regimens; collectively, they are termed *HIV-associated neurocognitive disorders* (HAND). Rather than having a specific pathologic lesion as its correlate, HAND is most closely related to inflammatory activation of microglia and perivascular macrophages, some of which are infected by HIV, and T cells. A wide range of possible mechanisms for neuronal dysfunction and injury have been proposed, including the actions of inflammatory cytokines, CNS viral escape, a cascade of toxic effects of HIV-derived proteins, neurotoxic effects of anti-HIV therapy, accelerated aging (leading to Alzheimer disease and vascular dementia), and aberrant synaptic pruning.

Progressive Multifocal Leukoencephalopathy

Progressive multifocal leukoencephalopathy (PML) is an encephalitis caused by the JC polyomavirus; because the virus preferentially infects oligodendrocytes, demyelination is its principal pathologic effect. The disease occurs almost exclusively in immunosuppressed individuals in various clinical settings, including chronic lymphoproliferative or myeloproliferative illnesses, immunosuppressive chemotherapy (including monoclonal antibody therapy targeting certain integrins), granulomatous diseases, and AIDS.

Although most people have serologic evidence of exposure to JC virus by their teens, primary infection is asymptomatic; PML results

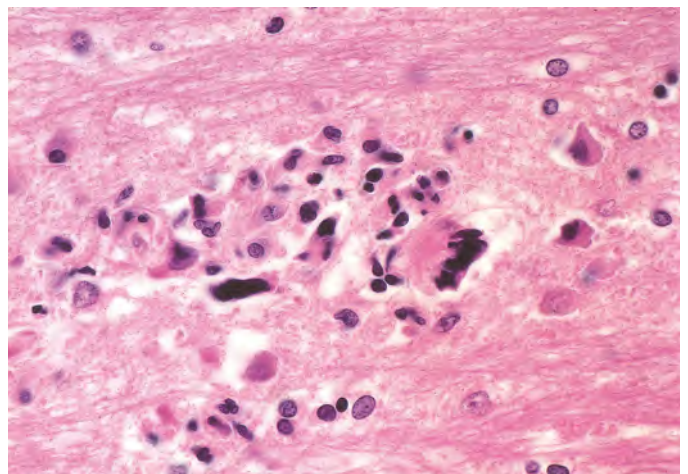


Fig. 28.28 HIV encephalitis. Note the microglial nodule and multinucleated giant cells.

from reactivation of virus in the setting of immunosuppression. Clinically, affected individuals develop focal and relentlessly progressive neurologic symptoms and signs, with imaging studies showing extensive, often multifocal, cerebral or cerebellar white matter lesions.

MORPHOLOGY

The lesions consist of patches of irregular, ill-defined white matter injury that range in size from millimeters to large confluent regions (Fig. 28.29A). Microscopically, a PML lesion shows an area of demyelination, most often in a subcortical location, in the center of which are sheets of lipid-laden macrophages and a reduced number of axons (Fig. 28.29B). Greatly enlarged oligodendrocyte nuclei containing glassy amphophilic viral inclusions (Fig. 28.29B, inset), which can be identified by immunohistochemistry, are typically found at the lesion edge. Bizarre giant astrocytes with one to several irregular, hyperchromatic nuclei are intermixed with more typical reactive astrocytes. Infection of granule cell neurons in the cerebellum has been demonstrated in rare instances.

Fungal Meningoencephalitis

Fungal infections of the CNS are encountered primarily in immunocompromised individuals. The brain is usually involved following widespread hematogenous dissemination of fungi; the most frequent offenders are *Candida albicans*, *Mucor* species, *Aspergillus fumigatus*, and *Cryptococcus neoformans*. In endemic areas, pathogens such as *Histoplasma capsulatum*, *Coccidioides immitis*, and *Blastomyces dermatitidis* may involve the CNS after a primary pulmonary or cutaneous infection; again, this often follows immunosuppression. Although most fungi reach the brain by hematogenous dissemination, direct extension may also occur, particularly in mucormycosis in the setting of diabetes mellitus.

The three main forms of injury in CNS fungal infection are chronic meningitis, vasculitis, and parenchymal invasion. Necrotizing vasculitis is most frequently seen with *Mucormycosis* and *Aspergillosis*, both of which invade blood vessel walls, but it occasionally also occurs

with other infections such as candidiasis. The resultant vascular thrombosis produces infarction that is often strikingly hemorrhagic.

Parenchymal infection, usually in the form of granulomas or abscesses, can occur with most of the fungi and often coexists with meningitis. The most commonly encountered fungi that invade the brain are *Candida* and *Cryptococcus*. Candidiasis usually produces multiple microabscesses, with or without granuloma formation; in contrast, parenchymal aggregates of *Cryptococcal* organisms are typically found within expanded perivascular (Virchow-Robin) spaces and are associated with minimal to no inflammation or gliosis.

Cryptococcal meningitis, a common opportunistic infection in the setting of AIDS, may be fulminant and fatal in as little as 2 weeks or indolent, evolving over months or years. The CSF may contain few cells but usually has a high concentration of protein. The mucoid-encapsulated yeasts can be visualized in the CSF with special stains or detected indirectly using assays for cryptococcal antigens (see Fig. 8.43).

Most cases of cryptococcal infection in immunosuppressed individuals are caused by *C. neoformans*. Recently, a second species, *C. gattii*, has been recognized and appears more likely to cause disease in immunocompetent individuals. Some case studies suggest that *C. gattii* involvement of the CNS is more likely to take the form of mass lesions ("cryptococcomas") and to present with symptoms related to increased CNS pressure due to mass effects.

Other Infectious Diseases of the Nervous System

Protozoal diseases (including malaria, toxoplasmosis, amebiasis, and trypanosomiasis), rickettsial infections (e.g., typhus and Rocky Mountain spotted fever), and metazoal diseases (especially cysticercosis and echinococcosis) may also involve the CNS and are discussed in Chapter 8.

- *Cerebral toxoplasmosis* is an opportunistic infection commonly found in the setting of HIV-associated immunosuppression, but also encountered occasionally in other immunocompromised and rarely even in immunocompetent patients. The clinical symptoms of infection of the brain by *Toxoplasma gondii* are subacute, evolving during a 1- or 2-week period, and may be both focal and diffuse. Computed tomography and magnetic resonance imaging studies may show multiple ring-enhancing lesions; however, this

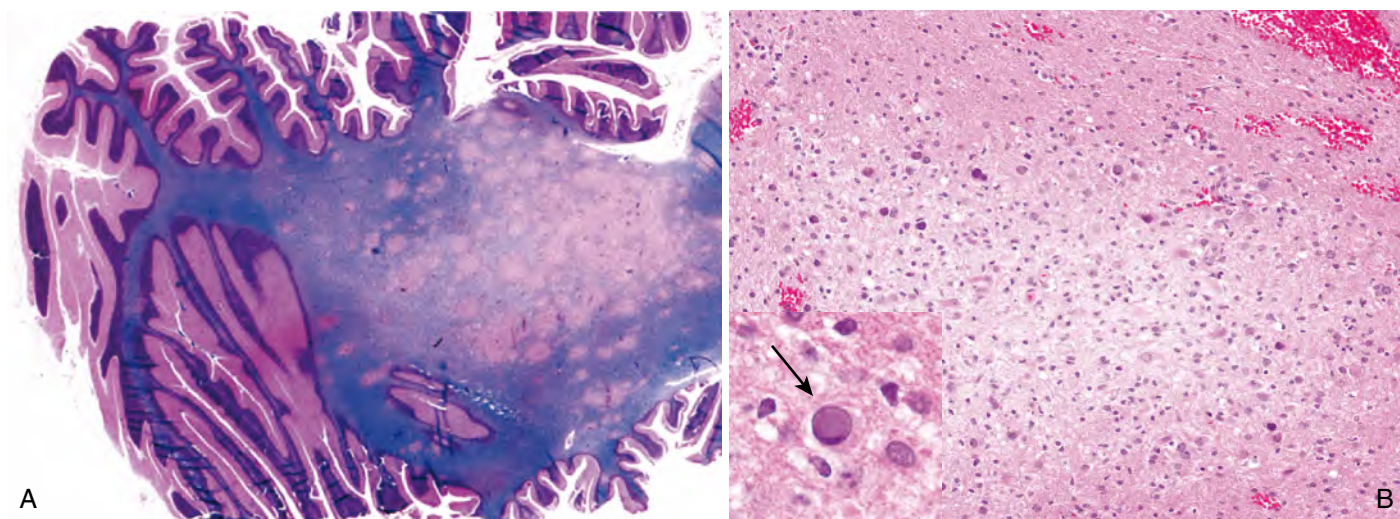


Fig. 28.29 Progressive multifocal leukoencephalopathy. (A) Section stained for myelin showing irregular, poorly defined areas of demyelination, which become confluent in places. (B) Microscopically, the lesions consist of areas of demyelination. Inset, Enlarged oligodendrocyte nucleus with plum-colored viral inclusion (arrow).

radiographic appearance is not specific, as glioblastoma, CNS lymphoma (in immunocompromised hosts), metastatic neoplasms, tuberculosis, and fungal infections produce similar findings. In immunocompetent hosts, the impact of toxoplasmosis on the brain is most often seen when primary maternal infection occurs early in pregnancy. Such infections often spread to the brain of the developing fetus and cause severe damage in the form of multifocal necrotizing lesions that may calcify. With early diagnosis, cerebral toxoplasmosis is often treatable with antibiotics, which may be administered empirically if this infection is in the differential diagnosis.

- **Cerebral amebiasis.** A rapidly fatal necrotizing encephalitis results from infection with *Naegleria* and *Balamuthia* species, while a chronic granulomatous meningoencephalitis has been associated with infection with *Acanthamoeba*. The amoebas may be difficult to distinguish morphologically from macrophages (Fig. 28.30). Methenamine silver or PAS stains are occasionally helpful in visualizing the organisms, although definitive identification ultimately depends on immunohistochemistry, culture, and molecular methods.
- **Cerebral malaria.** Cerebral malaria is a complication of infection by *Plasmodium falciparum*, the species with the highest mortality. Most likely the result of sticking of infected red cells to inflamed vascular endothelium, cerebral involvement by malaria is accompanied by reduced cerebral blood flow and results in ataxia, seizures, and coma in the acute phase as well as long-term cognitive deficits in up to 20% of affected children (Chapter 8).

MORPHOLOGY

Toxoplasmosis of the CNS produces brain abscesses, which are found most often in the cerebral cortex (near the gray-white junction) and deep gray nuclei, less often in the cerebellum and brainstem, and rarely in the spinal cord (Fig. 28.31A). Acute lesions exhibit central necrosis, punctate hemorrhages surrounded by acute and chronic inflammation, macrophage infiltration, and vascular proliferation. Encephalitis-like foci are also common. Both free

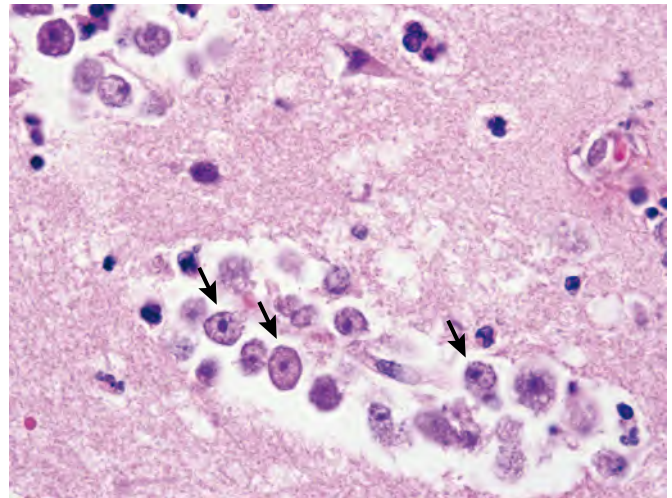


Fig. 28.30 Amebic meningoencephalitis; several amoebas are highlighted by the arrows. (*Acanthamoebae* resemble macrophages on routine stains, but are slightly smaller and less granular.)

tachyzoites and encysted bradyzoites may be found at the periphery of the necrotic foci. The organisms are often seen on routine H&E or Giemsa stains but are more easily recognized by immunohistochemical methods (Fig. 28.31B). The blood vessels near these lesions often show intimal proliferation or vasculitis with fibrinoid necrosis and thrombosis.

KEY CONCEPTS

Infections

- Pathogens from viruses through parasites can infect the brain. Different pathogens use distinct routes to reach the brain and cause different patterns of disease.

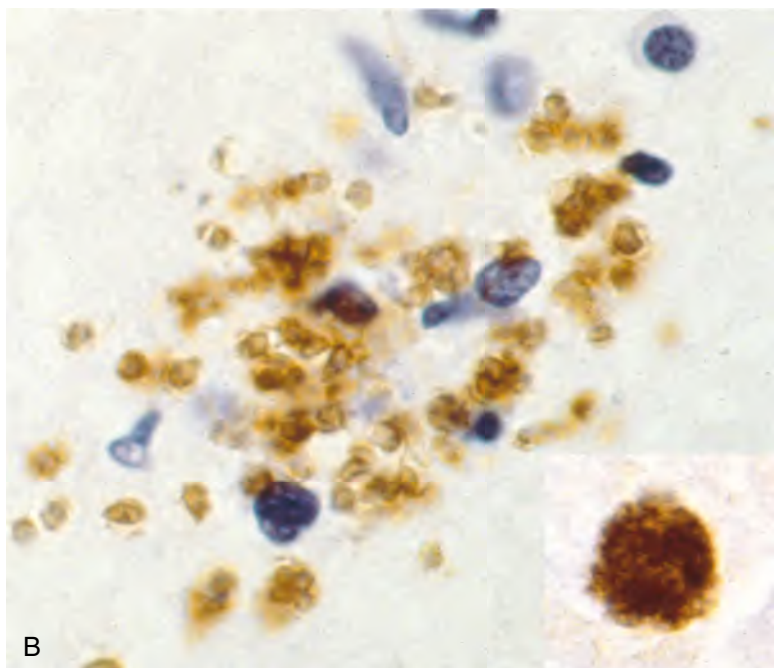
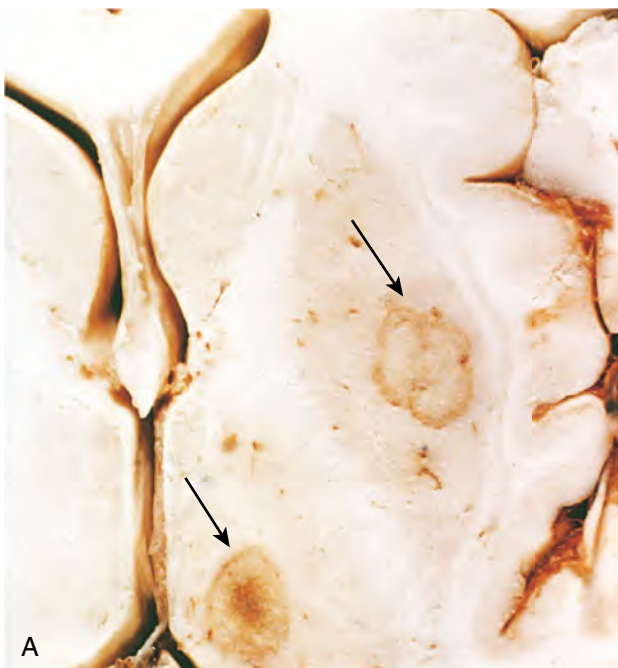


Fig. 28.31 Cerebral toxoplasmosis. (A) Toxoplasma abscesses in the putamen and thalamus (arrows). (B) Free tachyzoites demonstrated by immunostaining; inset, *Toxoplasma* pseudocyst with bradyzoites highlighted by immunostaining.

- Routes of access of organisms to the brain include: hematogenous spread (e.g., abscess formation in the setting of endocarditis), direct extension (following trauma or with extension from the sinuses), and retrograde transport along nerves (e.g., rabies).
- Bacterial infections may cause meningitis, cerebral abscesses, or a chronic meningoencephalitis. The distribution of pathogens is influenced by various host factors, such as age and level of immune function.
- Viral infections can cause meningitis or meningoencephalitis. Some viruses have characteristic patterns of infection (HSV-1 in the temporal lobes, polio in the anterior horn).
- HIV can directly cause meningoencephalitis, or indirectly affect the brain by increasing the risk of opportunistic infections (toxoplasmosis, CMV) and CNS lymphoma.

DEMYELINATING DISEASES

Demyelinating diseases of the CNS are acquired conditions characterized by damage to myelin with relative preservation of axons. The clinical deficits, at least initially, are due to the effect of myelin loss on the transmission of electrical impulses along axons. The natural history of demyelinating diseases is determined, in part, by the limited capacity of the CNS to regenerate normal myelin and by the degree of secondary damage to axons that occurs as the disease runs its course.

Several pathologic processes can cause loss of myelin. These include immune-mediated destruction of myelin (as in multiple sclerosis [MS]) and infections (as in PML, described earlier). In addition, inherited disorders may affect synthesis or turnover of myelin components; these are termed *leukodystrophies* and are discussed later in this chapter.

Multiple Sclerosis

Multiple sclerosis (MS) is an autoimmune demyelinating disorder characterized by distinct episodes of neurologic deficits that are separated in time and are attributable to patchy white matter lesions that are separated in space. It is the most common of the demyelinating disorders, having a prevalence of approximately 1 per 1000 persons in most of the United States and Europe. The disease may become clinically apparent at any age, although onset in childhood or after 50 years of age is relatively rare. Females are affected more than twice as often as are males.

In most individuals with MS, the clinical course takes the form of relapsing and remitting episodes of variable duration (weeks to months to years) marked by neurologic defects, followed by gradual and partial recovery of neurologic function (*relapsing-remitting MS*). In some individuals with relapsing-remitting MS, the frequency of relapses tends to decrease over time, but there is a steady neurologic deterioration in most affected individuals (*secondary progressive MS*) that occurs even when acute relapses are prevented by immunosuppressive therapy. Some MS patients, however, develop progressive disability from the outset, with no or minor improvements (*primary progressive MS*).

Pathogenesis

MS is an autoimmune disease, but it remains unclear which self-antigen is the target of the autoimmune response. As in other autoimmune disorders, the pathogenesis involves both genetic and environmental factors (Chapter 6). A genetic predisposition may be suggested from the fact that the incidence of MS is 15-fold higher when the disease is present in a first-degree relative and roughly 150-fold higher with an affected monozygotic twin. Despite a series of well-powered studies, only a portion of the genetic basis of the disease has been explained. There is a strong association with a DR haplotype of

the major histocompatibility complex. Genome-wide association studies have identified additional associations with the IL-2 and IL-7 receptor genes, and with several other genes encoding proteins involved in the immune response, including B cell genes, cytokines and their receptors, costimulatory molecules, and cytoplasmic signaling molecules; many of these loci are also associated with other autoimmune diseases. These genetic studies have not explained why the clinical course for individuals with MS is so variable.

Environmental factors are also important. For example, there is geographic variation in the prevalence of MS, with a higher number of cases diagnosed away from the equator; the mechanisms underlying this association are not established. In addition, recent large epidemiologic studies strongly suggest that prior infection with Epstein-Barr virus (EBV) is necessary (but not sufficient) for MS development. Notably, late infection (during adolescence but not in early childhood) increases MS risk 40-fold.

Multiple immune mechanisms have been implicated in the destruction of myelin that is the hallmark of MS.

- For many years, largely based on analysis of animal models in which the disease is induced by immunization with myelin proteins with strong adjuvants, it was believed that MS is primarily a T cell-mediated disease in which CD4⁺ T cells (Th1 and Th17) and CD8⁺ T cells, both reacting against myelin antigens, produce cytokines that activate macrophages, which then destroy the myelin sheath. However, it is now appreciated that these animal models overestimate the contribution of T cells and may not accurately mimic the human disease.
- More recent studies and the results of clinical trials showing the efficacy of B cell depletion strongly suggest that B cells have a critical role in disease pathogenesis through several possible mechanisms: they produce autoantibodies that react against glial cell adhesion molecule (GlialCAM) and other glial proteins; they may present self antigens to T cells; and they may produce inflammatory cytokines. Somatic mutated and clonally expanded B cells have been detected in the CSF of patients and the autoantibodies may contribute to disease. These autoreactive B cells may mature in lymphoid follicles that appear in the meninges of many MS patients. Of note, the presence and number of these follicles correlate with disease severity.
- There is also evidence for immunological cross-reactivity between EBV proteins such as EBNA1 (EBV nuclear antigen 1) and some human glial proteins, a phenomenon known as molecular mimicry (see Chapter 6). Cross-reactive T cells and antibodies may both contribute to disease. EBV-activated B cells may produce increased amounts of autoantibodies, and both T and B cells may secrete inflammatory cytokines.
- Progression seems to be related to neurodegeneration, with slowly enlarging white matter lesions, low-grade diffuse white matter inflammation, and cortical damage.

MORPHOLOGY

MS is a largely white matter disease that is best appreciated in sections of the brain and spinal cord. In the fresh state, the lesions are firmer than the surrounding white matter ("**sclerosis**") and appear as well circumscribed, somewhat depressed, glassy, gray-tan, irregularly shaped **plaques** (Fig. 28.32A). The area of demyelination often has sharply defined borders, a feature best appreciated with stains for myelin (Fig. 28.32B). The size of lesions varies considerably, from small foci that are only recognizable microscopically to large confluent plaques. Plaques commonly occur adjacent to the lateral ventricles and are also frequent in the corpus callosum, optic nerves and chiasm, brainstem, ascending and descending fiber tracts, cerebellum, and spinal cord. In addition, there is

also extensive, largely subpial demyelination of gray matter, which contains thin myelinated fibers. The **subpial cortical lesions** are the hallmark of MS (they are not seen in any other brain disease) and are typically associated with B cell rich inflammatory foci in the leptomeninges, which may include lymphoid follicles.

Microscopically, in an **active plaque**, there is ongoing myelin breakdown associated with abundant foamy macrophages (Fig. 28.33A); lymphocytes are also present, mostly as perivascular cuffs, especially at the outer edge of the lesion. Active lesions are often centered on small veins; myelin is usually completely absent (Fig. 28.33B), but axons are relatively preserved (Fig. 28.33C). In time, astrocytes undergo reactive changes. As lesions become quiescent, the inflammatory cells slowly disappear. Within **inactive plaques**, there are no macrophage-rich infiltrates, little to no myelin is found, and there is a reduction in the number of oligodendrocyte nuclei; instead, reactive gliosis is prominent. Axons in old gliotic plaques are usually greatly diminished in number. **Chronic smoldering plaques** are seen only in progressive forms of MS and have a mixed appearance (hypocellular center reminiscent of inactive plaques and a rim of activated microglia/macrophages at the lesion border).

In some MS plaques (shadow plaques), the white matter pallor is less severe, and abnormally thinned-out myelin sheaths can be demonstrated, especially at the outer edges. This phenomenon is interpreted as evidence of partial and incomplete remyelination by surviving oligodendrocytes. Abnormally myelinated fibers have also been observed at the edges of typical plaques. Although these histologic findings suggest a limited potential for remyelination in the CNS, the remaining axons within most MS plaques remain unmyelinated.

Clinical Features

Although MS lesions can occur anywhere in the CNS and consequently may produce a wide range of clinical manifestations, certain patterns of neurologic symptoms and signs are more common. Unilateral visual impairment due to involvement of the optic nerve (*optic neuritis*) is a frequent initial manifestation of MS. However, only a minority of individuals (10% to 50%, depending on the population studied) with an episode of optic neuritis go on to develop MS (which requires multiple episodes to support the diagnosis). Involvement of the brainstem produces cranial nerve signs (ataxia, nystagmus, and internuclear ophthalmoplegia from interruption of the fibers of the medial longitudinal fasciculus). Spinal cord lesions give rise to motor and sensory impairment of trunk and limbs, spasticity, and loss of bladder and bowel control.

Examination of the CSF in individuals with MS shows a normal or only mildly elevated total protein level and in one-third of cases a moderate pleocytosis. IgG levels in the CSF are increased, and oligoclonal IgG bands are usually observed on immunoelectrophoresis, indicating the presence of expanded B cell clones. Magnetic resonance imaging has taken on a prominent role in assessing disease progression. As already mentioned, treatment consists of several types of immunosuppressive or immunomodulatory agents, with B cell-depleting agents, notably anti-CD20 antibodies, currently the first line of therapy. These immunomodulatory therapies are highly effective against relapses, but only partially control the late neurodegenerative stage of the disease. Accumulating evidence indicates that for most patients, early initiation of effective immunomodulatory therapy provides maximum protection against late disability.

Neuromyelitis Optica

Neuromyelitis optica (NMO) is a syndrome with synchronous (or near-synchronous) bilateral optic neuritis and spinal cord demyelination. NMO has an even greater skewing toward females than MS, is more commonly associated with poor recovery from the first attack,

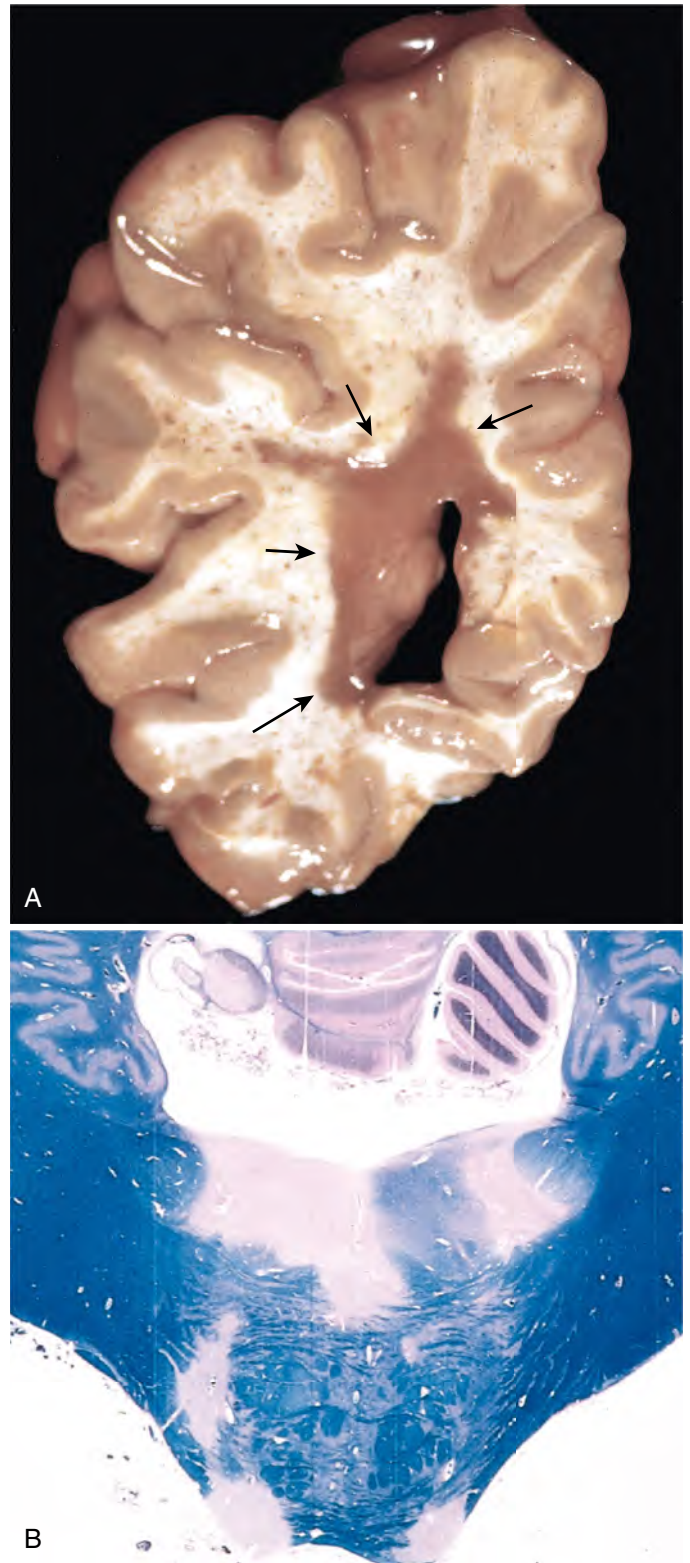


Fig. 28.32 Multiple sclerosis. (A) Section of fresh brain shows gray-brown plaque around the occipital horn of the lateral ventricle (arrows). (B) Regions of demyelination (MS plaques) around the fourth ventricle lack the normal blue staining of myelin (Luxol fast blue periodic acid–Schiff stain).

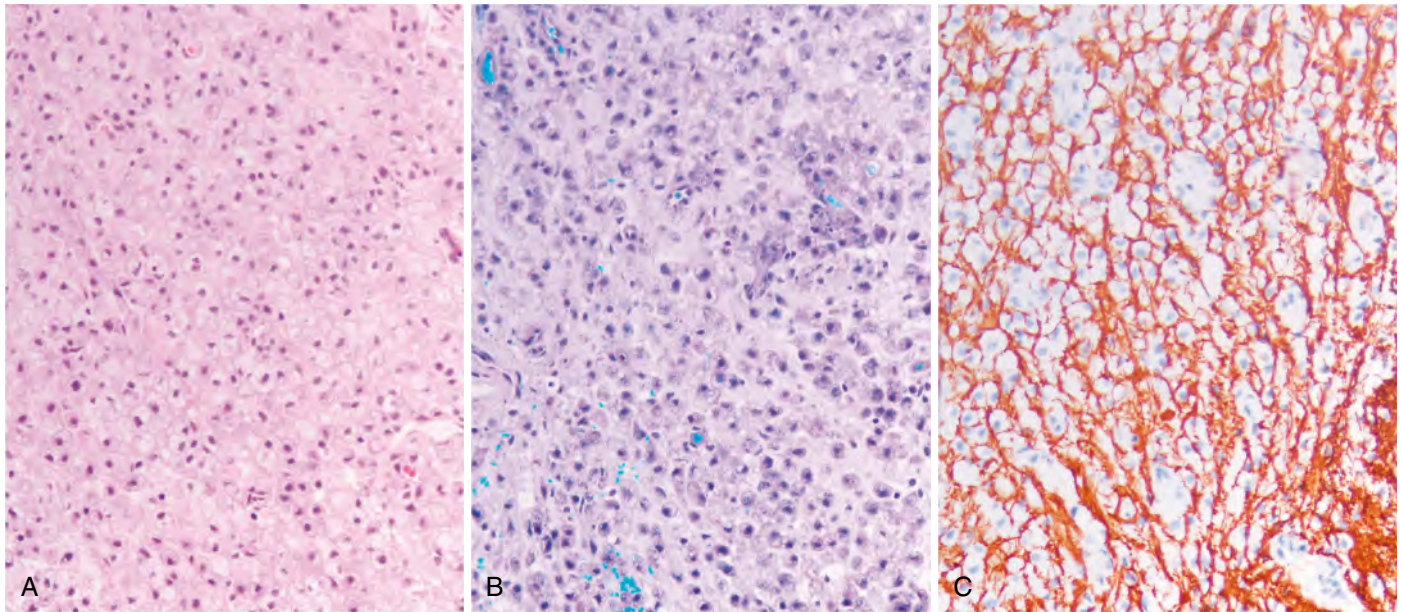


Fig. 28.33 Multiple sclerosis. (A) Active demyelinating plaques are very cellular due to the presence of numerous lipid-laden macrophages. (B) The same lesion stained with Luxol fast blue periodic acid-Schiff shows a complete lack of myelin (which would stain blue). (C) Relative preservation of axons is seen on a neurofilament immunostain (*brown*).

and is most often characterized by the presence of aquaporin-4 (AQP4) antibodies, which are pathogenic. Less commonly, myelin oligodendrocyte glycoprotein (MOG) antibodies are present instead, consistent with *MOG antibody-associated disease (MOGAD)*; in those cases, there is no female predominance and the disease is more likely to be monophasic. Aquaporin-4 is the major water channel of astrocytes. In AQP4-NMO, AQP4 antibodies damage astrocytes and the blood-brain-barrier, resulting in primary inflammatory astrocytopathy with secondary demyelination. White blood cells (often including neutrophils) are common in the CSF. Within damaged areas of white matter, there is typically necrosis, an inflammatory infiltrate including neutrophils and eosinophils, vascular immunoglobulin and complement deposition, and loss of aquaporin-4 expression. Acute attacks are treated with glucocorticoids or plasma exchange, while long-term treatments include agents that decrease antibody titers (e.g., by depleting B cells) or that inhibit complement (e.g., inhibitors of the C5–C9 membrane attack complex).

Acute Disseminated Encephalomyelitis and Acute Necrotizing Hemorrhagic Encephalomyelitis

Acute disseminated encephalomyelitis is a diffuse, monophasic demyelinating disease that follows a viral infection or, rarely, a viral immunization. Symptoms typically develop 1 or 2 weeks after the antecedent event and include headache, lethargy, and coma rather than focal findings, as seen in MS; in contrast to MS, all brain lesions appear similar, consistent with the clinically monophasic nature of the disorder. The clinical course is rapid, and as many as 20% of those affected die; the remaining patients usually recover completely, and there is no secondary progression. Acute disseminated encephalomyelitis is the most common clinical presentation of MOGAD; as with other presentations of MOGAD, the incidence is similar in males and females, with predilection for children and young adults. The lesions are similar to those seen in experimental autoimmune encephalomyelitis animal models, in which animals

develop MOG-specific antibodies and T cells after immunization with myelin components.

Acute necrotizing hemorrhagic encephalomyelitis (also known as acute hemorrhagic leukoencephalitis or Weston-Hurst disease) is a fulminant syndrome of CNS demyelination. The illness is almost invariably preceded by a recent episode of upper respiratory infection, most often of unknown cause. This disorder shows histologic similarities with acute disseminated encephalomyelitis; however, the damage is more severe and rapidly progressive. The disease is fatal in many patients, with significant deficits present in most survivors.

Central Pontine Myelinolysis

Central pontine myelinolysis is an acute disorder characterized by loss of myelin in the base of the pons and portions of the pontine tegmentum, typically in a roughly symmetric pattern. It most commonly arises 2 to 6 days after rapid correction of hyponatremia, although it can also be associated with other severe electrolyte disturbances or osmolar imbalances; it is known clinically as *osmotic demyelination syndrome*. It appears that rapid increases in osmolality damage oligodendrocytes through uncertain mechanisms. Inflammation is absent from the lesions, and neurons and axons are well preserved. Because of the synchronous onset of damage, all lesions appear to be at the same stage of myelin loss and reaction. Although originally described in the pons, extra-pontine lesions with similar appearance and apparent etiology may also occur.

Although this disorder can involve most parts of the brain, periventricular and subpial regions are spared, and it is extremely rare for the process to extend below the pontomedullary junction. The clinical presentation is a rapidly evolving quadriplegia, which may be fatal or lead to severe long-term deficits, including the “locked-in” syndrome, in which patients are fully conscious yet unresponsive. It is imperative that hyponatremia be corrected slowly and carefully to prevent this serious complication.

KEY CONCEPTS

Demyelinating Diseases

- Because of the critical role of myelin in nerve conduction, diseases of myelin can lead to widespread and severe neurologic deficits.
- Demyelinating diseases show evidence of breakdown and destruction of previously normal myelin, often by inflammatory processes. Secondary injury to axons typically emerges over time.
- Multiple sclerosis (MS), an autoimmune demyelinating disease primarily affecting young adults, is the most common disorder of myelin, in which antibodies and T cells may react with antigens of glial cells and myelin. It often pursues a relapsing-remitting course, with eventual progressive accumulation of neurologic deficits that is thought to reflect cortical demyelination and secondary axon loss.
- Neuromyelitis optica (NMO) is a severe immune-mediated demyelinating disease that involves bilateral optic nerves and the spinal cord that is most commonly caused by antibodies against the water channel aquaporin 4.
- Acute disseminated encephalomyelitis and acute necrotizing hemorrhagic encephalomyelitis generally follow infections and have an acute, monophasic clinical course that can lead to death in severe cases. These diseases are caused by autoantibodies and T cells that target myelin oligodendrocyte protein.

NEURODEGENERATIVE DISEASES

Neurodegenerative diseases are disorders characterized by the progressive loss of groups of neurons that often have shared functions. Thus, the different diseases tend to involve particular neural systems and have relatively stereotypic presenting signs and symptoms.

The pathologic process that is common across most of the neurodegenerative diseases is the accumulation of protein aggregates (hence the occasional use of the term *proteinopathy*). Protein aggregates may arise because of mutations that alter the affected protein's conformation or disrupt the pathways that are involved in the processing or clearance of an otherwise normal protein. In other situations, there may be imbalances between protein synthesis and clearance (due to genetic, environmental, or unknown causes) that allows gradual accumulation of misfolded and/or aggregated proteins.

Regardless of how they arise, the protein aggregates typically are resistant to degradation and show aberrant localization within neurons. They are recognized histologically as inclusions, which serve as diagnostic hallmarks. The aggregates may cause cell injury by multiple mechanisms. First, accumulation of misfolded proteins (usually in small, oligomeric aggregates) may trigger cell death by apoptosis (Chapter 2). Second, as more and more protein is shunted into the aggregates, the normal function of the protein may be lost, and this may also contribute to cell injury. Other recent evidence suggests that protein aggregates may spread from cell to cell (see later). The data supporting this concept are largely derived from experimental animal studies.

The basis for aggregation varies from one disease to another. It may be directly related to an intrinsic feature of a mutated protein (e.g., expanded polyglutamine repeats in Huntington disease [HD]), an intrinsic feature of a peptide derived from a larger precursor protein (e.g., A β in Alzheimer disease [AD]), or an unexplained alteration of a normal cellular protein (e.g., α -synuclein in sporadic Parkinson disease [PD]).

Neurodegenerative diseases differ with respect to the anatomic localization of involved areas and the specific morphologic and biochemical abnormalities (e.g., tangles, plaques, Lewy bodies). Accordingly, they can be classified using two different approaches:

- *Symptom-based/anatomic*: based on the anatomic regions that are most affected, which is typically reflected in the clinical manifestations (e.g., neocortical involvement results in cognitive impairment and dementia)
- *Pathologic/molecular*: based on the types of inclusions or abnormal structures observed (e.g., all diseases with tau inclusions are classified as *tauopathies*)

It must be acknowledged, however, that within the spectrum of degenerative diseases there is remarkable overlap in terms of characteristic neurologic deficits, functional/anatomic distribution of lesions, and cellular pathology (Tables 28.3 and 28.4). With this caveat, for the sake of simplicity, we will follow the time-honored classification based on the original description of these diseases.

Prion Diseases

Prion diseases are rapidly progressive neurodegenerative disorders caused by aggregation and intercellular spread of a misfolded prion protein (PrP); they may be sporadic, familial, or transmitted. Prion diseases include Creutzfeldt-Jakob disease (CJD), Gerstmann-Sträussler-Scheinker syndrome, fatal familial insomnia, and kuru in humans; scrapie in sheep and goats; mink-transmissible encephalopathy; chronic wasting disease of deer and elk; and bovine spongiform encephalopathy. All these diseases are characterized morphologically by “spongiform change” caused by intracellular vacuoles in neurons and glia, and clinically by a rapidly progressive dementia.

Pathogenesis

Prion diseases are caused by “spreading” of misfolded proteins, a remarkable phenomenon that allows a pathogenic protein to acquire some of the characteristics of an infectious organism. Normal PrP is a 30-kD cytoplasmic protein of unknown function. Disease occurs when PrP undergoes a conformational change from its normal α -helix-containing isoform (PrP^C) to an abnormal β -pleated sheet isoform, usually termed PrP^{Sc} (for scrapie) (Fig. 28.34). Associated with this conformational change, PrP acquires resistance to digestion by proteases, such as proteinase K. Accumulation of PrP^{Sc} in neural tissue seems to be the cause of the pathologic changes in these diseases, but how this material induces the development of cytoplasmic vacuoles and eventual neuronal death is still unknown. Immunostaining for PrP after partial digestion with proteinase K allows detection of PrP^{Sc}, which is diagnostic.

The conformational change resulting in PrP^{Sc} may occur spontaneously at an extremely low rate (resulting in sporadic cases of CJD) or at a higher rate if various mutations are present in PrP^C, such as occurs in familial forms of CJD, Gerstmann-Sträussler-Scheinker syndrome, and fatal familial insomnia. Regardless of how it originates, PrP^{Sc} then facilitates, in a cooperative fashion, the conversion of other PrP^C molecules to PrP^{Sc} molecules; it is this propagation of PrP^{Sc} that accounts for the transmissible variants of prion diseases, which include iatrogenic CJD, variant CJD, and kuru. The suggestion that, at least within an individual, there may be cell-to-cell spread of disease-associated protein aggregates provides a link between prion diseases and other neurodegenerative disorders such as Alzheimer and Parkinson disease.

Creutzfeldt-Jakob Disease

Creutzfeldt-Jakob disease (CJD), the most common prion disease, is a rare disorder that manifests clinically as a rapidly progressive dementia. The sporadic form of CJD has an annual incidence of approximately 1 per 1 million people and accounts for over 90% of CJD cases. The less common familial forms are caused by mutations in *PRNP*, the gene that

Table 28.3 Features of the Major Neurodegenerative Diseases

Disease	Clinical Pattern	Protein Inclusions	Genes Altered in Inherited Disease Forms
Prion diseases	Rapidly progressive dementia	Kuru plaques and diffuse PrP ^{Sc} deposits	<i>PRNP</i> (mutations)
Alzheimer disease (AD)	Dementia	A β (plaques) and tau (tangles)	<i>APP</i> (mutations; three wt copies in trisomy 21), <i>PSEN1</i> (mutations), <i>PSEN2</i> (mutations), <i>APOE</i> (ϵ 4 allele)
Frontotemporal lobar degeneration (FTLD)	Behavioral changes, language disturbance	tau TDP-43 FUS	<i>MAPT</i> (mutations) <i>TARDBP</i> (mutations), <i>GRN</i> (mutations), <i>C9orf72</i> (hexanucleotide repeat expansion) <i>FUS</i> (mutations)
Parkinson disease (PD)	Hypokinetic movement disorder with or without dementia	α -synuclein tau α -synuclein or none	<i>SNCA</i> (mutations; gene amplifications), <i>LRRK2</i> (mutations), <i>GBA1</i> (heterozygous mutations) <i>LRRK2</i> (mutations) <i>PARK7</i> (encodes DJ-1; mutations), <i>PINK1</i> (mutations), <i>PRKN</i> (mutations)
Progressive supranuclear palsy (PSP)	Parkinsonism with abnormal eye movements	tau	<i>MAPT</i> (mutations)
Corticobasal degeneration (CBD)	Parkinsonism with asymmetric movement disorder	tau	None
Multiple system atrophy (MSA)	Parkinsonism, cerebellar ataxia, autonomic failure	α -synuclein	None
Huntington disease (HD)	Hyperkinetic movement disorder	Huntington (polyglutamine containing)	<i>HTT</i> (trinucleotide repeat expansion)
Spinocerebellar ataxias (SCA-1, 2, 3, 6, 7, 8, 17 and DRPLA)	Cerebellar ataxia	Various proteins (polyglutamine containing)	Multiple loci (trinucleotide repeat expansion)
Amyotrophic lateral sclerosis (ALS)	Weakness with upper and lower motor neurons signs	SOD1 TDP-43 FUS	<i>SOD1</i> (mutations) <i>TARDBP</i> (mutations), <i>C9orf72</i> (hexanucleotide repeat expansion) <i>FUS</i> (mutations)
Spinal bulbar muscular atrophy (SBMA)	Lower motor neuron weakness, diminished androgen	Androgen receptor (polyglutamine containing)	<i>AR</i> (trinucleotide repeat expansion)

The proteins encoded by the affected genes are the following (in alphabetical order): *APOE*, apolipoprotein E; *APP*, amyloid precursor protein; *AR*, androgen receptor; *C9orf72*, chromosome 9 open reading frame 72, encodes multiple proteins; *FUS*, RNA-binding protein fused in sarcoma; *GBA1*, glucocerebrosidase; *GRN*, progranulin; *HTT*, huntingtin; *LRRK2*, leucine-rich repeat kinase 2; *MAPT*, microtubule-associated protein tau; *PARK7*, DJ-7 protein; *PRKN*, parkin; *PINK1*, PTEN-induced kinase-1; *PRNP*, PrP (prion protein); *PSEN1*, *PSEN2*, presenilin-1 and -2; *SNCA*, α -synuclein; *SOD1*, superoxide dismutase-1; *TARDBP*, transactive response DNA-binding protein.

Table 28.4 Relationship Between Proteins and Neurodegenerative Diseases

Protein	Diseases With Inclusions
PrP ^{Sc}	Prion diseases (CJD, vCJD, kuru, etc.)
A β	Alzheimer disease
Tau	Alzheimer disease Frontotemporal lobar degeneration Parkinson disease (with <i>LRRK2</i> mutations) Progressive supranuclear palsy Corticobasal degeneration Chronic traumatic encephalopathy
TDP-43	Frontotemporal lobar degeneration Amyotrophic lateral sclerosis
FUS	Frontotemporal lobar degeneration Amyotrophic lateral sclerosis
α -synuclein	Lewy body disease (Parkinson disease/dementia with Lewy bodies spectrum) Multiple system atrophy
Polyglutamine aggregates (different protein in each disease)	Huntington disease Some forms of spinocerebellar ataxia Spinal bulbar muscular atrophy

encodes PrP. The disease has a peak incidence in the seventh decade. There are also well-established cases of iatrogenic transmission, notably following corneal or dural transplantation, deep implantation of electrodes in the brain, and administration of contaminated preparations of cadaveric human growth hormone. The onset of the disease is marked by subtle changes in memory and behavior followed by a rapidly progressive dementia, often associated with pronounced involuntary jerking muscle contractions on sudden stimulation (startle myoclonus). Signs of cerebellar dysfunction, usually manifested as ataxia, are present in a minority of affected individuals. The disease is uniformly fatal; the average survival is only 7 months after the onset of symptoms. A few patients have lived for several years, and these long-surviving cases show extensive atrophy of involved gray matter.

Variant Creutzfeldt-Jakob Disease

Starting in 1995, a new CJD-like illness came to medical attention in the United Kingdom. This illness was different from classic CJD in several important respects: the disease affected young adults, behavioral disorders figured prominently in the early stages of the disease, and the neurologic syndrome progressed more slowly than in individuals with other forms of CJD. Multiple lines of evidence indicated that the new variant of CJD (vCJD) was linked to exposure to bovine spongiform encephalopathy, either through consumption of contaminated food or transfusion of

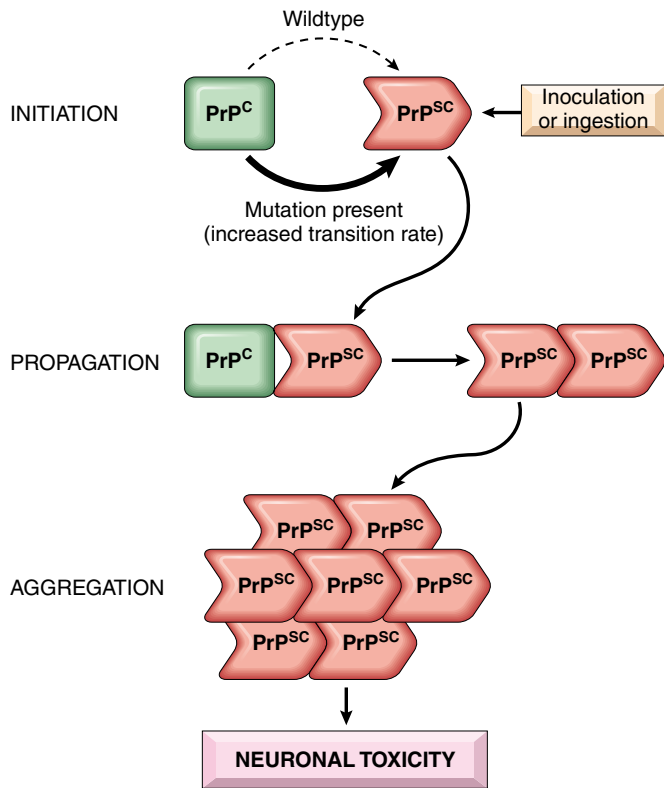


Fig. 28.34 Pathogenesis of prion disease. α -helical PrP^{C} may spontaneously shift to the β -sheet PrP^{Sc} conformation, an event that occurs at a much higher rate in the familial form of the disease associated with germline PrP mutations. PrP^{Sc} may also be acquired from exogenous sources, such as contaminated food, medical instrumentation, or medicines. Once present, PrP^{Sc} converts additional molecules of PrP^{C} into PrP^{Sc} through physical interaction, eventually leading to the formation of pathogenic PrP^{Sc} aggregates.

blood from patients in the asymptomatic/preclinical stage of vCJD. Recognition of this link led to public health measures that have been effective in limiting the spread of vCJD; the incidence in the United Kingdom peaked in 2000 and has dropped precipitously since that time.

MORPHOLOGY

The progression of the dementia in CJD is usually so rapid that there is little if any grossly evident brain atrophy. The pathognomonic finding is a **spongiform** transformation of the cerebral cortex and deep gray matter structures (caudate, putamen); this multifocal process results in the uneven formation of small and apparently empty microscopic vacuoles of varying sizes within the neuropil and sometimes in the perikaryon of neurons (Fig. 28.35A). In advanced cases, there is severe neuronal loss, reactive gliosis, and sometimes expansion of the vacuolated areas into cyst-like spaces ("status spongiosus"). Inflammation is notably absent. Electron microscopy shows the vacuoles to be membrane-bound and located within the cytoplasm of neuronal processes.

Kuru plaques are extracellular deposits of aggregated abnormal PrP . They are Congo red- and PAS-positive and usually occur in the cerebellum (Fig. 28.35B) but are abundant in the cerebral cortex in cases of vCJD (Fig. 28.35C). In all forms of prion disease, immunohistochemical staining demonstrates the presence of proteinase K-resistant PrP^{Sc} in tissue.

Alzheimer Disease

Alzheimer disease (AD) is the most common cause of dementia in older adults, with increasing incidence as a function of age. The disease usually becomes clinically apparent as insidious impairment of higher cognitive functions. Patients rarely become symptomatic before 50 years of age; the incidence of the disease increases with age, and the prevalence roughly doubles every 5 years, starting from a level of 1% for the 60- to 64-year-old population and reaching 40% or more for the 85- to 89-year-old cohort. This progressive increase in incidence with increasing age has given rise to major medical, social, and economic concerns in countries with aging populations.

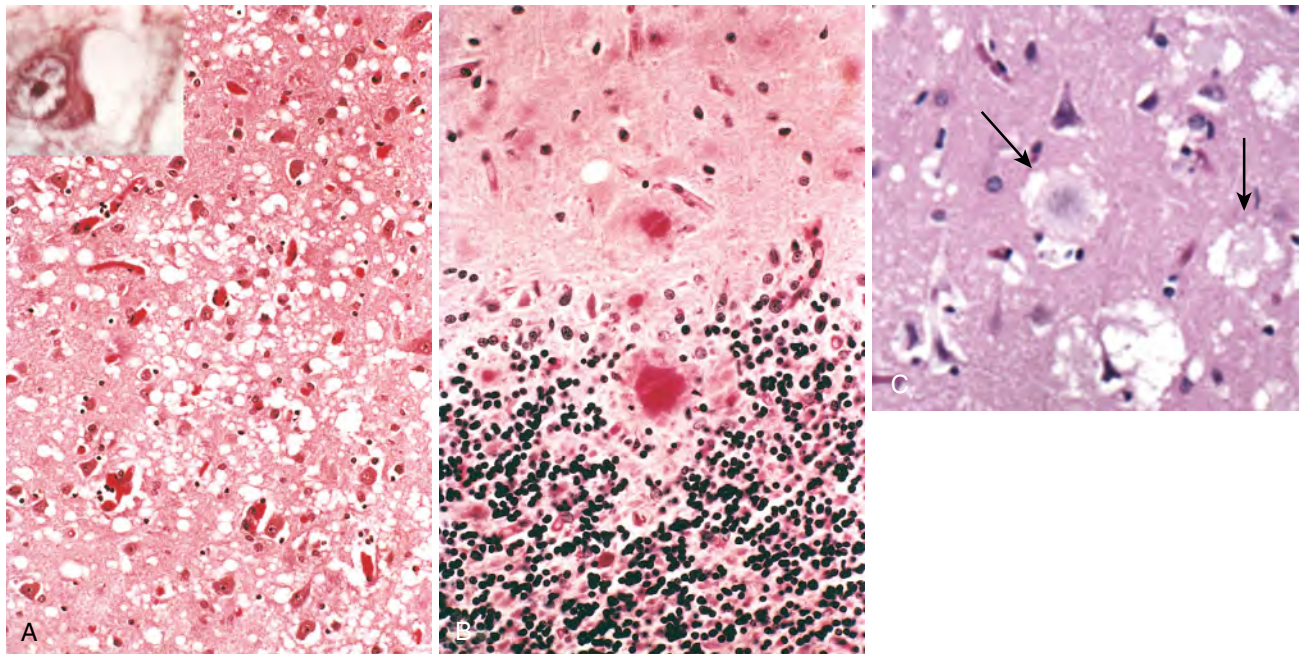


Fig. 28.35 Prion disease. (A) Spongiform change in the cerebral cortex. *Inset*, High magnification of neuron with vacuoles. (B) Cerebellar cortex showing kuru plaques (periodic acid-Schiff stain) that consist of aggregated PrP^{Sc} . (C) Cortical kuru plaques (arrows) surrounded by spongiform change in variant Creutzfeldt-Jakob disease.

About 5% to 10% of cases are familial; these have provided important insight into the pathogenesis of the more common sporadic form of the disease. Although pathologic examination of brain tissue obtained at autopsy remains necessary for definitive diagnosis of AD, the combination of clinical assessment and current radiologic imaging allows accurate premortem diagnosis in 80% to 90% of cases. Assays for plasma levels of β -amyloid peptides may also be diagnostically useful.

Pathogenesis

The fundamental abnormality in AD is the accumulation of two proteins ($A\beta$ and tau) in specific brain regions, likely because of excessive production and defective removal (Fig. 28.36). The two pathologic hallmarks of AD, particularly evident in the end stages of the illness, are *neuritic amyloid plaques* and *neurofibrillary tangles*. Plaques contain deposits of aggregated $A\beta$ peptides in the neuropil, while tangles are aggregates of the microtubule binding protein tau, which develop intracellularly and then persist extracellularly after neuronal death. Both plaques and tangles appear to contribute to the neural dysfunction, and the interplay between the processes that lead to the accumulation of these two types of abnormal protein aggregates is a

critically important aspect of AD pathogenesis that has yet to be fully unraveled.

Several lines of evidence strongly support a model in which $A\beta$ generation is the critical initiating event for the development of AD. First, there are diseases (such as chronic traumatic encephalopathy [CTE, discussed earlier], some variants of frontotemporal lobar degeneration, progressive supranuclear palsy, and corticobasal degeneration) in which there are abundant tau deposits but no $A\beta$ deposits, and clinical features of AD do not develop. This suggests that having abnormal deposits of tau in the brain is not a sufficient stimulus to elicit deposition of $A\beta$. Additionally, multiple lines of genetic evidence (discussed later) point to the likely importance of altered $A\beta$ catabolism. In contrast, mutations in the *MAPT* (microtubule-associated protein tau) gene, which encodes tau, do not give rise to AD but rather cause frontotemporal lobar degeneration (discussed later). Finally, treatment with a monoclonal antibody targeting $A\beta_{42}$ slows the rate of cognitive decline and reduces the total amyloid burden in patients with early AD, although the benefit is modest.

The pathogenesis of AD thus involves $A\beta$, tau, and several other genetic and host factors:

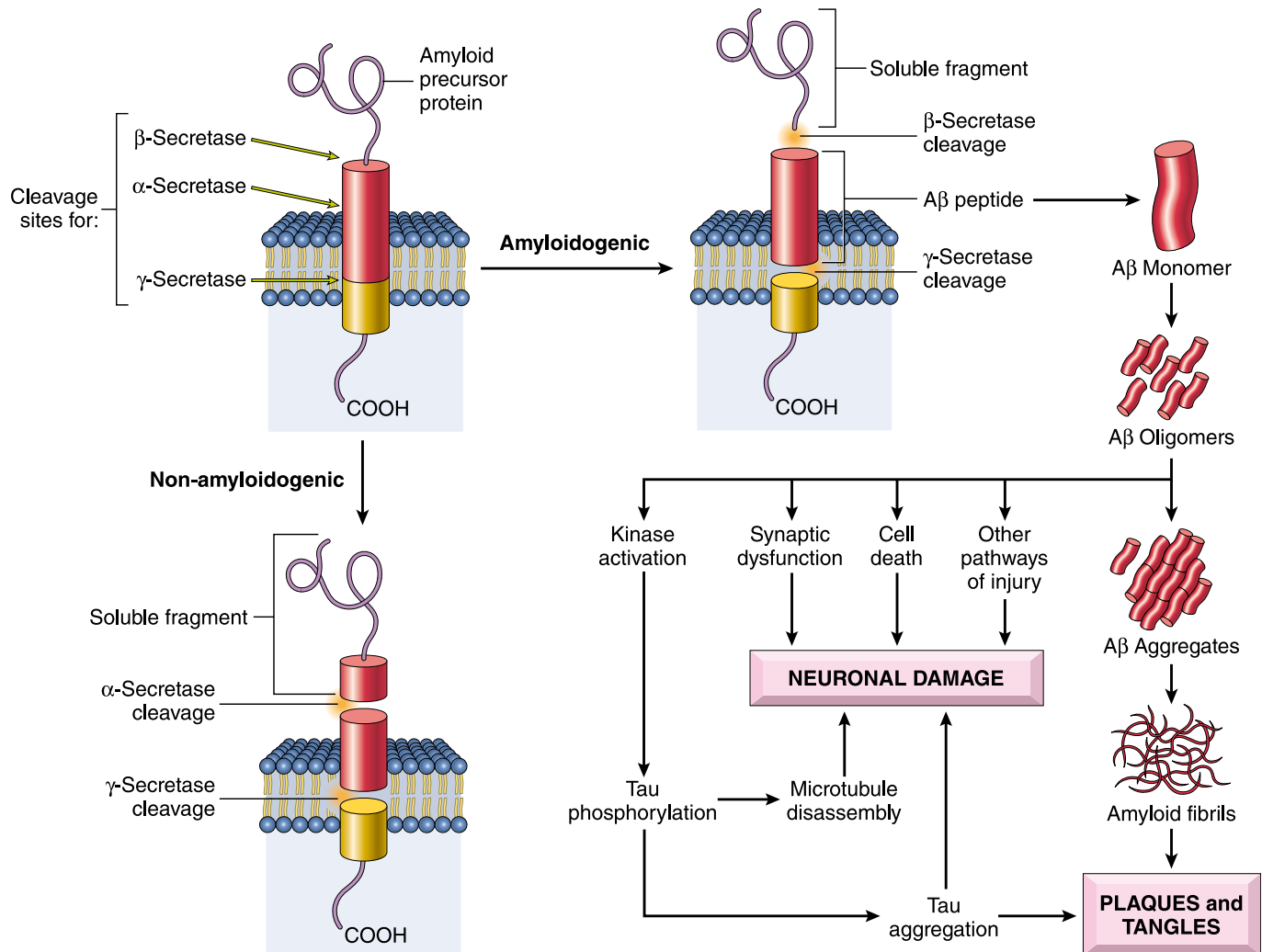


Fig. 28.36 Protein aggregation in Alzheimer disease. Amyloid precursor protein cleavage by α -secretase and γ -secretase produces a harmless soluble peptide, whereas amyloid precursor protein cleavage by β -amyloid-converting enzyme and γ -secretase releases $A\beta$ peptides, which form pathogenic aggregates and contribute to the characteristic plaques and tangles of Alzheimer disease.

- **Role of A β .** Amyloid β precursor protein (APP) is a cell surface protein with a single transmembrane domain that may function as a receptor, possibly for PrP^c among other ligands. The A β portion of the protein extends from the extracellular region into the transmembrane domain (see Fig. 28.36). Processing of APP begins with cleavage in the extracellular domain, followed by an intramembranous cleavage. There are two potential extracellular sites of cleavage, which may be carried out by two different classes of proteases, α -secretase and β -secretase. If the first cut occurs at the α -secretase site, then A β is not generated (the nonamyloidogenic pathway); this cleavage event mostly occurs at the cell surface because proteases with α -secretase activity are involved in the shedding of surface proteins. Surface APP may also be endocytosed into vesicles, where it can be cleaved by β -secretase, which cuts at a slightly more N-terminal site within APP (the amyloidogenic pathway). Following cleavage of APP at either of these sites, the γ -secretase complex performs an intramembranous cleavage. Cleavage by β -secretase and γ -secretase releases A β 42, which is prone to aggregation and amyloid formation. The γ -secretase complex, which contains presenilin-1 or presenilin-2, nicastrin, PEN2, and APh1, is also responsible for processing of Notch receptors and many other membrane proteins. Once generated, A β 42 aggregates, first into small oligomers (which appear to be the toxic form responsible for neuronal dysfunction), and eventually into large aggregates and fibrils that can be visualized by microscopy. Recent data suggest that at least some A β aggregates form intracellularly within abnormally alkalinized lysosomes, and are then released into the extracellular space after the affected neurons die.

Familial forms of AD support the central role of A β generation as a critical step for initiation of AD pathogenesis. The two loci identified as causes of the majority of early-onset familial AD (*PSEN1* on chromosome 14 and *PSEN2* on chromosome 1) encode presenilin-1 and presenilin-2. These mutations lead to a gain of function, such that the γ -secretase complex generates increased amounts of A β , particularly A β 42. Point mutations in *APP* are another cause of familial AD. Some mutations lie near the β -secretase and γ -secretase cleavage sites, and others sit in the A β sequence and increase its propensity to aggregate. The gene encoding APP lies in the region of chromosome 21 that is abnormal in patients with Down syndrome. These patients develop cognitive impairment, with appearance of histopathologic findings typical of AD, in the second and third decades, followed by neurologic decline about 20 years later. A similar gene dosage effect is produced by localized chromosome 21 duplications that span the *APP* locus in some patients with familial AD. Interestingly, some people carry the A673T variant of the *APP* gene that reduces the amount of A β that is generated; these individuals are protected from AD, with absent or significantly reduced amyloid brain burden and a lower likelihood of age-associated cognitive decline.

- **Role of tau.** Because neurofibrillary tangles contain the tau protein, there has been much interest in the role of this protein in AD. Tau is a microtubule-associated protein present in axons in association with the microtubular network. With the development of tangles in AD, it shifts to a somatic-dendritic distribution, becomes hyperphosphorylated, and loses the ability to bind to microtubules. How toxic A β oligomers lead to hyperphosphorylation of tau has not yet been fully established. The formation of tangles is an important component of AD, and the increased tangle burden in the brain over the course of the illness eventually appears to become independent of the A β . How tangles injure neurons also remains poorly understood, but two possible pathways have been suggested, which are not mutually exclusive: (1) the aggregates of tau protein elicit a

stress response; and (2) the loss of tau protein destabilizes microtubules.

- **Other genetic risk factors.** The genetic locus on chromosome 19 that encodes ApoE has a strong influence on the risk of developing AD. Three alleles exist (ϵ 2, ϵ 3, and ϵ 4) based on two amino acid polymorphisms. The dosage of the ϵ 4 allele increases the risk of AD and lowers the age of onset of the disease, such that individuals with the ϵ 4 allele are overrepresented in populations of patients with AD. This ApoE isoform appears to affect multiple processes involved in AD pathogenesis: it promotes A β generation and deposition, exacerbates both A β -dependent and A β -independent tau-mediated neurodegeneration, enhances the neuroinflammatory response, and decreases brain resilience by impairing the lysosomal degradation pathway and astrocyte-dependent synapse maintenance. In addition, the ϵ 4 allele appears to be selectively toxic to GABAergic neurons, leading to hyperexcitability of neuronal networks early in the disease progression. Overall, this locus has been estimated to convey about one-quarter of the risk for development of late-onset AD. Genome-wide association studies have identified several other loci that contribute to the risk of AD; the connection between these genetic loci and the pathogenesis of AD remains to be explored.
- **Role of inflammation.** Both small aggregates and larger deposits of A β elicit an inflammatory response from microglia and astrocytes. This response probably assists in the clearance of the aggregated peptide, but it may also stimulate the secretion of mediators that cause damage. Additional consequences of the activation of these inflammatory cascades may include alterations in tau phosphorylation, oxidative injury to neurons, and aberrant pruning of synapses.
- **Basis for cognitive impairment.** Although there remains disagreement regarding the best morphological correlate of dementia in individuals with AD, it is clear that the presence of a large burden of plaques and tangles is highly associated with severe cognitive dysfunction. The number of neurofibrillary tangles correlates better with the degree of dementia than does the number of amyloid/neuritic plaques. Biochemical markers that have been correlated with the degree of dementia include loss of choline acetyltransferase, loss of synaptophysin immunoreactivity, and amyloid burden.
- **Biomarkers.** Among the more important recent developments in the understanding of AD is the discovery of possible biomarkers, which draw on the understanding of the biologic processes discussed earlier. It is now possible to demonstrate A β deposition in the brain through imaging methods that rely on 18 F-labeled amyloid-binding compounds; this approach can identify asymptomatic patients who are in the early stages of AD. Additional evidence of neuronal degeneration associated with AD-related pathologic processes includes a decreased plasma A β 42:A β 40 ratio. Together, these biomarkers have allowed for the identification of preclinical stages of AD, well in advance of the development of dementia or other signs and symptoms. This in turn has enabled the focus of pharmacologic trials to shift toward individuals in the earliest stages of the illness, in whom it is hoped interventions will slow or prevent disease progression and limit disability.

MORPHOLOGY

Grossly, the brain shows variable **cortical atrophy** marked by gyral narrowing and sulcal widening that is usually most pronounced in the frontal, temporal, and parietal lobes (Fig. 28.37). With significant atrophy, there is compensatory ventricular enlargement (hydrocephalus *ex vacuo*) secondary to reduced brain volume. Structures of the medial temporal lobe, including the hippocampus,



Fig. 28.37 Alzheimer disease Cortical atrophy is most evident on the right, where meninges have been removed. (Courtesy Dr. E.P. Richardson Jr, Massachusetts General Hospital, Boston.)

entorhinal cortex, and amygdala, are involved early in the disease course, and are usually severely atrophied in the later stages.

The major microscopic abnormalities of AD are **neuritic (senile) plaques** and **neurofibrillary tangles**. There is progressive, eventually severe, neuronal loss and reactive gliosis in the same regions that bear the burden of plaques and tangles.

Neuritic plaques are focal, spherical collections of dilated, tortuous, axonal and dendritic processes (dystrophic neurites) around a central amyloid core, which may be surrounded by a clear halo (Fig. 28.38A). Neuritic plaques range in size from 20 to 200 μm in diameter; microglial cells and reactive astrocytes are present at their periphery. Plaques are found throughout the brain, although there is usually relative sparing of primary motor and sensory cortices (this also applies to neurofibrillary tangles). The amyloid core, which can be stained by Congo red or a beta amyloid immunostain, contains several abnormal proteins. The dominant component of the amyloid plaque core is A β (Fig. 28.38B); other proteins, including components of the complement cascade, proinflammatory cytokines, α_1 -antichymotrypsin, and apolipoproteins, are present in lesser abundance. In some cases, there is deposition of A β peptides in the absence of the surrounding dystrophic neurites. These lesions, termed **diffuse plaques**, are found mainly in superficial portions of the cerebral cortex, the basal ganglia, and the cerebellar cortex.

Neurofibrillary tangles are tau-containing bundles of filaments in the cytoplasm of neurons that displace or encircle the nucleus. In pyramidal neurons, they often have an elongated “flame” shape; in rounder cells, the basket weave of filaments around the nucleus takes on a rounded contour (“globose” tangles). Neurofibrillary tangles are visible as basophilic fibrillar structures with H&E staining (Fig. 28.38C) but are demonstrated more clearly by silver (Bielschowsky) staining (Fig. 28.38D) and tau immunohistochemistry (Fig. 28.38E). They are commonly found in cortical neurons, first in the entorhinal cortex but eventually throughout the entire neocortex, as well as in

other sites such as pyramidal cells of the hippocampus, the amygdala, the basal forebrain, and the raphe nuclei. Neurofibrillary tangles are insoluble and apparently resistant to clearance in vivo, thus remaining visible in tissue sections as “ghost” or “tombstone” tangles long after the death of the parent neuron. Ultrastructurally, neurofibrillary tangles are composed predominantly of paired helical filaments along with some straight filaments that appear to have a similar composition. Aggregated tau is also present in dystrophic neurites that form the outer portions of neuritic plaques and in axons coursing through the affected gray matter as neuropil threads. In contrast to neuritic and diffuse plaques, tangles are found in other neurodegenerative diseases and are thus not specific for AD.

In addition to the diagnostic features of plaques and tangles, several other pathologic findings are seen in the setting of AD. **Cerebral amyloid angiopathy (CAA)** (see Fig. 28.14D) is an almost invariable accompaniment of AD; it can also be found in brains of individuals without AD. In contrast to the amyloid deposits in neuritic and diffuse plaques, which mainly consist of A β 42, the amyloid in CAA is predominantly comprised of A β 40.

Clinical Features

The progression of AD is slow but relentless, with a symptomatic course often running more than 10 years. Initial symptoms are forgetfulness and other memory disturbances; with progression, other symptoms emerge, including language deficits, loss of mathematical skills, and loss of learned motor skills. In the final stages, affected individuals may become incontinent, mute, and unable to walk; intercurrent disease, often pneumonia, is usually the terminal event. Current clinical trials are focused on treating subjects in early, preclinical stages of the illness, using strategies that include clearing A β from the brain through immunologic approaches (using antibodies targeting A β 42), disrupting the generation of A β with pharmacologic agents that target secretases, as well as approaches aimed at preventing alterations in tau.

Frontotemporal Lobar Degenerations

Frontotemporal lobar degenerations (FTLDs) are a heterogeneous group of disorders associated with focal degeneration of frontal and/or temporal lobes. They are clinically distinguished from AD by the pattern of cognitive deficits, with alterations in personality, behavior, and language (aphasias) preceding memory loss; however, as with other neurodegenerative diseases, postmortem pathologic evaluation is required for definitive diagnosis. Global dementia occurs with progressive disease, and a subset of patients develops extrapyramidal motor symptoms. Several clinical variants have been described based on whether the behavioral change or aphasias dominate, but they have overlapping features. FTLDs are one of the more common causes of early-onset dementia and occur at the same frequency as AD in those younger than 65 years. In the clinical setting, these diseases are commonly referred to as *frontotemporal dementias* (FTD); however, the preferred pathologic terminology highlights the anatomic pattern of involvement (lobar degeneration) rather than the clinical symptom (dementia).

Like most neurodegenerative diseases, FTLD is associated with cellular inclusions composed of specific proteins; the two most common patterns are those with tau inclusions (FTLD-tau) and those with TDP-43 inclusions (FTLD-TDP). Within each of these groups, there are heritable and sporadic forms. There is no clear relationship between the clinical subtypes of FTLD and the type of neuronal inclusions.

FTLD-tau

These are forms of FTLD in which the affected cortical regions demonstrate progressive neuronal loss and reactive gliosis, along

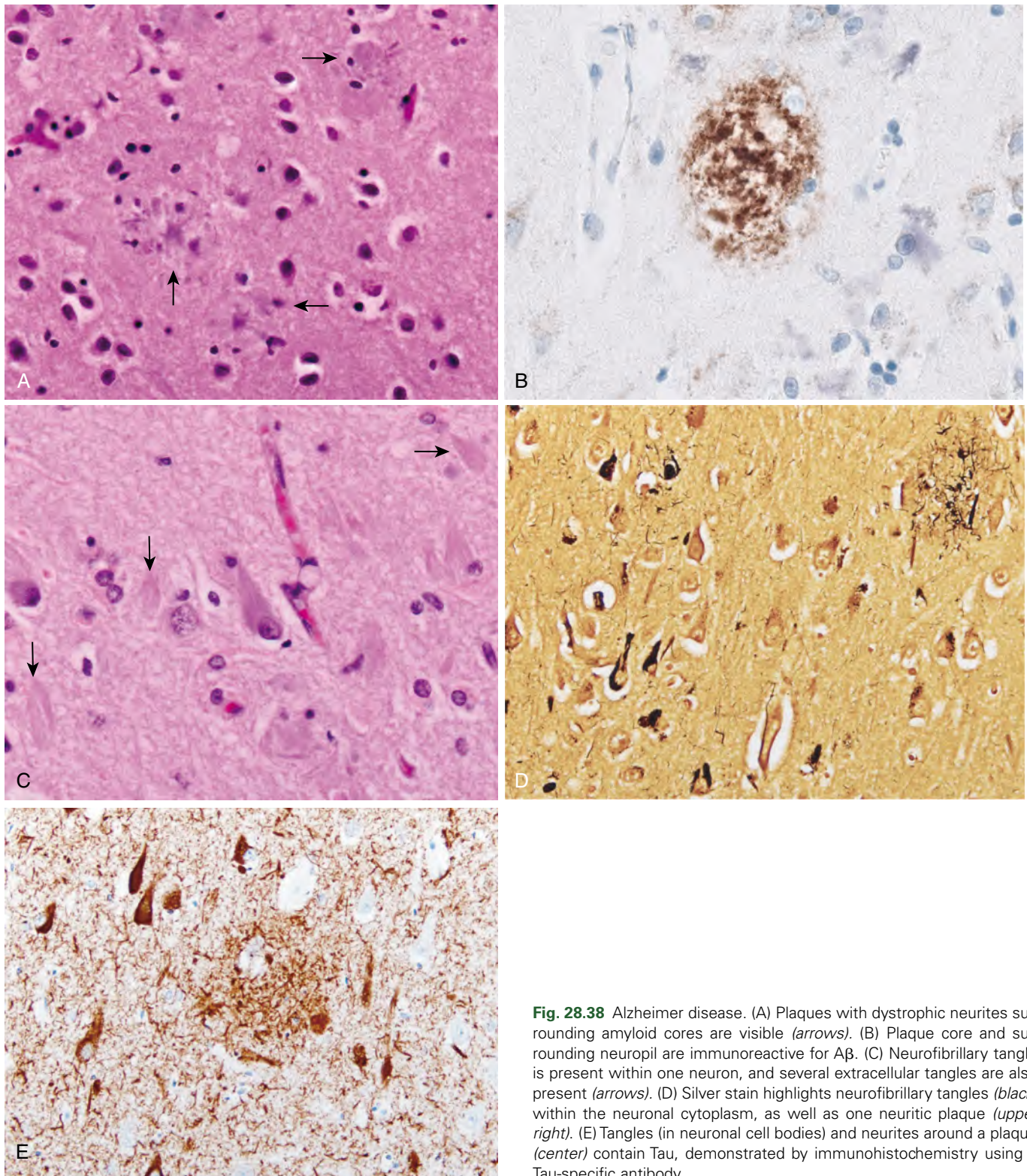


Fig. 28.38 Alzheimer disease. (A) Plaques with dystrophic neurites surrounding amyloid cores are visible (*arrows*). (B) Plaque core and surrounding neuropil are immunoreactive for A β . (C) Neurofibrillary tangle is present within one neuron, and several extracellular tangles are also present (*arrows*). (D) Silver stain highlights neurofibrillary tangles (*black*) within the neuronal cytoplasm, as well as one neuritic plaque (*upper right*). (E) Tangles (in neuronal cell bodies) and neurites around a plaque (*center*) contain Tau, demonstrated by immunohistochemistry using a Tau-specific antibody.

with the presence of cytoplasmic tau inclusions in neurons. Unlike AD, which is characterized by the combination of A β and tau deposition, FTLD-tau shows only tau aggregation and accumulation. In some cases, FTLD-tau inclusions resemble the tangles seen in AD, while in others the inclusions have smooth contours (called *Pick bodies*).

Pathogenesis

FTLD-tau may be associated with mutations in *MAPT*, the gene that encodes tau, or it may arise sporadically. As mentioned earlier, tau interacts with microtubules; its binding to microtubules may be regulated by phosphorylation. There is an inverse relationship between the degree of tau phosphorylation and its ability to bind microtubules. Tau,

particularly when hyperphosphorylated, has a propensity to aggregate. Interestingly, tau exists as a complex series of isoforms that are encoded by different mRNA splice variants; the balance between these isoforms appears to be critical for normal tau function in neurons, and disturbances in isoform ratio may also provoke tau aggregation.

Two different types of tau mutations have been described. Some missense point mutations affect tau phosphorylation, tipping the balance from microtubule binding to aggregation. Other mutations include point mutations that affect splicing; many of these are intronic and alter the loop-stem structures recognized by the spliceosome. The resulting change in isoform ratio is thought to lead to neuronal dysfunction and, as discussed earlier, may also enhance tau aggregation.

It remains unclear how abnormal tau injures neurons, although there appears to be both a loss-of-function component, as aggregation depletes neurons of tau, and a toxic gain-of-function component due to presence of aberrantly hyperphosphorylated aggregated protein.

MORPHOLOGY

There is atrophy of frontal and temporal lobes to variable extent and severity. The pattern of atrophy can often be predicted from the clinical symptoms. The atrophic regions of cortex are marked by neuronal loss, gliosis, and the presence of tau-containing fibrillary inclusions (Fig. 28.39A). Nigral degeneration occurs in some disease subtypes. Inclusions can also be found in glial cells in some forms of the disease.

In **Pick disease** (a subtype of FTLD-tau), the brain invariably shows a pronounced and frequently asymmetric atrophy of the frontal and temporal lobes, with conspicuous sparing of the posterior two-thirds of the superior temporal gyrus and only rare involvement of the parietal and occipital lobes. The atrophy can be severe, reducing gyri to a wafer-thin ("knife-edge") appearance. The neuronal loss is most severe in the outer three layers of the cortex. Some of the surviving neurons show a characteristic chromatolysis-like swelling (Pick cells), and others contain **Pick bodies**, cytoplasmic, round to oval, filamentous inclusions that stain strongly with silver stains (Fig. 28.39B) and tau immunohistochemistry.

FTLD-TDP

Some individuals with clinically diagnosed FTD and macroscopic changes of relatively localized cortical atrophy have inclusions that do not contain tau and instead contain TDP-43 (transactive response DNA-binding protein 43), an RNA-binding protein with roles in RNA processing and the formation of stress granules, which are RNA-protein aggregates that form to sequester free mRNAs when protein translation is halted under conditions of cellular stress. Individuals with FTLD-TDP typically present with either behavioral problems or language deficits, features similar to FTLD-tau.

Pathogenesis

Most cases of FTLD are sporadic, but the familial types have provided valuable insights into pathogenesis. Mutations in three different genes have been found in the inherited forms of FTLD-TDP.

- The most common familial form of FTLD-TDP is the result of an expansion of a hexanucleotide repeat in the 5' region of *C9orf72* (a gene encoding a protein of unknown function); the spectrum of disease associated with *C9orf72* expansion also includes ALS (discussed later). How the repeat expansion results in formation of aggregates of TDP-43 is a mystery at present, although recent evidence suggests that neuroinflammation secondary to *C9orf72* haploinsufficiency plays a role.
- Mutations in the *TARDBP* gene (encoding TDP-43) are less common and also occur in some familial cases of ALS. Both TDP-43

loss of function and toxic gain of function may contribute to this form of FTLD (and ALS).

- A third form of FTLD-TDP is the result of mutations in the *GRN* gene, which encodes a protein called progranulin; in contrast to *TARDBP* and *C9orf72*, mutations in *GRN* have not been linked to ALS. Progranulin mutations cause loss of function, and the disease is believed to stem from progranulin deficiency. Progranulin is a secreted protein that has been implicated in regulation of inflammation in the brain, and the proinflammatory state associated with *GRN* haploinsufficiency appears to promote accumulation of TDP-43-containing inclusions. However, the mechanism by which neuroinflammation promotes TDP-43 aggregation remains obscure.

MORPHOLOGY

The appearance of FTLD-TDP is similar to the other forms of FTLD, with variable atrophy of frontal and temporal lobes accompanied by neuronal loss and gliosis. Normally, TDP-43 is found diffusely in the nucleus but in FTLD-TDP, TDP-43 is found in inclusions (Fig. 28.39C) that may be in the cell body (neuronal cytoplasmic inclusions), in the nucleus (neuronal intranuclear inclusions), or in neuronal processes (TDP-43 threads). In the inclusions, TDP-43 is phosphorylated and ubiquitinated. Inclusions are most abundant in the frontal and temporal cortex, the striatum, and the dentate gyrus of the hippocampus. There is a strong correlation between the presence of needle-like intranuclear inclusions and progranulin mutations (Fig. 28.39D).

Parkinson Disease

Parkinson disease (PD) is a neurodegenerative disease marked by hypokinetic movement abnormalities caused by loss of dopaminergic neurons from the substantia nigra. The clinical syndrome of *parkinsonism* combines diminished facial expression (often termed masked facies), stooped posture, slowing of voluntary movement, festinating gait (progressively shortened, accelerated steps with forward leaning), rigidity, and a "pill-rolling" tremor. This type of motor disturbance is seen in several conditions that are associated with damage to the nigrostriatal dopaminergic system. Although PD is the most common cause of parkinsonism, similar symptoms can be induced by dopaminergic antagonists or by toxins that selectively damage the dopaminergic system. Also discussed later are other rare diseases that have parkinsonism as part of the clinical presentation.

An acute parkinsonian syndrome and destruction of neurons in the substantia nigra follows exposure to a neurotoxin MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine), discovered as a contaminant in illicitly synthesized batches of the opioid meperidine. This toxin has been used to create animal models of PD, which are used to test new therapies. Epidemiologic evidence has also suggested that pesticide exposure is a risk factor for PD, while caffeine and nicotine appear to be protective.

Pathogenesis

PD is associated with mutations in genes that lead to protein aggregation, mitochondrial abnormalities, and neuronal loss in the substantia nigra and elsewhere in the brain. As in other neurodegenerative diseases, most PD is sporadic, but identification of genetic causes has helped shed light on its pathogenesis.

- **Mutations affecting α -synuclein.** The first mutations identified as a cause of autosomal dominant PD involved *SNCA*, a gene that encodes α -synuclein, an abundant lipid-binding protein normally localized to synapses. This protein is a major component of the *Lewy body*, which is the diagnostic hallmark of PD. Mutations in *SNCA* are rare; they include inherited point mutations and somatic amplification of the region of chromosome 4q21 that contains the gene. Like A β in AD,

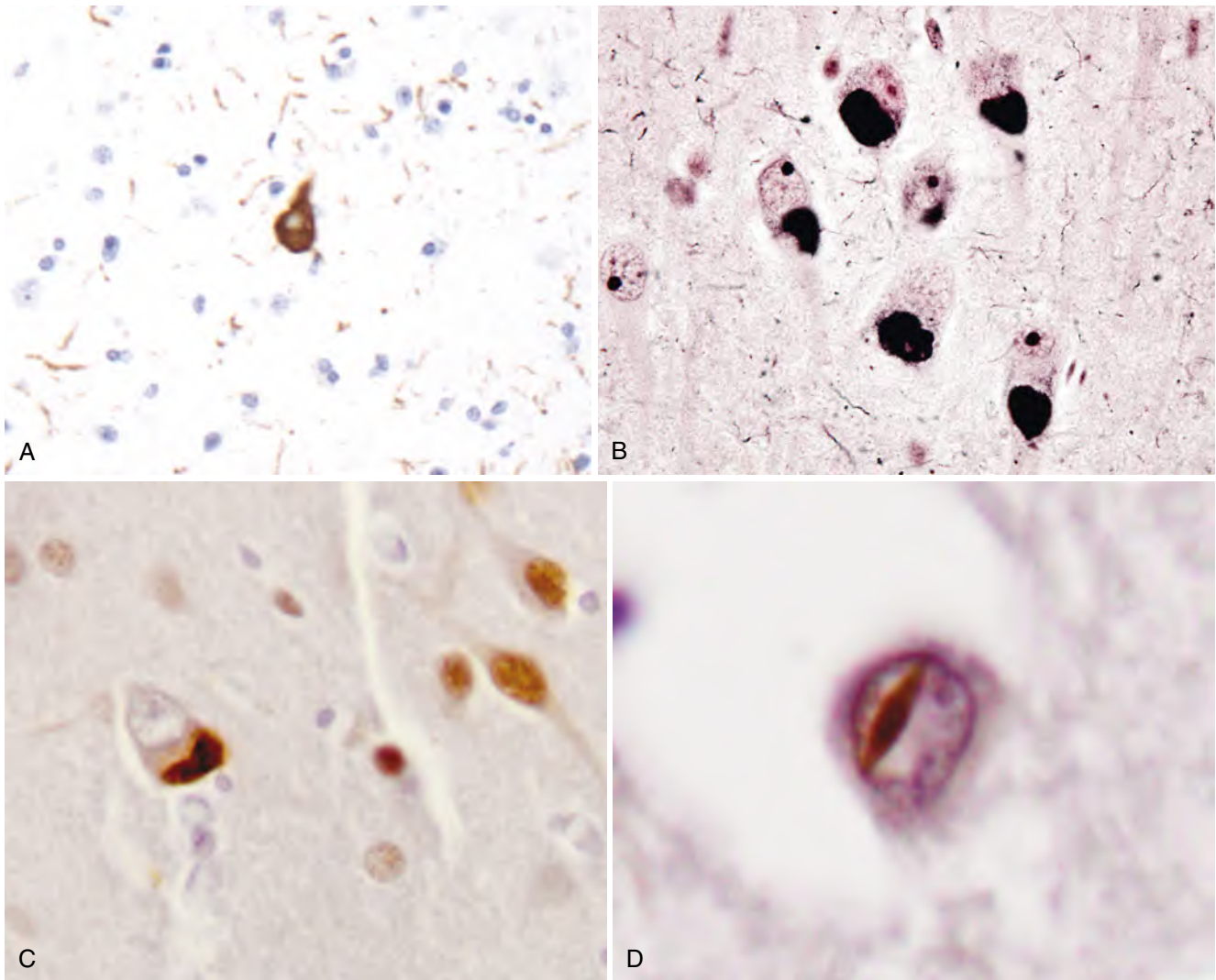


Fig. 28.39 Frontotemporal lobar degenerations (FTLDs). (A) FTLN-tau. A tangle is present along with numerous tau-containing neurites. (B) Pick disease. Pick bodies are round, homogeneous neuronal cytoplasmic inclusions that stain intensely with silver stains. (C) FTLN-TDP. Cytoplasmic inclusions containing TDP-43 are seen in association with loss of normal nuclear immunoreactivity. (D) FTLN-TDP. With progranulin mutations, the TDP-43-containing inclusions are commonly intranuclear and needle-like in shape.

α -synuclein forms aggregates; of these, small oligomers appear to be the most toxic to neurons. There is also evidence that aggregates can be released from one neuron and taken up by another, suggesting a prion-like pattern of spread within the brain. Consistent with this idea, α -synuclein-containing aggregates (Lewy bodies and Lewy neurites) are first seen in the medulla and then in contiguous areas of the brain, ascending through the brainstem and extending into limbic structures and finally the neocortex.

- **Mitochondrial dysfunction** has been implicated as a contributing factor based on autosomal recessive forms of PD that are caused by mutations in genes that encode the proteins DJ-1, PINK1, and Parkin. DJ-1 has multiple cellular roles, including acting as a transcriptional regulator, but in settings of oxidative stress, it can relocate to the mitochondria and have cytoprotective effects. PINK1 is a kinase that localizes to the outer membrane of dysfunctional mitochondria, where it recruits and phosphorylates parkin. This modification activates Parkin, an E3 ubiquitin ligase that ubiquitinates mitochondrial proteins that serve as key targets for the autophagic degradation of mitochondria (termed mitophagy). Under normal circumstances,

the activation of the PINK-Parkin Pathway results in clearance of dysfunctional mitochondria through mitophagy; this process is impaired by defects in PINK1 or Parkin. Intriguingly, levels of mitochondrial complex I, a component of the oxidative phosphorylation cascade, are reduced in the brains of patients with sporadic PD.

- **Heterozygous mutations in the *GBA1* gene encoding lysosomal enzyme glucocerebrosidase** are an important risk factor for development of PD, accounting for about 5% of PD cases. (Homozygous mutations in this enzyme cause Gaucher disease, a lysosomal storage disease described in more detail in Chapter 5). This genetic link suggests that dysfunction of the autophagy-lysosome pathway contributes to PD pathogenesis, most likely through abnormal metabolism of α -synuclein, impaired autophagy, activation of the unfolded protein response, and increased inflammation.
- **Mutations in the gene encoding LRRK2** (leucine-rich repeat kinase 2) give rise to autosomal dominant PD and are also found in some late-onset, apparently sporadic, cases of the disease. LRRK2 is a cytoplasmic kinase; several of the pathogenic mutations increase its kinase activity, and recent evidence suggests that hyperphosphorylation of small

GTPase Rab proteins, possibly together with other LRRK2 substrates, plays a key role in PD pathogenesis. However, the molecular mechanisms linking LRRK2-mediated hyperphosphorylation, α -synuclein accumulation, and neurodegeneration have not yet been elucidated.

MORPHOLOGY

A characteristic gross finding in PD is **pallor of the substantia nigra** (compare Fig. 28.40A and B) and locus ceruleus, which is due to loss of the pigmented, catecholaminergic neurons in these nuclei. **Lewy bodies** (Fig. 28.40C) are usually found in some of the remaining neurons; these are single or multiple, cytoplasmic, eosinophilic, round to elongated inclusions that often have a dense core surrounded by a pale halo. Ultrastructurally, Lewy bodies are composed of fine filaments, densely packed in the core but loose at the rim, containing α -synuclein. Lewy bodies can also be found in the cholinergic neurons of the basal nucleus of Meynert, as well as in other brainstem nuclei including the locus ceruleus and the dorsal motor nucleus of the vagus. Areas of neuronal loss also typically show gliosis. Lewy neurites are dystrophic processes that contain aggregated α -synuclein.

Clinical Features

The diagnosis of PD is based on the triad of tremor, rigidity, and bradykinesia (slow movement) in the absence of a toxic or other known underlying etiology. Although these clinical features reflect the decreased dopaminergic innervation of the striatum, the disease is not restricted to dopaminergic neurons or to the basal ganglia; in fact, degeneration of the substantia nigra (which results in the motor symptoms) represents a mid-stage in a progressive disease that begins lower in the brainstem and eventually progresses to involve the cerebral cortex, leading to cognitive impairment.

The severity of the motor symptoms in PD is proportional to the dopamine deficiency, which can, at least initially, be corrected by replacement therapy with L-DOPA (the immediate precursor of dopamine). However, treatment does not arrest the progression of the disease; with time, drug therapy becomes less effective, and symptoms more difficult to manage. Deep brain stimulation has emerged over the past decade as another therapy for the motor symptoms of PD.

Dementia with Lewy Bodies

About 10% to 15% of individuals with PD develop dementia, particularly with advancing age, characterized by a fluctuating course, visual hallucinations, and prominent frontal signs. Although some individuals with dementia have pathologic evidence of Alzheimer disease (or, less frequently, other degenerative diseases), in others the most prominent histologic correlate is the presence of widespread Lewy bodies containing α -synuclein in neurons of the cortex and brainstem; hence the name *dementia with Lewy bodies* (DLB). In some cases, DLB represents an advanced stage of PD in which protein aggregates have “spread,” possibly through propagation of misfolded proteins, to neurons in the cerebral cortex. In other individuals, however, dementia is the first presenting symptom, and motor symptoms appear later, suggesting a descending rather than ascending pattern of disease spread.

Cortical Lewy bodies are less distinctive than those in the brainstem but are also composed predominantly of α -synuclein. Immunohistochemical staining for α -synuclein also reveals the presence of abnormal neurites that contain aggregated protein, called *Lewy neurites*. In DLB, the gross pathologic findings typically include depigmentation of the substantia nigra and locus ceruleus, paired with relative preservation of the cortex, hippocampus, and amygdala. The burden of cortical Lewy bodies is usually low, and the mechanism by which this disease wreaks havoc on cognitive functioning is not clear.

Despite their distinct clinical presentations, PD and DLB are currently considered to be part of the same disease spectrum: not only are

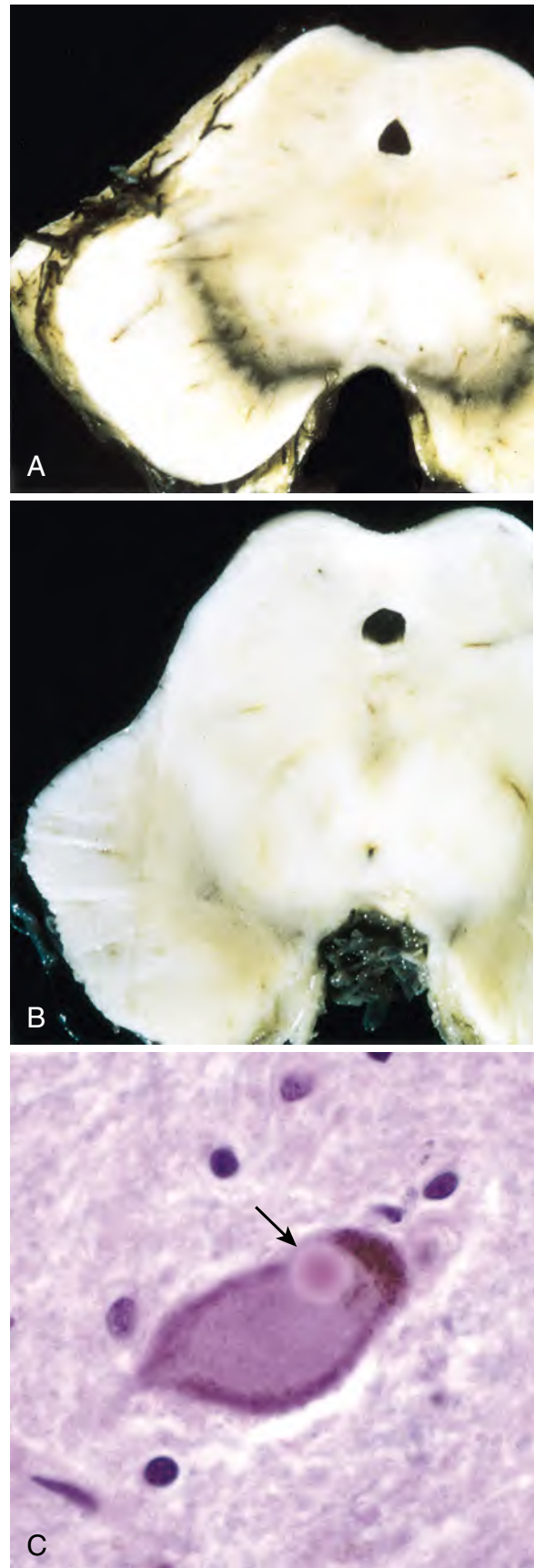


Fig. 28.40 Parkinson disease. (A) Normal substantia nigra. (B) Depigmented substantia nigra in idiopathic Parkinson disease. (C) Lewy body in a substantia nigra neuron, staining bright pink (arrow).

they caused by many of the same genetic alterations, but they also show the same α -synuclein molecular fold, which is distinct from the α -synuclein fold seen in multiple-system atrophy (discussed later).

Atypical Parkinsonian Syndromes

As discussed earlier, the clinical syndrome of parkinsonism, with bradykinesia and rigidity, reflects dysfunction of the dopaminergic nigrostriatal pathway. In addition to the forms of PD already discussed, there are a variety of disorders that include parkinsonism as a component of the symptoms. These diseases, in general, are minimally responsive to treatment with L-DOPA; they are also distinguished from PD through the presence of additional signs and symptoms. For these reasons, they are considered to be *atypical parkinsonian syndromes* or *Parkinson-plus syndromes*. These disorders include progressive supranuclear palsy (PSP) and corticobasal degeneration (CBD), both of which are tauopathies, and multisystem atrophy, a synucleinopathy. While PSP and CBD have parkinsonian features, they also show frontotemporal degeneration and are currently classified as subtypes of FTLT-tau.

Progressive Supranuclear Palsy

Progressive supranuclear palsy (PSP) is a tauopathy in which affected individuals develop progressive truncal rigidity, disequilibrium with frequent falls, and difficulty with voluntary eye movements. Other common symptoms include nuchal dystonia, pseudobulbar palsy, and a progressive dementia. The onset is usually between the fifth and seventh decades, with males affected approximately twice as frequently as females. The disease is often fatal within 5 to 7 years of onset.

The pathologic hallmark of PSP is the presence of tau-containing inclusions in neurons and glia, with the greatest burden being located in the brainstem and deep gray matter. Although causative mutations in the tau gene have been identified in only a few cases, risk for sporadic PSP is linked to polymorphisms near the tau gene locus that enhance expression of tau isoforms that are more likely to form inclusions. Other risk alleles have been identified through genome-wide association studies, but how they influence development of PSP is unclear.

Corticobasal Degeneration

Corticobasal degeneration (CBD) is a progressive tauopathy that is most often characterized by extrapyramidal rigidity, asymmetric motor disturbances (jerking movements of limbs), and impaired higher cortical function (typically in the form of apraxia). As with PSP, cognitive decline may occur, typically later in the illness. The same tau gene variants linked to PSP are also highly associated with CBD. Overall, CBD and PSP share many clinical and pathologic features; in CBD a greater fraction of Tau inclusions localize to the cerebral cortex. Interestingly, tau inclusions in each of these diseases show a unique molecular fold in vitro.

MORPHOLOGY

In **PSP**, there is widespread neuronal loss in the globus pallidus, subthalamic nucleus, substantia nigra, colliculi, periaqueductal gray matter, and dentate nucleus of the cerebellum. Tangles with a globular appearance (as compared to the more filamentous tangles seen in Alzheimer disease) are found in the affected regions in neurons and in glia. In **CBD**, the brain shows cortical atrophy, mainly of the motor, premotor, and anterior parietal lobes. In affected regions of cortex, there is severe loss of neurons, gliosis, and “**ballooned**” neurons. Tau immunoreactivity has been found in astrocytes (“tufted astrocytes”), oligodendrocytes (“coiled bodies”), basal ganglionic neurons, and, variably, cortical neurons. Clusters of Tau-positive processes around astrocytes (“astrocytic plaques”) and the presence of Tau-positive threads in gray and white matter may be the most specific pathologic findings of CBD. The substantia nigra and locus ceruleus show neuronal loss, ballooned neurons, and tangles.

Multiple System Atrophy

Multiple system atrophy (MSA) is a sporadic disorder that primarily affects white matter and is characterized by α -synuclein inclusions in the cytoplasm of oligodendrocytes. In contrast to other neurodegenerative diseases, the histopathologic changes in MSA are mainly observed in glial cells and are associated with degeneration of white matter tracts. There is also accompanying neuronal degeneration, typically without neuronal inclusions. The “multiple” in the term *multiple system atrophy* refers to three distinct neuroanatomic circuits that are commonly involved: the striatonigral circuit (leading to parkinsonism), the olivopontocerebellar circuit (leading to ataxia), and the autonomic nervous system including the central elements (leading to autonomic dysfunction, with orthostatic hypotension as a prominent component). In a given individual, one of these components may predominate at the onset of the illness, but typically the other systems become affected as MSA progresses.

Pathogenesis

As in PD, α -synuclein is the major component of the inclusions. While no causative mutations in the gene encoding α -synuclein have been identified, polymorphisms near this gene appear to confer increased risk. The relationship between glial cytoplasmic inclusions and disease is supported by observations showing that the burden of inclusions increases as the disease progresses and that glial cytoplasmic inclusions can occur in the absence of neuronal loss, suggesting that they are the primary pathologic event. However, the origin of the α -synuclein in oligodendrocytes remains perplexing because α -synuclein is a neuronal protein associated with synaptic vesicles. Several studies have shown that there is no up-regulation of α -synuclein expression in white matter or in oligodendrocytes in MSA, leading to the suggestion that oligodendrocytes acquire α -synuclein aggregates secondarily from injured or dying neurons. When α -synuclein is present in oligodendrocytes, they become more sensitive to oxidative stress and show impaired interaction with the extracellular matrix.

MORPHOLOGY

The pathologic findings in MSA match the clinical presentation in a given case. In **cerebellar forms**, there is atrophy of the cerebellum, including the cerebellar peduncles, pons (especially the base, Fig. 28.41A), and medulla (especially the inferior olive); in **parkinsonian forms**, the atrophy involves the substantia nigra and striatum (especially the putamen). Autonomic symptoms are related to loss of the catecholaminergic nuclei of the medulla and the intermediolateral cell column of the spinal cord. Atrophic brain regions show evidence of neuronal loss.

The diagnostic **glial cytoplasmic inclusions** were originally demonstrated in oligodendrocytes with silver impregnation methods and contain α -synuclein as well as ubiquitin (see Fig. 28.41B).

Huntington Disease

Huntington disease (HD) is an autosomal dominant disease caused by degeneration of striatal neurons that is characterized by a progressive movement disorder and dementia. Jerky, hyperkinetic, sometimes dystonic movements involving all parts of the body (*chorea*) are characteristic; affected individuals may later develop bradykinesia and rigidity. The disease is relentlessly progressive and uniformly fatal, with an average course of about 15 years.

Pathogenesis

HD is a prototypic polyglutamine trinucleotide repeat-expansion disease (Chapter 5). The gene for HD, *HTT*, located on chromosome

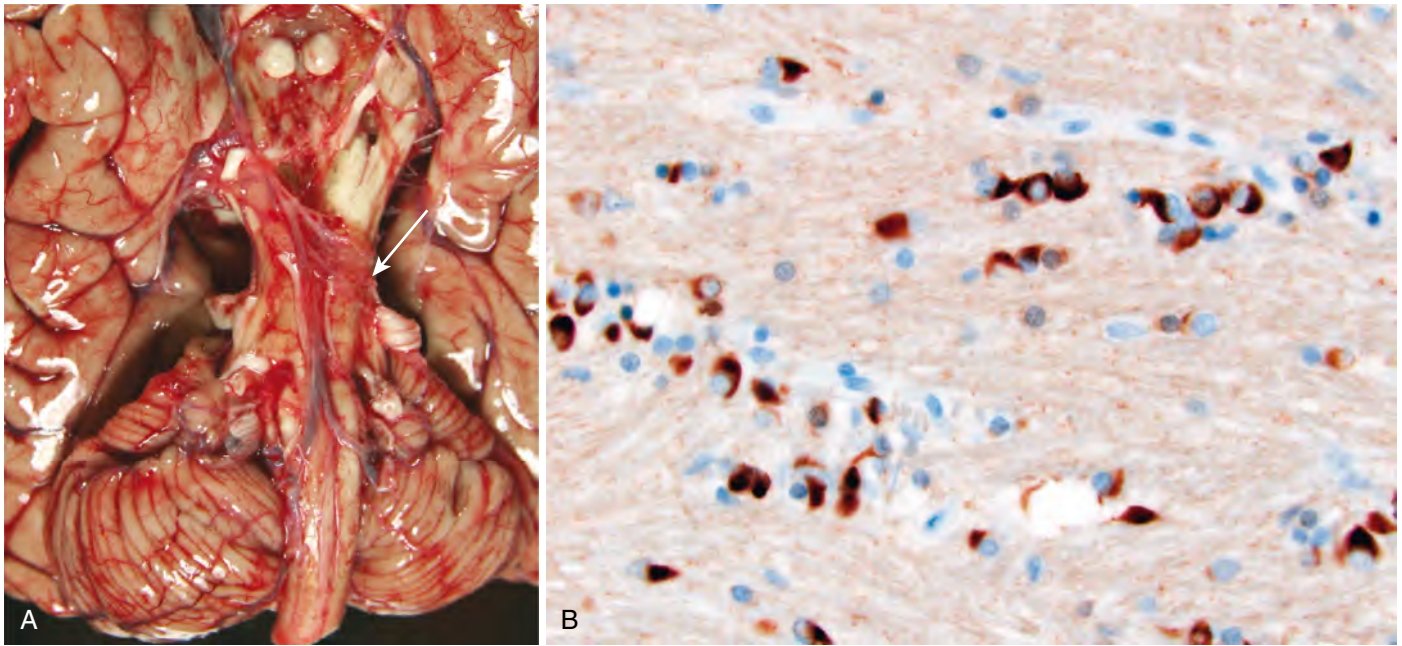


Fig. 28.41 Multiple system atrophy (MSA). (A) Severe atrophy of the basis pontis (arrow) in a case of MSA. (B) Inclusions in oligodendrocytes contain α -synuclein (brown stain).

4p16.3, encodes a 348-kD protein known as *huntingtin*. In the first exon of the gene, there is a stretch of CAG repeats that encodes a polyglutamine region near the N terminus of the protein. The normal *HTT* gene contains 6 to 35 copies of the repeat; if the number of repeats increases beyond this range, disease occurs. There is an inverse relationship between the number of repeats and the age of onset, such that higher numbers of repeats are associated with earlier onset. Repeat expansions occur during spermatogenesis, so that paternal transmission is associated with earlier onset in the next generation, a phenomenon termed *anticipation*. In contrast to many other neurodegenerative diseases, there is no sporadic form of HD. New mutations are uncommon; most apparently sporadic cases are explained by nonpaternity, the death of a parent before the disease develops, or an unaffected father with a mild repeat expansion that enlarges to a pathogenic size during spermatogenesis.

The biologic function of normal huntingtin remains unknown, and it appears that the expansion of the polyglutamine region bestows a toxic gain-of-function on huntingtin. For this reason, various approaches to silencing expression of the mutant allele are being investigated as potential therapies. It is interesting to note that although huntingtin is expressed in all tissues of the body, the deleterious effects of mutant huntingtin occur only in selected parts of the central nervous system.

Although development of intranuclear inclusions containing huntingtin is a pathologic hallmark of HD, this process does not appear to be directly involved in cell injury and may actually be neuroprotective, as striatal neurons overexpressing mutant huntingtin that form inclusions survive the longest. Although the mechanism for this neuroprotective effect is not clear, it is generally thought that it stems from the sequestration of neurotoxic forms of mutated huntingtin. Transcriptional dysregulation has been implicated in HD, based on the observation that mutant forms of huntingtin bind various transcriptional regulators. Certain transcription factors that interact with mutant huntingtin are involved in mitochondrial biogenesis and protection against oxidative injury, and their reduced activity may result in increased susceptibility to oxidative stress. Other alterations that

may contribute to the pathogenesis of HD include reduced expression of the growth factor brain-derived neurotrophic factor (BDNF), and disruption of proteasomal and autophagic degradation pathways. Interestingly, emerging evidence suggests that aggregated huntingtin can be transferred between cells, another example of possible prion-like spread of a pathogenic protein.

MORPHOLOGY

The brain is small and shows striking **atrophy of the caudate nucleus and putamen**, components of the dorsal striatum; The globus pallidus may atrophy secondarily, and the lateral and third ventricles are dilated (Fig. 28.42). Atrophy is frequently also seen in the frontal lobe, less often in the parietal lobe, and occasionally throughout the entire cortex. On microscopic examination, there is **profound loss of striatal neurons**; the most marked changes are found in the caudate nucleus, especially in the tail and in portions nearer the ventricle. Pathologic changes develop in a medial-to-lateral direction in the caudate and from dorsal to ventral in the putamen. Medium-sized, spiny neurons that use γ -aminobutyric acid as their neurotransmitter, along with enkephalin, dynorphin, and substance P, are especially affected. Fibrillary gliosis is more extensive than in the usual reaction to neuronal loss. **Protein aggregates containing huntingtin** can be found in neurons in the striatum and cerebral cortex (Fig. 28.42, inset).

Clinical Features

The loss of striatal neurons, which function to dampen motor activity, results in increased motor output, often manifested as choreoathetosis. The cognitive changes associated with the disease are probably related to the loss of neurons in the cerebral cortex. The age at onset is most commonly in the fourth and fifth decades and, as stated earlier, is related to the length of the CAG repeat in the *HTT* gene. Motor symptoms often precede the cognitive impairment. The movement disorder of HD is choreiform, with increased and involuntary jerky movements of all parts of the body; writhing movements of the extremities are typical. Early symptoms of higher cortical dysfunction include forgetfulness as well as

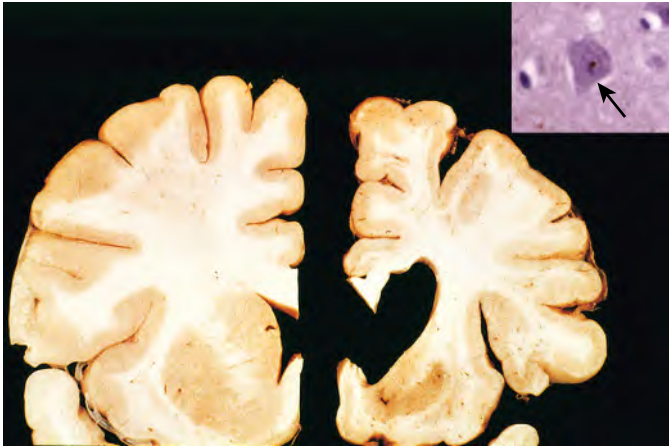


Fig. 28.42 Huntington disease. Normal brain hemisphere on the left compared with the hemisphere with Huntington disease on the right showing atrophy of the striatum and ventricular dilation. *Inset*, Intranuclear inclusions in neurons are highlighted by an immunohistochemical stain for ubiquitin (arrow). (Courtesy Dr. J.-P. Vonsattel, Columbia University, New York.)

cognitive and affective dysfunction; as the disease progresses, severe dementia appears..

The most common immediate cause of death in HD patients is pneumonia, typically occurring at an advanced stage of the disease. Tragically, HD patients resort to suicide at a rate that is approximately twice that of the general population. Given the ability to screen for disease-causing mutations, one might assume that genetic screening of individuals at risk would be routine; however, in the absence of effective therapy and given the devastating nature of the disease, the ethics of screening is a matter of debate.

Spinocerebellar Degenerations

The shared feature of this group of degenerative diseases is that they cause neuronal loss and dysfunction in the cerebellum, the spinal cord, and peripheral nerves. In other regards, they are quite heterogeneous, with differences in genetics, age at onset, and signs and symptoms. Genetic analysis has impacted the classification of these diseases but has yet to lead to clearer insight into their pathogenesis or to effective treatments.

The term *spinocerebellar ataxia* (SCA) is applied to a group of autosomal dominant inherited spinocerebellar degenerations. We will also briefly discuss Friedreich ataxia and ataxia-telangiectasia, two of the more common autosomal recessive disorders that are characterized by spinocerebellar degeneration.

Spinocerebellar Ataxias

This group of spinocerebellar degenerations is remarkably heterogeneous; current classifications that rely on a combination of genetic, clinical, and pathologic findings include >40 distinct subtypes. Here we focus on the common varieties of pathogenic mutations that are found in spinocerebellar ataxia (SCA).

Pathogenesis

The genes responsible for over half of the numerous SCAs have been identified. Three distinct types of pathogenic mutations are recognized:

- **CAG repeat expansions** leading to inclusion of polyglutamine in encoded proteins, similar to HD. Eight types of SCA are caused by this type of mutation, and in each the mutated protein accumulates

in neurons as nuclear inclusions, suggesting the involvement of pathogenic mechanisms previously discussed for HD. The group of polyglutamine diseases includes SCA1, SCA2, SCA3 (also known as *Machado-Joseph disease*), SCA6, SCA7 (relatively unique in that it includes visual impairment), SCA8, SCA17, and dentatorubral-pallidoluysian atrophy (DRPLA).

- **Expansion of noncoding repeats**, similar to myotonic dystrophy. This mechanism is involved in six forms of SCA (SCA10, SCA12, SCA27B, SCA31, SCA36, and SCA37). In most, the genetic locus underlying the disease has been defined. How these noncoding repeat expansions lead to disease is unknown.
- **Conventional mutations**. The remaining SCAs are associated with copy number variations, deletions, insertions, and/or point mutations in a variety of genes that alter proteins involved in signal transduction (e.g., receptors, ion channels, kinases).

Friedreich Ataxia

Friedreich ataxia is an autosomal recessive disease characterized by progressive ataxia, spasticity, weakness, sensory neuropathy, and cardiomyopathy. It generally begins in the first decade of life with gait ataxia, followed by hand clumsiness and dysarthria. Deep tendon reflexes are depressed or absent, but an extensor plantar reflex is typically present. Joint position and vibratory sense are impaired, and there is sometimes loss of pain, temperature sensation, and light touch; many affected individuals develop pes cavus and kyphoscoliosis. Most patients become wheelchair users within about 5 years of onset, and life expectancy is typically limited to 40 or 50 years of age. The accompanying cardiomyopathy is associated with a high incidence of arrhythmias and congestive heart failure, which contribute to the deaths of most affected individuals. Concomitant diabetes is found in up to 25% of patients.

Friedreich ataxia is caused by expansion of a GAA trinucleotide repeat in the first intron of *FXN*, a gene on chromosome 9q13 that encodes *frataxin*, a protein found in the mitochondrial inner membrane. Frataxin is involved in the assembly of the iron-sulfur cluster enzymes of mitochondrial complexes I and II. The principal effect of the GAA repeat expansion is to induce changes in chromatin structure that lower the level of frataxin expression. With reduced frataxin, there is decreased mitochondrial oxidative phosphorylation (similar to the defect in mitochondrial encephalomyopathies) as well as increased free iron; the presence of free iron within the mitochondria may contribute to oxidative stress. Most cases of Friedreich ataxia are associated with GAA-repeat expansion in both alleles, but sometimes one allele has a repeat expansion and the other harbors a point mutation that causes a loss of function.

Ataxia-Telangiectasia

Ataxia-telangiectasia (Chapter 7) is an autosomal recessive disorder characterized by an ataxic-dyskinetic syndrome beginning in early childhood, with the subsequent development of telangiectasias in the conjunctiva and skin, along with immunodeficiency. The ataxia-telangiectasia mutated (*ATM*) gene on chromosome 11q22-q23 encodes a kinase with a critical role in orchestrating the cellular response to double-stranded DNA breaks. In addition to this critical cellular role, signals mediated through ATM also regulate apoptosis, maintenance of telomeres, mitochondrial homeostasis, response to oxidative stress, and the ubiquitin-proteasomal degradation system. It remains unclear which of these pathways contribute to the degenerative phenotype observed in the setting of loss of ATM protein in neurons. Intriguingly, there are comparable patterns of neurodegeneration associated with other diseases in which repair of single-stranded DNA breaks is disrupted.

Clinical Features

The disease is relentlessly progressive, with death typically occurring in early adulthood. Common initial symptoms are recurrent sinopulmonary infections and unsteadiness in walking; later, speech is noted to become dysarthric and eye movement abnormalities develop. Many affected individuals develop lymphoid neoplasms, which are most often T-cell leukemias.

Amyotrophic Lateral Sclerosis

Amyotrophic lateral sclerosis (ALS) is a progressive disorder in which there is loss of upper motor neurons in the cerebral cortex and lower motor neurons in the spinal cord and brainstem. Loss of these neurons results in denervation of muscles, producing weakness that becomes profound as the disease progresses. The disease has an overall incidence of about 2 per 100,000 people and commonly emerges in the fifth decade or later. Sporadic ALS is more common than familial ALS, which may account for up to 20% of cases. Notably, the genetics, pathogenesis, neuropathology, and clinical features of ALS demonstrate significant overlap with FTLT. Hence, although these two disorders are considered distinct diseases, it is becoming increasingly appreciated that FTLT and ALS may represent variations across a common disease spectrum.

Pathogenesis

Both sporadic and familial forms of ALS are associated with degeneration of upper and lower motor neurons, often in association with toxic protein accumulation. Over two dozen genetic loci have been identified as causing familial ALS, with nearly all being autosomal dominant disorders.

One of the earliest discovered hereditary forms of ALS is caused by mutations in the gene encoding copper-zinc superoxide dismutase (*SOD1*) on chromosome 21; this variant accounts for about 20% of familial cases. Initially, identification of *SOD1* mutations suggested that neuronal injury in ALS might reflect an impaired capacity to detoxify free radicals, but it is now believed that mutations lead to an adverse gain-of-function phenotype associated with mutant *SOD1* protein. It appears that mutated *SOD1* protein misfolds and forms aggregates (which can include wildtype protein) that cause cellular injury through a variety of postulated mechanisms, including dysregulation of proteasomal and autophagic catabolism, disruption of axonal transport and mitochondrial function, and sequestration of other proteins within the aggregates. Accumulation of protein aggregates can eventually trigger the unfolded protein response, with subsequent initiation of apoptosis. The overall importance of protein degradation pathways, and particularly autophagy (Chapter 1), is reinforced by the discovery of mutations in genes implicated in protein degradation that are also associated with familial ALS.

The most common mutation that gives rise to ALS and FTLT simultaneously is an expansion of a hexanucleotide repeat in the 5'-untranslated region of a transcript of unknown function, *C9orf72*. This mutation is estimated to cause up to 40% of familial ALS. Non-AUG-initiated translation (in all three reading frames) can occur from these expanded repeats, and neuronal deposits of the encoded peptides have been found in the brains of individuals bearing this mutation; however, it is unknown whether these aggregates contribute to disease.

Other genetic loci that cause ALS and FTLT encode RNA binding proteins located at stress granules, such as TDP-43 and FUS; proteins that mediate selective autophagy, such as sequestosome (SQSTM1, also known as p62) as well as optineurin (OPTN) and ubiquilin-1 (UBQN1); and cytoskeletal proteins, such as tubulin alpha 4a (*TUBA4A*) and profilin 1 (*PFN1*). The underlying link between alterations in RNA-binding

proteins and manifestations of motor neuron disease remains incompletely understood, but seems to involve aberrant RNA processing and expression of cryptogenic exons, while the aggregation of the protein in the cytoplasm leads to abnormal proteostasis and autophagy defects that are common to many of the proteinopathies.

MORPHOLOGY

The anterior roots of the spinal cord are thin (Fig. 28.43A) due to **loss of lower motor neuron axons**, and the precentral motor gyrus in the cortex may be atrophic in especially severe cases. There is a reduction in the number of anterior horn neurons throughout the length of the spinal cord, associated with reactive gliosis. Similar findings are seen in the hypoglossal, ambiguous, and motor trigeminal cranial nerve nuclei. Remaining neurons often contain PAS-positive cytoplasmic inclusions called Bunina bodies (which appear to be remnants of autophagic vacuoles) and TDP-43-positive cytoplasmic inclusions (which are seen in sporadic cases and familial cases caused by mutations in *C9orf72* and the TDP-43 gene [*TARDBP*], but are absent in *SOD1*-mutant and *FUS*-mutant cases). Skeletal muscles innervated by the degenerated lower motor neurons show neurogenic atrophy. Loss of the upper motor neurons leads to degeneration of the corticospinal tracts, resulting in volume loss and absence of myelinated fibers, which may be particularly evident at the lower segmental levels (Fig. 28.43B).

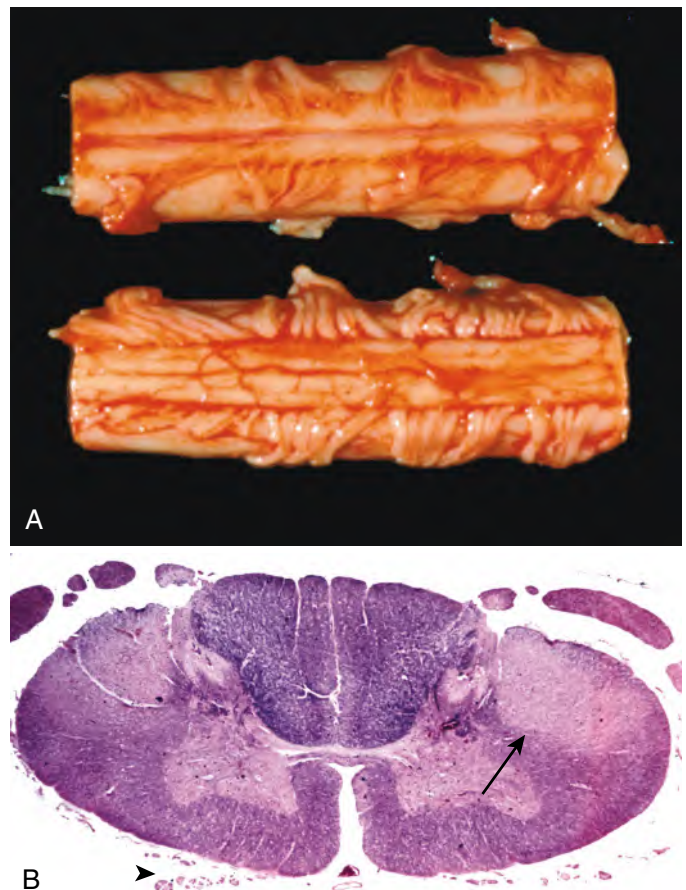


Fig. 28.43 Amyotrophic lateral sclerosis. (A) Segment of spinal cord viewed from anterior (upper) and posterior (lower) surfaces showing attenuation of anterior (motor) roots compared with posterior (sensory) roots. (B) Spinal cord showing loss of myelinated fibers (lack of stain) in corticospinal tracts (best seen on the right side of this specimen; arrow) as well as degeneration of anterior roots (arrowhead).

Clinical Features

Early symptoms of ALS include asymmetric weakness of the hands, manifested as dropping of objects and difficulty performing fine-motor tasks, and cramping and spasticity of the arms and legs. As the disease progresses, muscle strength and bulk diminish, and involuntary contractions of individual motor units, termed fasciculations, occur. The disease eventually involves the respiratory muscles, leading to recurrent bouts of pneumonia. Although most affected individuals have a combination of upper and lower motor neuron involvement, other patterns are observed. The term *progressive muscular atrophy* applies to uncommon cases in which lower motor neuron involvement predominates, while *primary lateral sclerosis* refers to cases with mostly upper motor neuron involvement. In some affected individuals, degeneration of the lower brainstem cranial motor nuclei occurs early and progresses rapidly, a pattern referred to as *progressive bulbar palsy* or *bulbar ALS*. In these individuals, abnormalities of swallowing and speech dominate, and the clinical course is inexorable during a 1- or 2-year period. When bulbar involvement is less severe, about half of affected individuals are alive 2 years after diagnosis. The motor neurons innervating extraocular muscles are among the last to be involved in ALS; with long survival, usually associated with ventilator support, even this form of motor output fails. Familial cases develop symptoms earlier than most sporadic cases, but the clinical course is comparable.

Although ALS is considered a disease of the motor system, many affected individuals have evidence of more widespread cerebral cortical disease. The clinical presentation of cerebral disease is usually a frontotemporal dementia, with pathologic findings most often matching those of FTLT associated with TDP-43 inclusions. The mechanistic link between these two processes is further strengthened by the presence of TDP-43-containing inclusions in many cases of ALS as well as overlap of genetic alterations in ALS and FTLT.

Other Motor Neuron Diseases

In addition to ALS, motor neurons are the primary target in a few other neurodegenerative diseases with distinct etiology and pathogenesis.

Spinal and Bulbar Muscular Atrophy (Kennedy Disease)

This X-linked polyglutamine repeat-expansion disease is characterized by distal limb amyotrophy and bulbar signs, such as atrophy and fasciculations of the tongue and dysphagia. These clinical findings are due to degeneration of lower motor neurons in the spinal cord and brainstem. The expanded CAG repeat occurs in the first exon of the androgen receptor and results in androgen insensitivity, gynecomastia, testicular atrophy, and oligospermia. The basis for the selective motor neuron involvement is unclear, but as in other polyglutamine expansion diseases, there are intranuclear inclusions that contain the involved protein. Progression of the disease is slow, and most affected individuals remain ambulatory until late in the disease course. Life span is normal.

Spinal Muscular Atrophy

Spinal muscular atrophy (SMA) includes a group of genetically linked disorders of childhood characterized by marked loss of lower motor neurons that results in progressive weakness. The most common variants of SMA are linked to chromosome 5q and are caused by a loss of SMN, a protein that is involved in the assembly of the spliceosome. The most common and severe form of 5q-SMA (SMA type I, Werdnig-Hoffmann disease) has onset during the first year of life, and is usually fatal by the age of 2 years. Other forms of 5q-SMA, with later onset, have more gradual courses; in SMA type III (Kugelberg-Welander disease), motor disability usually emerges during later childhood and adolescence.

Most SMN protein comes from mRNA transcripts derived from the *SMN1* gene. There is an adjacent *SMN2* gene that differs by a few base pairs; these changes alter mRNA splicing and increase the production of an SMN isoform that is unstable and rapidly degraded. *SMN2* is an example of a gene that shows copy number variation, such that normal individuals may have one or multiple copies of *SMN2* on each chromosome 5. All forms of 5q-SMA are caused by loss-of-function mutations in *SMN1* (usually deletions), with the differences in clinical phenotype being determined by the number of *SMN2* gene copies. A novel gene therapy for 5q-SMA is now available that uses an adeno-associated virus to express SMN in motor neurons, albeit at a cost of over \$2 million. Genetic therapies that alter the splicing of *SMN2*-derived transcripts have also been approved for treatment of this devastating disease.

KEY CONCEPTS

Neurodegenerative Diseases

- Neurodegenerative diseases are characterized by progressive neuronal loss involving specific neuronal circuits and brain regions. Most of these diseases are associated with accumulation of abnormal protein aggregates, typically in the form of intracellular or extracellular inclusions. The clinical disease phenotype reflects the patterns of brain involvement rather than the type of inclusion.
- These diseases can be grouped by clinical presentation into dementias, hypokinetic movement disorders, hyperkinetic movement disorders, cerebellar ataxias, and motor neuron diseases.
- Prion diseases may be sporadic, familial, or transmissible. Regardless of the cause, the disease is driven by the conversion of a normal cellular protein into an abnormal conformation, with the acquisition of distinct characteristics including relative resistance to protease digestion, self-propagation, and the ability to spread. The most common clinical presentation is that of rapidly progressive dementia.
- Among dementias, Alzheimer disease (with plaques of A β and tangles of tau) is the most common; other predominantly dementing diseases include the various forms of frontotemporal lobar degeneration (usually with either TDP-43- or tau-containing inclusions) and dementia with Lewy bodies (with α -synuclein inclusions).
- Among the hypokinetic movement disorders, Parkinson disease is the most common (α -synuclein-containing inclusions); other diseases with features of parkinsonism include progressive supranuclear palsy and corticobasal degeneration (both forms of tauopathy).
- Amyotrophic lateral sclerosis is the most common form of motor neuron disease, with diverse genetic causes as well as sporadic forms.

GENETIC METABOLIC DISEASES

Genetic disorders that disrupt metabolic processes in neurons and glia typically present early in life and are often relentlessly progressive. Some of these disorders manifest in the immediate postnatal period, while others emerge later in life. Overall, metabolic diseases with earlier onset tend to have a more rapid and aggressive clinical course.

This heterogeneous group of disorders can be organized based on the organelles or the type of cell that is affected, and by the anatomic distribution of disease:

- *Neuronal storage diseases* are predominantly autosomal recessive disorders caused by the deficiency of an enzyme or trafficking protein that is required for the catabolism of sphingolipids (including the gangliosides), mucopolysaccharides, or mucolipids.
- *Leukodystrophies* are mostly autosomal recessive disorders caused by mutations in genes encoding enzymes involved in myelin synthesis or catabolism.

- *Mitochondrial encephalomyopathies* are disorders of oxidative phosphorylation, often affecting multiple tissues including skeletal muscle (Chapter 27). These disorders may be caused by mutations in the mitochondrial or the nuclear genomes.

Neuronal Storage Diseases

These disorders are characterized by the accumulation of storage material within neurons, typically followed by neuronal death. The neurologic manifestations that result from this neuronal dysfunction or death are most often seizures and generalized loss of neurologic function. Although many are associated with deficits of specific enzymes, others appear to be caused by defects in protein or lipid trafficking within neurons. This is a large class of disorders, with genetic heterogeneity even within clinically homogeneous entities. A few diseases in this category (Tay-Sachs and Niemann-Pick diseases, mucopolysaccharidoses, and others) are described in more detail in Chapter 5.

Leukodystrophies

These disorders are caused by mutations of genes whose products are involved in the generation, turnover, or maintenance of myelin. Unlike demyelinating diseases such as multiple sclerosis, the leukodystrophies typically present with an insidious and progressive loss of cerebral function, often at young ages, associated with diffuse and symmetric changes on imaging studies. Although many leukodystrophies are caused by single enzyme defects resulting in altered metabolism of myelin-associated lipids, a variety of other genetic alterations can lead to white matter diseases. The following examples are representative of this spectrum of disorders.

- *Krabbe disease* is an autosomal recessive leukodystrophy resulting from a deficiency of glucocerebrosidase β -galactosidase, the enzyme required for the catabolism of glucocerebroside to ceramide and galactose. As a consequence of the impaired catabolism of glucocerebroside in the brain, an alternative catabolic pathway shunts glucocerebroside to galactosylsphingosine, which is toxic to both oligodendrocytes and astrocytes. The clinical course is rapidly progressive, with onset of symptoms (dominated by motor signs such as stiffness and weakness) between 3 and 6 months of age; survival beyond 2 years of age is uncommon. The brain shows loss of myelin and oligodendrocytes (Fig. 28.44); a similar process affects peripheral nerves. Neurons and axons are relatively spared. A unique, diagnostic feature of Krabbe disease is the aggregation of engorged macrophages (*globoid cells*) in the brain parenchyma and around blood vessels (Fig. 28.44, inset).
- *Metachromatic leukodystrophy* is an autosomal recessive disease that results from a deficiency of arylsulfatase A, a lysosomal enzyme. This enzyme, present in a variety of tissues, cleaves sulfate from sulfate-containing lipids (sulfatides) as the first step in their degradation. The enzyme deficiency thus leads to an accumulation of sulfatides, especially cerebroside sulfate, that are toxic to white matter. The most striking histologic finding is demyelination/dysmyelination with resulting gliosis. Macrophages with vacuolated cytoplasm are scattered throughout the white matter. The membrane-bound vacuoles contain complex crystalloid structures composed of sulfatides; when bound to certain dyes such as toluidine blue, sulfatides shift the absorbance spectrum of the dye, a property called *metachromasia*. Similar metachromatic material can be detected in peripheral nerves and in urine, the latter being a sensitive method of establishing the diagnosis.
- *Adrenoleukodystrophy* is an X-linked recessive disease caused by loss-of-function mutations in *ABCD1*, which encodes an ATP-binding cassette transporter protein that is required for transport of

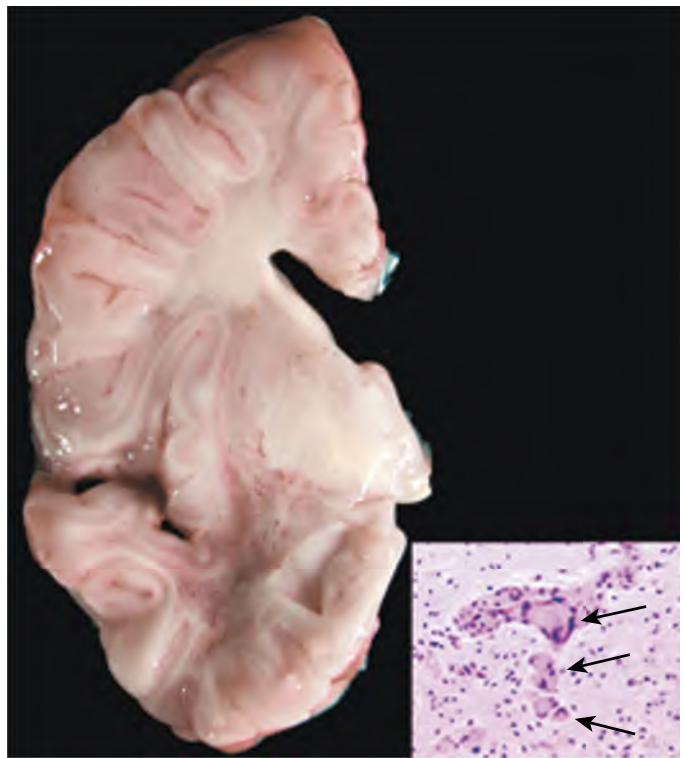


Fig. 28.44 Krabbe disease. Much of the white matter is gray/yellow because of the loss of myelin. Inset, “Globoid” cells (arrows) are the hallmark of the disease.

fatty acids into peroxisomes. In the typical form, young males present with behavioral changes and adrenal insufficiency. The loss of *ABCD1* leads to an inability to catabolize very-long-chain fatty acids within peroxisomes, resulting in elevated serum levels of these lipids. The symptoms result from progressive CNS, peripheral nerve, and adrenal disease. In the white matter, myelin loss is accompanied by gliosis and extensive lymphocytic infiltration. Adrenal cortical atrophy explains the hypocortisolism.

A variety of other leukodystrophies, with diverse mechanisms of white matter injury, are recognized, including those associated with pathogenic mutations in genes that encode the intermediate filament protein GFAP (*Alexander disease*), the translation initiation factor eIF2B (*vanishing white matter leukoencephalopathy*), and a major myelin protein PLP1 (*Pelizaeus-Merzbacher disease*).

Mitochondrial Encephalomyopathies

Disorders of energy generation can cause a range of neurologic disease, often in association with abnormalities in other tissues. Although many of the inherited disorders of mitochondrial oxidative phosphorylation present as skeletal muscle diseases (Chapter 27), the critical dependence of neurons on oxidative phosphorylation for generation of ATP also leads to frequent CNS involvement.

The mitochondrial genome, which is inherited from the mother, encodes only 13 proteins, 22 transfer RNAs (tRNAs), and two ribosomal RNAs (rRNAs). The remaining proteins that are required for mitochondrial structure and function are encoded by the nuclear genome. Thus, some mitochondrial disorders show maternal transmission because the affected genes lie in the mitochondrial genome, but many others do not. There is a complex genotype-phenotype relationship in these disorders: the same mutation may manifest as

different phenotypes, and the same phenotype may be caused by different mutations.

A critical mitochondrial disease concept is *heteroplasmy*, defined as the presence of both wildtype and mutated mitochondrial genomes within different populations of mitochondria in single cells. Because mitochondria are distributed in a random fashion when “mother cells” divide, daughter cells may inherit different proportions of functional and dysfunctional mitochondria, leading to wide cell-to-cell variation in disease expression. In general, mitochondrial disorders in the CNS selectively target neurons; the disruption of energy generation is often reflected in elevated tissue lactate levels, which can be demonstrated by spectroscopic imaging methods. Another marker of mitochondrial dysfunction is loss of enzymatic activity of cytochrome C oxidase, which can be assessed with special histochemical stains in nonfixed tissue.

Some of the better recognized mitochondrial encephalopathies include:

- *Mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (MELAS)* is characterized by recurrent episodes of acute neurologic dysfunction, cognitive changes, muscle weakness, and lactic acidosis. Pathologically, there may be infarcts, sometimes with vascular proliferation and focal calcification. The most common mutation observed in MELAS is in the gene encoding mitochondrial tRNA-leucine (*MTTL1*).
- *Myoclonic epilepsy with ragged red fibers (MERRF)* is a maternally transmitted disease in which affected individuals have myoclonus (a seizure disorder) and a myopathy that is characterized by ragged red fibers on muscle biopsy (Chapter 27). Ataxia, associated with neuronal loss in the cerebellum and degeneration of white matter tracts in the spinal cord, is also common. Most cases of MERRF are associated with mutations in mitochondrial tRNA genes.
- *Leigh syndrome* is a disease of infancy characterized by lactic acidemia, arrest of psychomotor development, feeding problems, seizures, extraocular muscle palsies, and weakness with hypotonia; death usually occurs within 1 to 2 years. On histologic examination, there are multifocal regions of destruction of brain tissue associated with a spongiform appearance and vascular proliferation. Brainstem nuclei, the thalamus, and the hypothalamus are typically involved, usually in a symmetric manner. A wide spectrum of nuclear and mitochondrial DNA mutations has been identified.

KEY CONCEPTS

Genetic Metabolic Diseases

- Mutations that disrupt metabolic or synthetic pathways can affect the nervous system. These pathways can involve general cellular processes or processes that are relatively specific to the nervous system.
- Diseases with earlier onset are typically more severe in the degree of damage and pace of illness.
- Neuronal storage diseases are usually autosomal recessive disorders. The characteristic finding is accumulation of material within neurons, along with evidence of neuronal death.
- Leukodystrophies are also typically autosomal recessive disorders associated with disruption of the synthesis or turnover of myelin components.
- Mitochondrial encephalomyopathies are pleiotropic disorders that involve neurons and tissues outside of the nervous system and can be caused by mutations in the nuclear or mitochondrial genomes.

TOXIC AND ACQUIRED METABOLIC DISEASES

Toxic and acquired metabolic diseases are relatively common causes of neurologic illnesses. These diseases are discussed in Chapter 9; only aspects that are relevant to CNS pathology are presented here.

Vitamin Deficiencies

Neural tissue has a very high metabolic demand and is thus significantly affected by deficiencies of B vitamins, which play a key role in cellular metabolism. Two distinct clinicopathologic syndromes involving the CNS that are associated with thiamine (vitamin B₁) and vitamin B₁₂ deficiency merit discussion, in part because their recognition and appropriate treatment can prevent permanent neurologic damage.

Thiamine (Vitamin B₁) Deficiency

The classic neurologic disease resulting from thiamine deficiency is called *beriberi* and consists of several related manifestations. *Wernicke encephalopathy* is characterized by acute psychosis and ophthalmoplegia. These symptoms are reversible when treated with thiamine. However, if unrecognized and untreated, they may be followed by a prolonged and largely irreversible condition (*Korsakoff syndrome*) that is marked by disturbances of short-term memory and confabulation. Patients also develop peripheral neuropathy, which can be so severe as to cause acute flaccid paralysis mimicking Guillain-Barré syndrome. The syndrome is particularly common in the setting of chronic alcohol use, but it may also be encountered in individuals with thiamine deficiency resulting from gastric disorders, including carcinoma, chronic gastritis, or persistent vomiting, and patients with eating disorders such as anorexia or bulimia.

MORPHOLOGY

In **Wernicke encephalopathy**, there are foci of hemorrhage and necrosis in the mammillary bodies and the walls of the third and fourth ventricles. Early lesions show dilated capillaries with prominent endothelial cells; subsequently, the capillaries become leaky, producing hemorrhagic areas. With time, there is infiltration of macrophages and development of a cystic space containing hemosiderin-laden macrophages. These chronic lesions predominate in individuals with **Korsakoff syndrome**. Lesions in the dorsomedial nucleus of the thalamus seem to correlate best with memory disturbance and confabulation. Peripheral nerves show axon loss and acute axonal degeneration (Chapter 27).

Vitamin B₁₂ Deficiency

Subacute combined degeneration of the spinal cord is caused by deficiency of vitamin B₁₂ and is marked by degeneration of the ascending and descending spinal tracts. The lesions result from a defect in myelin formation; the mechanism likely involves a blockage in one of the several enzymatic reactions that require vitamin B₁₂ (Chapter 14), but the details remain obscure. Symptoms may present over a few weeks, initially with bilaterally symmetric numbness, tingling, and slight ataxia in the lower extremities, which may progress to include spastic weakness of the lower extremities. Paraplegia may occur, usually only later in the course. With prompt vitamin B₁₂ replacement therapy, clinical improvement occurs; however, once paraplegia has developed, recovery is poor. On microscopic examination, there is swelling of myelin layers, producing vacuoles in the affected tracts; with time, axons degenerate as well. In the early stages of the disease, the mid-thoracic level of the spinal cord is preferentially affected, from which the process may extend proximally and distally.

Neurologic Sequelae of Metabolic Disturbances Hypoglycemia

Because the brain requires glucose and oxygen for its energy production, the cellular effects of diminished glucose resemble those of oxygen deprivation, described earlier. Some regions of the brain are more sensitive to hypoglycemia than others. Glucose deprivation initially leads to selective injury to large cortical pyramidal neurons, which, if

severe, may result in pseudolaminar necrosis of the cortex, predominantly involving deep layers. The Sommer sector (area CA1) of the hippocampus is also vulnerable, as are cerebellar Purkinje cells.

Hepatic Encephalopathy

The encephalopathy found in the setting of impaired liver function (Chapter 18) is accompanied by a glial response within the CNS; critical mediators appear to include elevated ammonia levels as well as proinflammatory cytokines. Astrocytes with enlarged nuclei and minimal cytoplasm, known as Alzheimer type II cells, appear in the cerebral cortex, basal ganglia, and other subcortical gray matter regions.

Toxic Disorders

Cellular and tissue injury from toxic agents is discussed in Chapter 9. Aspects of several important toxic disorders that are of unique neurologic importance are discussed here.

Carbon Monoxide

Many of the pathologic findings that follow acute carbon monoxide (CO) exposure are the result of impaired oxygen-carrying capacity of hemoglobin; in addition, CO can interact with the heme of cytochrome C oxidase, inhibiting tissue utilization of oxygen by blocking electron transport in the mitochondria. Selective injury of the neurons in layers III and V of the cerebral cortex, Sommer sector of the hippocampus, and Purkinje cells is characteristic. A unique pattern of bilateral globus pallidus necrosis may also occur. Demyelination of white matter tracts may be a later event.

Ethanol

The effects of acute ethanol intoxication are reversible, but chronic alcohol use disorder is associated with a variety of neurologic sequelae, including Wernicke-Korsakoff syndrome from thiamine deficiency (see earlier). The toxic effects of chronic alcohol intake may be either direct or secondary to nutritional deficits. Cerebellar dysfunction occurs in about 1% of patients with alcohol use disorder and is associated with a clinical syndrome of truncal ataxia, unsteady gait, and nystagmus. The histologic changes are atrophy and loss of granule cells, predominantly in the superior anterior vermis (Fig. 28.45). In advanced cases, there is loss of Purkinje cells and proliferation of the adjacent



Fig. 28.45 Alcohol-induced cerebellar degeneration. The anterior and superior portion of the vermis (*upper portion of figure*) is atrophic, with widened spaces between the folia.

astrocytes (*Bergmann gliosis*) between the depleted granular cell layer and the molecular layer of the cerebellar cortex. Fetal alcohol syndrome is discussed in Chapter 10.

Radiation

Exposure of the brain to radiation can occur accidentally or as part of therapeutic regimens for tumors. As discussed in Chapter 9, exposure to very high doses of radiation (>10 Gy) can cause intractable nausea, confusion, convulsions, and rapid onset of coma, followed by death. Delayed effects of radiation can also present with rapidly evolving symptoms, including headaches, nausea, vomiting, and papilledema, which may appear months to years after irradiation.

The pathologic findings consist of large areas of coagulative necrosis, primarily in white matter. This is accompanied by marked edema and gliosis in the surrounding tissue, along with fibrinoid vascular necrosis, hyaline vascular thickening, telangiectasias, and other vascular malformations. Methotrexate, which is administered intrathecally to treat some brain tumors, can increase the damage caused by radiation. Axons and cell bodies in the vicinity of the radiation-induced lesions undergo dystrophic mineralization. Radiation can also induce tumors, which usually develop years after radiation therapy and include meningiomas, nerve sheath tumors, sarcomas, and gliomas.

KEY CONCEPTS

Toxic and Acquired Metabolic Diseases

- Certain vitamin deficiencies result in neurologic disease as well as systemic disorders, with B₁ and B₁₂ vitamin deficiencies being the most common.
- The metabolic demand of the CNS makes it highly susceptible to injury from hypoglycemia and carbon monoxide poisoning.
- Chronic alcohol exposure may result in several forms of brain injury; atrophy of the superior anterior cerebellar vermis is particularly characteristic.
- Metabolic disturbances may disrupt brain function, often without detectable morphologic changes.

TUMORS

The annual incidence of primary tumors of the CNS is roughly 23 per 100,000 persons for intracranial tumors and 2 per 100,000 persons for intraspinal tumors. Tumors of the CNS account for nearly 20% of all cancers of childhood. Seventy percent of childhood CNS tumors arise in the posterior fossa; a comparable number of tumors in adults arise within the cerebral hemispheres above the tentorium.

Although pathologists have developed classification schemes that distinguish between benign and malignant lesions based on histologic criteria, the clinical course of a patient with a brain tumor is strongly influenced by patterns of tumor growth and location. Thus, even benign tumors and low-grade malignancies can lead to serious clinical deficits and may prove fatal. Because of the capacity of diffuse gliomas to widely infiltrate the brain parenchyma, such tumors are not amenable to complete surgical resection without compromising neurologic function. Also, any CNS neoplasm, regardless of histologic grade or classification, may be lethal if situated in a critical brain region; for example, a benign posterior fossa meningioma may cause cardiorespiratory arrest by compressing vital centers in the medulla. Even the most highly malignant gliomas rarely metastasize outside the CNS. Some pediatric tumors (e.g., medulloblastoma) spread through the CSF when they encroach on the subarachnoid space, resulting in cerebrospinal dissemination far from the original tumor site.

With increased understanding of the events that drive the development of CNS tumors, classification schemes for these tumors have increasingly incorporated molecular and genetic criteria. Treatment protocols and clinical trials are currently based on the 2021 World Health Organization (WHO) classification, which segregates tumors into one of four grades according to their biologic behavior, ranging from CNS WHO grade 1 to grade 4. Like most cancers, malignant brain tumors tend to progress and become more aggressive with time due to clonal evolution, a change in behavior that is often reflected in a change to a higher tumor grade.

The major classes of primary brain tumors to be considered here include gliomas, neuronal/glioneuronal tumors, embryonal tumors, and a few less common intraparenchymal categories. In addition, meningiomas (the most common primary brain tumors), metastases (the most common tumors involving the CNS overall), and familial tumor syndromes will be covered.

Gliomas

Gliomas, the most common group of primary intraparenchymal brain tumors, include astrocytomas, oligodendrogliomas, and ependymomas. These tumor types have characteristic histologic and molecular features that form the basis for their classification. As is true of many hematologic malignancies (Chapter 13), it appears these tumors are mostly derived from multipotent progenitor cells that preferentially differentiate down a particular cellular lineage. Many CNS tumors have a predilection for specific anatomic regions within the brain and tend to occur within particular age groups.

Diffuse Astrocytomas (Astrocytoma, IDH-mutant and Glioblastoma, IDH-wildtype)

Although the diffusely infiltrating astrocytomas and glioblastomas were once considered part of a single continuous spectrum ranging from lowest to highest grade, it is now recognized that they represent two separate tumor types with unique genetic features and highly disparate clinical behavior; as such, they are considered distinct tumor entities in the 2021 WHO classification scheme. One important genetic criterion is the mutational status of the *IDH1* and *IDH2* genes, which you will recall are involved by acquired gain of function mutations in several cancers that result in expression of forms of isocitrate dehydrogenase with novel enzyme activities that catalyze the synthesis of 2-hydroxymethyl glutarate, an oncometabolite (Chapter 7). *Glioblastomas* account for roughly 14% of all primary CNS tumors (and over half of all CNS malignancies), being encountered mostly in the cerebral hemispheres of adults older than 55 years. *IDH-mutant astrocytomas* are less common, though precise epidemiologic estimates are not yet available; they are predominantly encountered in the cerebral hemispheres of young to middle-aged adults (median age of 38 years). The most common presenting signs and symptoms for both tumor types are seizures, headaches, and focal neurologic deficits related to

the anatomic site of involvement, though the clinical course for glioblastomas tends to be much shorter. The *IDH-mutant astrocytomas* (CNS WHO grades 2 to 4) have a much better prognosis than glioblastomas (CNS WHO grade 4), even when morphologically the astrocytomas qualify for grade 4 designation (Table 28.5).

Pathogenesis

- **Glioblastoma, IDH-wildtype.** Glioblastomas occur most commonly as a new onset (de novo) disease and do not evolve from low-grade astrocytomas. They typically occur in older individuals, and are *IDH-wildtype*, CNS WHO grade 4 by definition. They harbor several common genetic alterations, most commonly combined gain of chromosome 7 with loss of chromosome 10, *TERT*-promoter mutation, and *EGFR* gene amplification. Even if an adult diffuse astrocytoma lacks the usual high-grade histologic features (see later), it still qualifies for a designation of glioblastoma, as long as it is *IDH-wildtype* and has at least one of these three key genetic features. These and other less common genetic and genomic alterations contribute to the acquisition of cancer hallmarks (Chapter 7). For example, most glioblastomas have genetic aberrations that lead to evasion of senescence (telomerase reverse transcriptase [*TERT*] promoter region mutations); escape from normal growth controls (homozygous *CDKN2A* deletion); activation of growth factor signaling pathways (*EGFR* or *PDGFR* gene amplification); and resistance to apoptosis (*TP53* mutation).
- **Astrocytoma, IDH-mutant.** In contrast, *IDH-mutant astrocytomas* typically harbor three simultaneous molecular alterations, *IDH1* or *IDH2* mutation (considered the earliest tumorigenic alteration), *TP53* mutation, and *ATRX* inactivation via gene mutation or other mechanisms (mechanism of senescence evasion). *IDH-mutant astrocytomas* with biallelic deletions of *CDKN2A* are now considered CNS WHO grade 4 by definition, even in the absence of high-grade histologic features, due to the considerably worse prognosis compared to otherwise similar tumors.

MORPHOLOGY

IDH-mutant astrocytomas, CNS WHO grade 2 are poorly defined, gray, infiltrative tumors that expand and distort the involved brain (Fig. 28.46A). These tumors range in size from a few centimeters to lesions that replace nearly the entire brain. The cut surface of the tumor may be either firm or soft and gelatinous; cystic degeneration may be seen. The tumor may appear well demarcated, but is more commonly ill-defined due to infiltration beyond the perceived margins.

Microscopically, these tumors are hypercellular compared to normal white matter and feature enlarged, elongated or irregular, hyperchromatic nuclei embedded within a fibrillar background (Fig. 28.46B) that is often GFAP-immunoreactive. Individual tumor cells infiltrate brain tissue some distance away from the main lesion. An immunostain for the *IDH1* p.R132H mutant protein highlights the tumor cells in up to 90% of cases (Fig. 28.46B, inset); *IDH1* and *IDH2* sequencing is

Table 28.5 Features of Diffuse Gliomas

	Astrocytoma, IDH-mutant (CNS WHO Grades 2–4)	Oligodendroglioma, IDH-mutant and 1p/19q-codeleted (CNS WHO Grade 2 or 3)	Glioblastoma, IDH-wt (WHO Grade 4)
<i>IDH</i> status	Mutant	Mutant	Wildtype
Other genetics	<i>TP53</i> -mut, <i>ATRX</i> -mut <i>CDKN2A</i> -HD (grade 4)	1p/19q-codeleted	+7/–10, <i>pTERT</i> -mut, <i>EGFR</i> -amp
Typical morphology	Nuclear atypia; mitoses in grade 3, MVP and/or necrosis in grade 4	Round nuclei, clear halos; mitoses, MVP, and/or necrosis in grade 3	Nuclear atypia, mitoses, MVP, necrosis

amp, Amplification; mut, mutant; HD, homozygous deletion; MVP, microvascular proliferation; p, promoter; wt, wildtype; +, whole-chromosome gain; –, whole-chromosome loss.

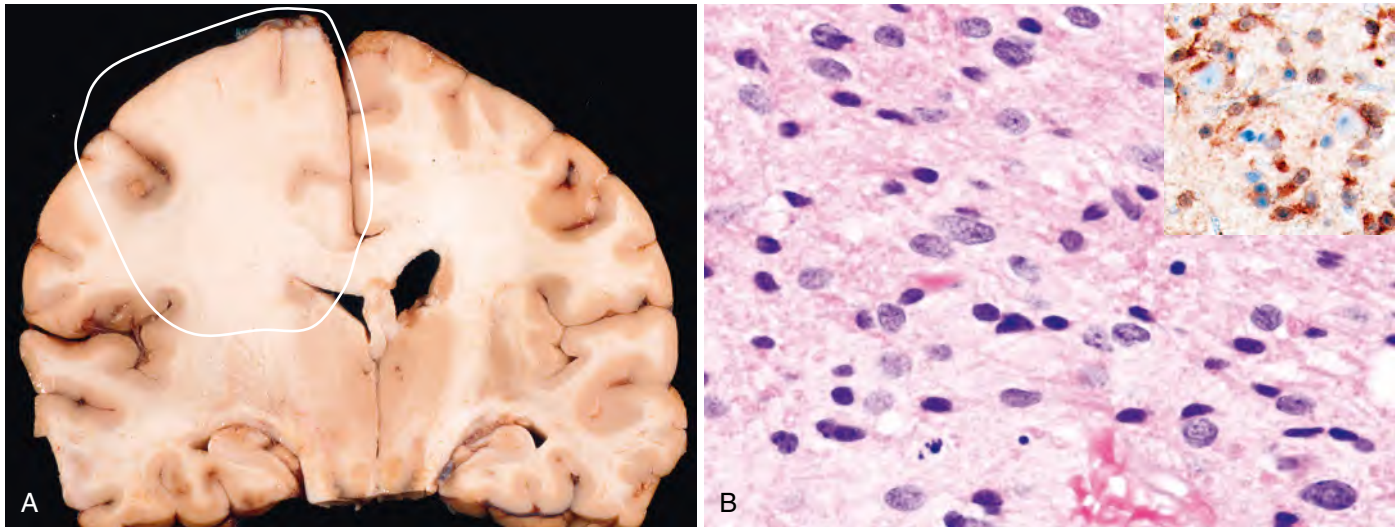


Fig. 28.46 Astrocytoma, IDH-mutant. (A) On coronal section at autopsy, the left frontal white matter is expanded, and there is blurring of the corticomedullary junction due to infiltrative tumor (*circled*). (B) This histologic section from the white matter shows enlarged, irregular, hyperchromatic nuclei that appear embedded within the native fibrillary matrix of the brain; the smaller round and oval nuclei are native oligodendrocytes and reactive astrocytes, respectively. *Inset*, An immunostain for IDH1 p.R132H is positive for mutant protein in tumor cells, some of which surround the larger immunonegative cortical neurons (“perineuronal satellitosis”).

required to identify less common pathogenic IDH mutations, and is done when the immunostain is negative. Nearly all cases also show ATRX loss and p53 overexpression by immunohistochemistry, reflecting the unique manifestations of those associated gene mutations.

CNS WHO grade 3 IDH-mutant astrocytomas have a similar appearance but are more densely cellular and have readily detectable mitotic activity. The CNS WHO grade 4 counterparts additionally show **microvascular proliferation** (multilayered small blood vessels), necrosis, and/or homozygous *CDKN2A* deletion.

In **glioblastomas**, IDH-wildtype, variation in the appearance of the tumor from region to region is characteristic; some areas are firm and gray-white, while others are soft and yellow due to necrosis or red due to hemorrhage and hypervascularity (Fig. 28.47). The histologic appearance is similar to IDH-mutant astrocytomas, but typically with a greater prominence of necrosis and/or microvascular proliferation. Necrosis in glioblastoma often occurs in a serpentine pattern with tumor hypercellularity along the edges of the necrotic regions, a histologic pattern referred to as **palisading** (Fig. 28.48). The microvascular cell proliferation produces tufts of cells that pile up and bulge into the lumen of small blood vessels (Fig. 28.48 *inset*); vascular endothelial growth factor (VEGF), produced by malignant astrocytes in response to hypoxia, contributes to this distinctive vascular change. Because histologic features can be extremely variable from one region to another, small biopsy specimens may not be representative, and the tumor may be undergraded if assessed by histology alone. In contrast, genetic, genomic (e.g., next generation sequencing), and/or epigenomic (e.g., DNA methylation profiling) studies often correctly identify the tumor as a glioblastoma even in the absence of necrosis and microvascular proliferation.

Clinical Features

The presenting symptoms of diffuse astrocytomas depend, in part, on the location and growth rate of the tumor. CNS WHO grade 2 IDH-mutant astrocytomas may remain stable or progress slowly; mean survival exceeds 10 years. Eventually, however, clinical deterioration occurs due to the emergence of a more rapidly growing subclone of higher histologic grade. Radiologic studies of grade 2 tumors show a nonenhancing mass and adjacent “edema,” which often includes a tumor

infiltrate. High-grade (i.e., grade 3 or 4) IDH-mutant astrocytomas and IDH-wildtype glioblastomas have abnormal vessels that are “leaky,” resulting in an abnormally permeable blood-brain barrier; this leads to contrast enhancement on magnetic resonance imaging, which is often ring-like in glioblastomas (Fig. 28.47A) and more patchy in CNS WHO grade 4 IDH-mutant astrocytomas. The prognosis for individuals with IDH-wildtype glioblastomas is very poor, although the use of newer chemotherapeutic agents has provided some benefit. Epigenetic silencing of the promoter for the gene encoding the DNA repair enzyme MGMT (methylguanine-DNA methyltransferase) predicts responsiveness to DNA alkylating drugs (e.g., temozolomide), as would be expected given that MGMT is critical for the repair of chemotherapy-induced DNA modifications. With current treatment, which consists of resection followed by radiation therapy and chemotherapy, the mean length of survival after diagnosis has increased to 15 months; 25% of such patients are alive after 2 years. Survival is substantially shorter in older patients and in patients with lower performance status or with large unresectable lesions. As stated earlier, the CNS WHO grade 4 IDH-mutant astrocytomas are also aggressive but have a better prognosis overall, with an average survival of 2 to 4 years.

Oligodendroglioma, IDH-mutant and 1p/19q-codeleted

Oligodendroglioma is the other major subtype of diffuse or infiltrating glioma; it is composed of cells that resemble oligodendrocytes. When corrected for tumor grade, the oligodendrogliomas have the best prognosis among glial tumors; as with their astrocytic counterparts, they are now defined using both morphologic and genetic features. These tumors constitute 5% to 15% of gliomas and are most common in the fourth and fifth decades. Patients may have had several years of neurologic complaints, often including seizures. The lesions are found mostly in the cerebral hemispheres and have a predilection for neocortex and superficial white matter.

Pathogenesis

Oligodendroglioma is molecularly defined by an IDH mutation in combination with whole-arm chromosomal codeletion of 1p and 19q.

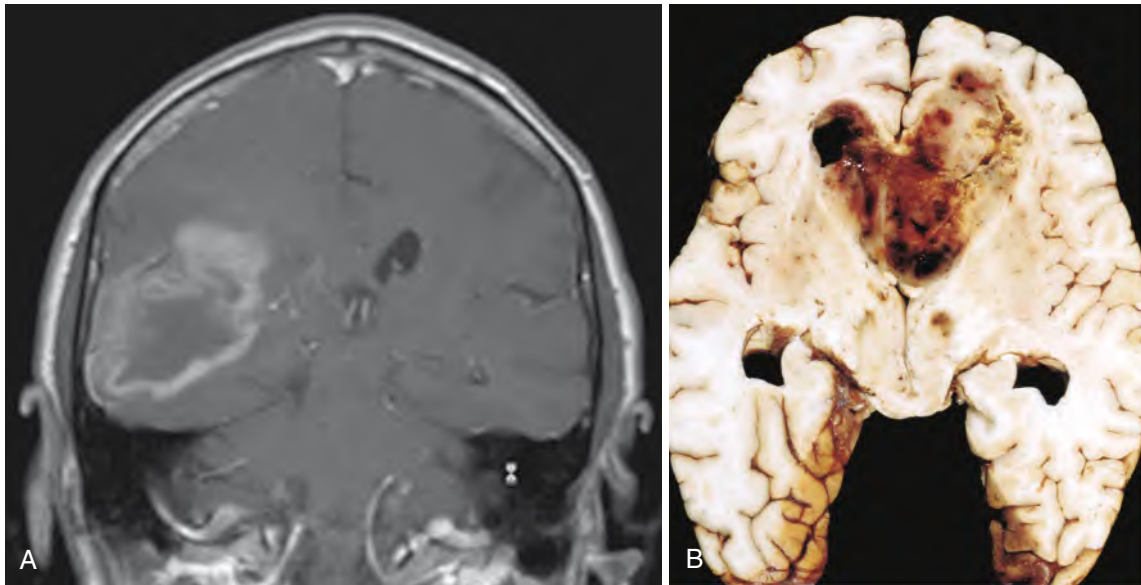


Fig. 28.47 Glioblastoma, IDH wildtype. (A) Contrast T1-weighted coronal magnetic resonance image shows a large mass in the right temporal lobe with “ring” enhancement. (B) Glioblastoma appearing as a necrotic, hemorrhagic, “butterfly” mass involving the corpus callosum and bihemispheric white matter.

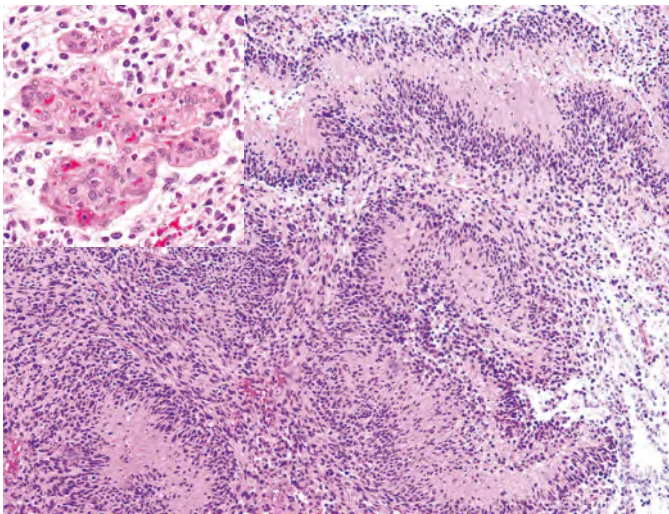


Fig. 28.48 Glioblastoma, IDH-wildtype. Serpiginous foci of palisading necrosis (tumor nuclei lined up around the pink anucleate zones of necrosis). *Inset*, Microvascular proliferation.

Additional genetic alterations occur with progression to higher grade oligodendroglioma; one of these includes losses of chromosome 9p, which result in deletion of the *CDKN2A* tumor suppressor gene, though this is much less common than in IDH-mutant astrocytomas. In contrast to high-grade astrocytic tumors, *EGFR* gene amplification is not seen. The vast majority of oligodendrogliomas feature the same *TERT*-promoter mutations encountered in glioblastoma.

MORPHOLOGY

Oligodendrogliomas are gelatinous, gray masses, often with cysts, focal hemorrhage, and calcification. The tumors are composed of sheets of regular cells with spherical nuclei containing finely granular chromatin (similar to normal oligodendrocytes) surrounded by a clear perinuclear halo (Fig. 28.49). The tumor

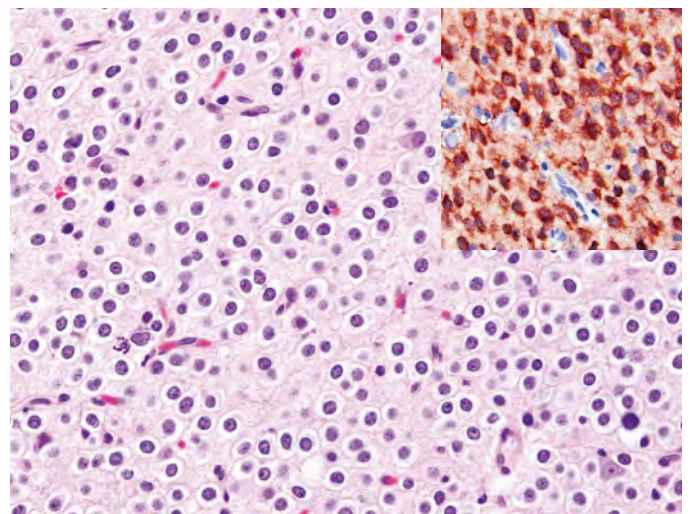


Fig. 28.49 Oligodendroglioma, IDH-mutant and 1p/19q codeleted. Tumor nuclei are round, with cleared cytoplasm forming “halos” and vasculature composed of thin-walled capillaries. Similar to astrocytoma, IDH-mutant (see Fig. 28.46B), tumor cells are usually positive for IDH1 p.R132H-mutant protein (*inset*). By definition, these tumors should also show whole arm 1p and 19q codeletion by genetic testing.

typically contains a delicate network of anastomosing capillaries, resembling “chicken wire.” Calcification, present in as many as 90% of these tumors, ranges from microscopic foci to massive deposits. Tumor cells that infiltrate the cerebral cortex often collect around neurons (perineuronal satellitosis). Mitotic activity and proliferation indices are low in low-grade (CNS WHO grade 2) oligodendroglioma.

CNS WHO grade 3 oligodendrogliomas are characterized by a higher cell density, nuclear anaplasia, increased mitotic activity, and occasionally necrosis. Such cases may arise *de novo*, but more often progress from CNS WHO grade 2 oligodendrogliomas.

Clinical Features

In general, individuals with oligodendrogliomas have a better prognosis than do those with astrocytomas. Current treatment with surgery, chemotherapy, and radiation therapy has yielded average survivals of 15 to 20 and 10 to 15 years for CNS WHO grades 2 and 3 tumors, respectively. Progression from low- to higher-grade lesions occurs, typically over a period of 5 or more years.

Pilocytic Astrocytoma

Pilocytic astrocytomas (CNS WHO grade 1) are distinguished from infiltrative astrocytomas by their more discrete borders, unique histopathologic and molecular characteristics, and relatively benign behavior. They typically occur in children and young adults and are commonly located in the cerebellum, but they may also appear in the region around the third ventricle, optic nerves, spinal cord, and occasionally in the cerebral hemispheres. The histologic separation of these tumors from other astrocytomas is supported by their relative circumscription, unique histopathologic features, and distinct genetic alterations of the MAP kinase pathway, the most common being a *KIAA1549-BRAF* gene fusion/duplication. Pilocytic astrocytomas grow very slowly and can often be treated by resection alone. Symptomatic recurrence and incompletely resected lesions may be treated with repeat surgery, chemotherapy, or targeted therapies.

MORPHOLOGY

Pilocytic astrocytoma is often radiologically and grossly well demarcated and cystic (Fig. 28.50A), with a contrast-enhancing mural nodule. Histologically, it differs from infiltrative astrocytoma of any grade by showing only limited brain invasion; often, there is biphasic architecture combining loose “microcystic” and compact, densely fibrillar areas (Fig. 28.50B). The tumor is composed of bipolar cells with long, thin “hairlike” processes that are GFAP-positive and form a dense fibrillary meshwork; **Rosenthal fibers** (eosinophilic corkscrew-shaped inclusions) and **eosinophilic granular bodies** (mulberry-like inclusions) are characteristic findings (Fig. 28.50C). In contrast to diffuse gliomas, the presence of microvascular proliferation or necrosis does not imply an unfavorable prognosis.

Ependymoma

Ependymomas are tumors that most often arise in proximity to the ependyma-lined ventricular system, including the oft-obiterated central canal of the spinal cord. In the first 2 decades of life, they typically occur near the fourth ventricle and constitute 5% to 10% of the primary brain tumors in this age group; less commonly, they are found in the cerebral hemispheres, where some lack an obvious connection to the ventricles. In adults, the spinal cord is the most common location; tumors in this site are more frequent in the setting of neurofibromatosis type 2 (NF2). In the most recent WHO classification scheme, the location (supratentorial, posterior fossa, spinal cord) is included in the name of the tumor and some are defined by specific molecular alterations.

Pathogenesis

Given the association of spinal ependymomas with NF2, it is not surprising that the *NF2* gene on chromosome 22 is commonly mutated in ependymomas arising in the spinal cord but not at other sites. Supratentorial tumors are often associated with fusions of the *ZFTA* (zinc finger translocation associated) gene, thought to be a transcriptional regulator, most commonly with the *RELA* (REL-associated protein, involved in NF- κ B activation) gene (*ZFTA::RELA* fusion). A rarer, favorable prognosis type is defined by fusions involving *YAP1*, a gene that may encode a transcriptional regulator and signaling molecule in many cellular processes. In the posterior fossa, the main stratification is the poor prognosis posterior fossa A (PFA) group versus the better prognosis posterior fossa B (PFB) group; these groups can be distinguished by DNA methylation profiling, although loss of H3K27me3 expression by immunohistochemistry is highly sensitive and specific for the PFA ependymoma. A rare poor prognosis spinal ependymoma is now defined by *MYCN* gene amplification.

MORPHOLOGY

Grossly and on MRI, **ependymomas** are solid (i.e., noninfiltrative) masses. In the posterior fossa, they typically arise from the floor of the fourth ventricle

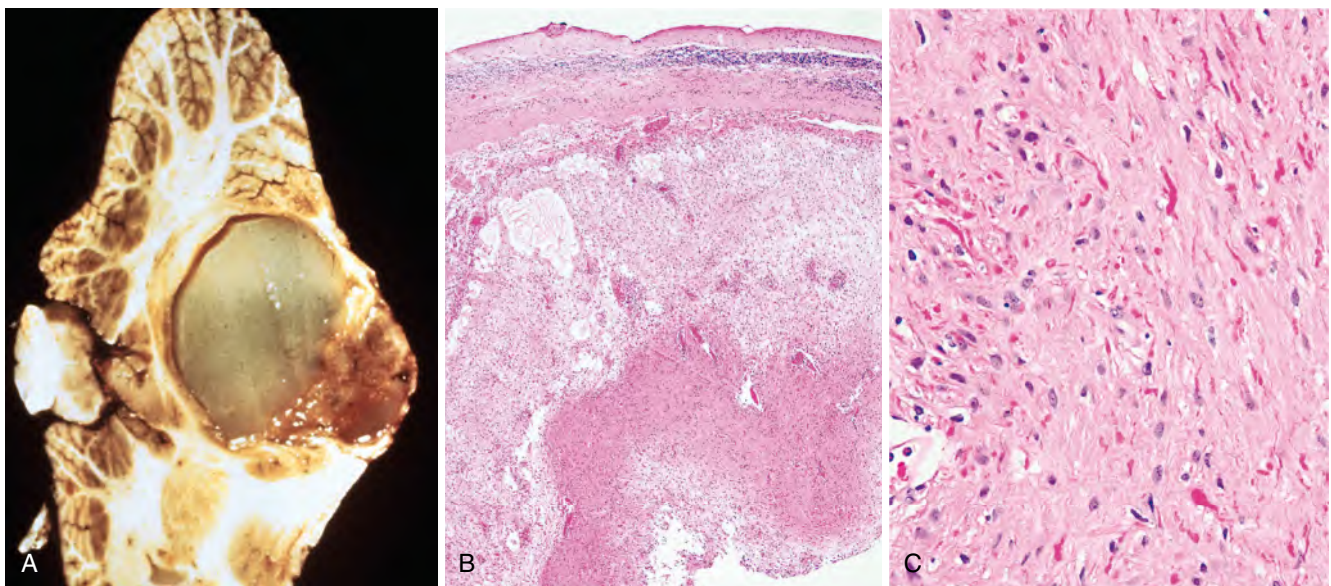


Fig. 28.50 Pilocytic astrocytoma. (A) Grossly, this cerebellar tumor forms a mural nodule within a cyst. (B) At low magnification, one can appreciate atrophic cerebellar cortex (*top*), sharp circumscription, and a biphasic pattern with alternating loose (*middle*) and compact (*bottom*) regions of tumor growth. (C) Higher magnification reveals oval to irregular tumor nuclei resembling those of IDH-mutant astrocytoma, but with numerous Rosenthal fibers (brightly eosinophilic corkscrew-shaped inclusions) in the background.

(Fig. 28.51A), occasionally growing through the foramina of Magendie or Luschka and extending below the foramen magnum into the cervical spinal canal. Although ependymomas are moderately well demarcated from adjacent brain, the proximity to vital structures often makes complete removal impossible; in the spinal cord, however, total resection is more feasible. Ependymomas are composed of cells with fibrillary processes that contain regular, round to oval nuclei with granular chromatin. Tumor cells may form glandlike round or elongated structures (true **rosettes**) that resemble the embryologic ependymal canal, with long, delicate processes extending into a lumen (Fig. 28.51B). More frequently present are **perivascular pseudorosettes** (Fig. 28.51B) in which there is a perivascular nuclear-free zone consisting of thin ependymal processes radiating toward the central blood vessel. Although most ependymomas are well differentiated and behave as CNS WHO grade 2 lesions, CNS WHO grade 3 counterparts feature increased cell density, high mitotic rates, areas of palisading necrosis, and/or microvascular proliferation. However, the tumor grade is less predictive of outcome than either the extent of surgical resection or the molecular subtype.

Two other variants of ependymoma merit brief mention. **Myxopapillary ependymomas** (CNS WHO grade 2) are distinct lesions that occur in the filum terminale of the spinal cord. Cuboidal tumor cells are arranged in a variably papillary architecture around mucin-rich fibrovascular cores. The prognosis is similar to conventional ependymoma and depends on completeness of surgical resection.

Subependymomas (CNS WHO grade 1) are solid, sometimes calcified, slow-growing nodular tumors that are usually found protruding into the lateral or fourth ventricle by growing under the ependyma. They are usually asymptomatic and are incidental findings at autopsy or imaging. However, if they are sufficiently large or strategically located, they can cause obstructive hydrocephalus. Subependymomas have a characteristic microscopic appearance, consisting of clusters of ependymal-appearing nuclei scattered in a dense fibrillar background of glial processes.

Clinical Features

Posterior fossa ependymomas often manifest with hydrocephalus secondary to progressive obstruction of the fourth ventricle. PFA tumors predominate in infants/young children, whereas PFB tumors mainly afflict adolescent and adult patients. Despite the relationship of ependymomas to the ventricular system, CSF dissemination is uncommon.

The clinical outcome for completely resected ependymomas is better than for their subtotaly resected counterparts.

Choroid Plexus Tumors

Benign and malignant choroid plexus tumors are recognized, both of which are rare. *Choroid plexus papillomas* (CNS WHO grade 1) are intraventricular tumors that are most common in children, in whom they tend to occur in the lateral ventricle; in adults, they more frequently involve the fourth ventricle. These papillary growths recapitulate the structure of the normal choroid plexus, wherein epithelioid tumor cells cover fibrovascular stalks. Clinically, choroid plexus papillomas usually present with hydrocephalus due to obstruction of the ventricular system by tumor and/or overproduction of CSF. The far rarer *choroid plexus carcinomas* (CNS WHO grade 3) resemble poorly differentiated papillary carcinomas; these tumors are almost always found in young children.

Neuronal and Glioneuronal Tumors

Much less common than glial tumors are those that exhibit neuronal differentiation. In general, neuronal tumors are more often seen in children and young adults with epilepsy.

- **Gangliogliomas** (CNS WHO grade 1) are tumors composed of a mixture of mature neuronal and glial cells. They are typically superficial lesions that present with seizures and are the most common neuronal tumors in the CNS. Most of these tumors are slow growing. When gangliogliomas present with medically refractory epilepsy, surgical resection is usually effective in controlling the seizures. Approximately 20% to 50% of these tumors have an activating mutation in the *BRAF* gene (V600E), which occasionally serves as the basis of targeted therapy. Gangliogliomas are most commonly found in the temporal lobe and often have a cystic component. The neoplastic ganglion cells are irregularly clustered and often have random orientation of processes; dysmorphic and/or binucleate forms are found. The glial component of these lesions most often resembles a pilocytic astrocytoma.
- **Dysembryoplastic neuroepithelial tumor** is a rare, benign (CNS WHO grade 1) tumor often associated with epilepsy. It has a good prognosis following surgical resection, with both low recurrence rates and favorable seizure control. These lesions are typically

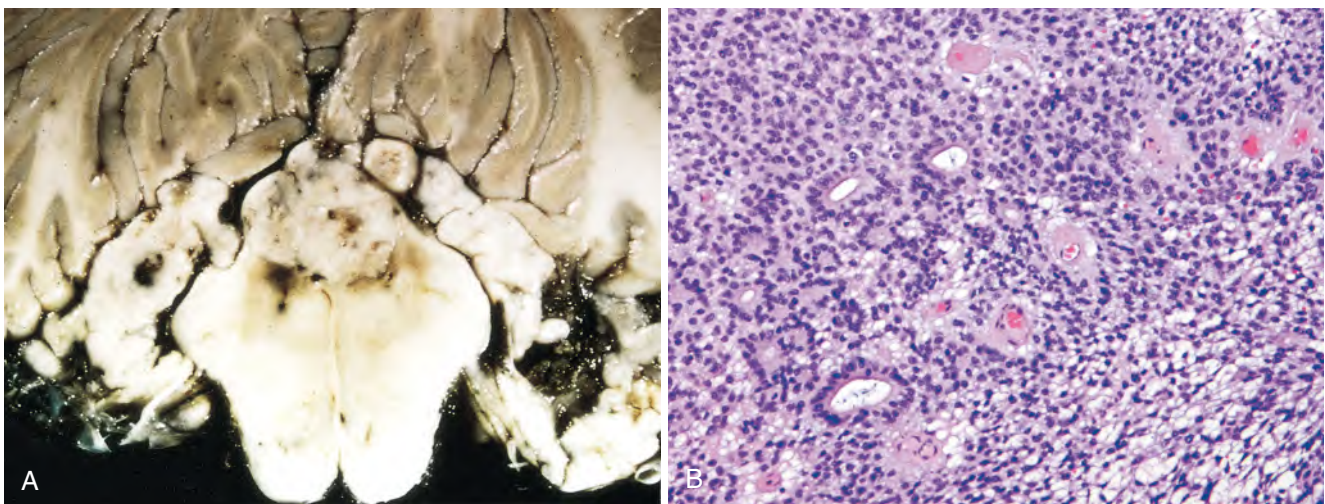


Fig. 28.51 Posterior fossa ependymoma. (A) Tumor of the fourth ventricle, distorting, compressing, and infiltrating surrounding structures. (B) The microscopic appearance includes both true rosettes (with a glandlike central lumen) and perivascular pseudorosettes (nuclear-free zone composed of fibrillary processes radiating toward a central blood vessel).

located in the superficial temporal lobe, although other sites may be seen. They usually form multiple discrete, mucin-rich, intracortical nodules of small, oligodendrocyte-like cells arranged in columns around central axonal bundles or capillaries. There are also “float-neurons” that sit in basophilic, mucin-rich pools.

Embryonal Tumors

A large category of mostly pediatric tumors is described as embryonal, meaning that they appear “primitive” or undifferentiated, although a limited degree of neuronal and less commonly glial differentiation is often encountered. The most common embryonal neoplasm is *medulloblastoma*, which accounts for 20% of pediatric brain tumors.

Medulloblastoma

This malignant embryonal tumor occurs predominantly in children and exclusively in the cerebellum (by definition). Although histologically, molecularly, and prognostically distinct subtypes have been identified, all are considered CNS WHO grade 4 tumors. Rapid growth may occlude the flow of CSF, leading to hydrocephalus.

Pathogenesis

Based on genetic alterations, medulloblastoma is classified into at least four groups. Some show alterations in genes involved in cerebellar development, such as the sonic hedgehog (SHH) pathway (which is involved in proliferation of cerebellar granule cells) and the Wnt/ β -catenin pathway, which is involved in embryogenesis, tissue homeostasis, and tumor development (Chapter 7). Two other tumor types are groups 3 and 4. SHH-activated/*TP53*-mutant and group 3 tumors have the worst prognosis and the Wnt tumors the best prognosis overall. Medulloblastomas in which the Wnt pathway is activated are often located around the anterior fourth ventricle or middle cerebellar peduncle, while SHH-activated medulloblastomas most often involve the vermis or lateral cerebellum.

MORPHOLOGY

Medulloblastoma is often well circumscribed, gray, and friable, and may extend to the surface of the cerebellar folia and involve the leptomeninges (Fig. 28.52A). On microscopic examination, the tumor is very densely cellular, with sheets of small primitive-appearing cells (Fig. 28.52B); individual tumor cells have scant cytoplasm and hyperchromatic nuclei that are frequently elongated or crescent-shaped. Mitoses are abundant, and markers of cellular proliferation, such as Ki-67, are positive in a high percentage of the cells. In the **classic** medulloblastoma, tumor cells form **Homer Wright rosettes** (Fig. 28.52B; also seen in neuroblastoma described in Chapter 10) and often express neuronal markers such as synaptophysin; the expression of glial markers (e.g., GFAP) is less common. They usually belong to the Wnt-activated group. The **desmoplastic/nodular tumor** is characterized by internodular areas of stromal response, marked by collagen and reticulin deposition (i.e., desmoplasia), and “pale islands” or nodules that have more neuropil and show more advanced neuronal differentiation (Fig. 28.52C). Most of these tumors are in the SHH-activated group. The **large cell/anaplastic tumor** is characterized by large and irregular vesicular nuclei, prominent nucleoli, cell wrapping (a cell-in-cell appearance), and high mitotic/apoptotic indices (Fig. 28.52D).

Medulloblastomas have a propensity to spread to the subarachnoid space. Dissemination through the CSF is a common complication, giving rise to either extensive growth along the surface likened to “icing” or nodular “drop metastases,” which tend to involve the cauda equina.

Clinical Features

Medulloblastomas are highly malignant, and the prognosis for untreated patients is dismal; however, they are exquisitely sensitive to

radiation and chemotherapy. The molecular and histologic subtype dramatically influences prognosis, with Wnt-activated medulloblastomas associated with nearly 100% survival at 5 years, whereas more aggressive subtypes fall into the 20% to 30% survival range. To account for these differences, ongoing clinical trials are exploring the option of either less aggressive or more aggressive treatment regimens based on subtype. Similarly, targeted therapies are being explored for those with actionable molecular alterations.

Other rare embryonal neoplasms that have unique clinicopathologic and genetic features are now being recognized throughout the CNS; their classification continues to evolve.

Primary CNS Lymphoma

Primary CNS lymphoma accounts for 2% of extranodal lymphomas and 1% of intracranial tumors. It is the most common CNS neoplasm in immunosuppressed individuals, including those with AIDS and immunosuppression after transplantation. In non-immunosuppressed populations, the age spectrum is relatively wide, but the frequency increases after 60 years of age.

The term *primary* emphasizes the distinction between these lesions and secondary involvement of the CNS by lymphoma arising elsewhere in the body (Chapter 13). Primary brain lymphoma is often multifocal and may also involve the eye, yet involvement outside of the CNS (in lymph nodes or bone marrow) is a rare and late complication. Conversely, lymphoma arising outside the CNS rarely involves the brain parenchyma; in this situation, secondary CNS involvement usually affects the meninges or the CSF, the latter sometimes diagnosed by the presence of malignant cells in a lumbar puncture specimen.

Most primary CNS tumors are diffuse large B-cell lymphomas; they are aggressive and generally have worse outcomes than tumors of comparable histology that occur at non-CNS sites (Chapter 13). In the setting of immunosuppression, the malignant B cells are usually latently infected by Epstein-Barr virus. In those who are not immunosuppressed, primary CNS large B-cell lymphomas often have amplification and overexpression of the *PDL1* gene, which encodes an important immune checkpoint protein that inhibits T-cell responses (Chapter 7). Trials of immune checkpoint inhibitors are ongoing.

MORPHOLOGY

Primary CNS lymphomas are single or frequently multiple, soft gray-white masses in the deep subcortical parts of the brain; periventricular spread is common. The malignant cells infiltrate the parenchyma of the brain and accumulate around blood vessels, express B-cell markers such as CD20, and have high proliferative indices. When tumors arise in the setting of immunosuppression, necrosis is often prominent and the Epstein-Barr virus is detectable by in situ hybridization for EBERS, small nuclear RNAs that are encoded by the viral genome.

Meningiomas

Meningiomas are predominantly benign tumors of adults that arise from the meniothelial cells of the arachnoid and are usually attached to the dura. Meningiomas may be found along any of the external surfaces of the brain as well as within the ventricular system, where they arise from the stromal arachnoid cells of the choroid plexus. Prior radiation therapy to the head and neck, typically decades earlier, is a risk factor for development of meningioma. Other tumors such as metastases, solitary fibrous tumors, and a range of poorly differentiated sarcomas may also grow as dura-based masses.

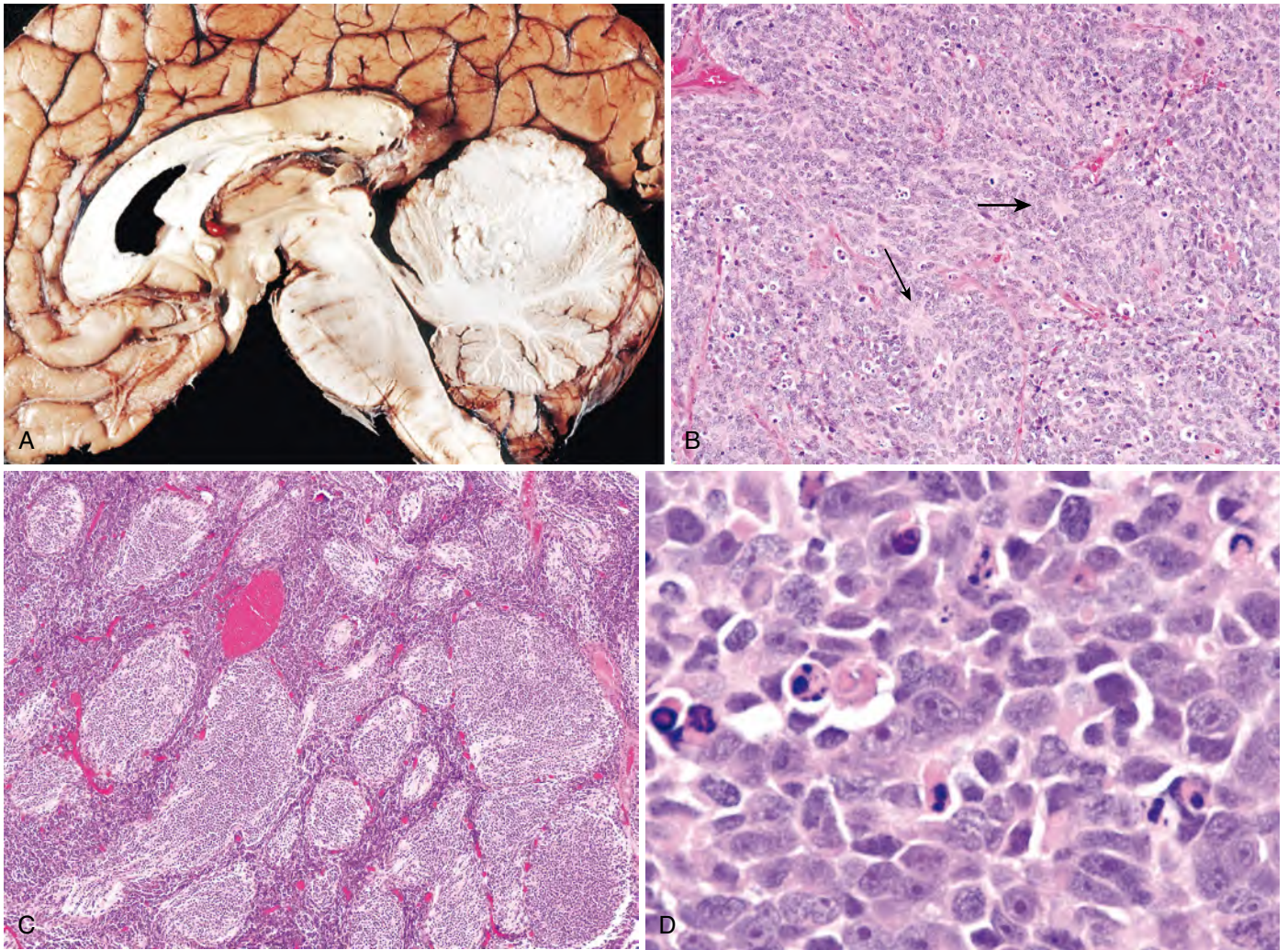


Fig. 28.52 Medulloblastoma. (A) Sagittal section of brain showing medulloblastoma replacing part of the superior cerebellar vermis. (B) The microscopic appearance of classic medulloblastoma includes primitive appearing “small blue cells” that form sheets and Homer Wright (neuroblastic) rosettes with central neuropil (arrows). (C) The desmoplastic/nodular tumor includes reticulin-rich internodular zones with more primitive appearing cells and “pale nodules” representing centers of partial neuronal differentiation; this variant is always the Hedgehog-activated molecular subtype. (D) The large cell/anaplastic tumor features increased cell size, large cherry red nucleoli, and cell wrapping (wherein tumor cells wrap around one another or surround pyknotic nuclei of dying tumor cells).

Pathogenesis

The most common cytogenetic abnormality is loss of chromosome 22, especially the long arm (22q). The deletions include the region of 22q12 that harbors the *NF2* gene, which encodes the protein merlin; as expected, meningiomas are a common lesion in the setting of *NF2* (see later). Of sporadic meningiomas, 50% to 60% harbor mutations in the *NF2* gene; these are most often seen in convexity tumors, including those that eventually become higher grade. In meningiomas without *NF2* mutations, the most common mutations occur in *TRAF7*, *KLF4*, *AKT1*, *SMO*, and *PIK3CA*; most such tumors involve the skull base and have a lower risk of malignant progression. In contrast, the rare and more aggressive clear cell and rhabdoid subtypes are often associated with mutations of the *SMARCE1* and *BAP1* genes respectively; both are also associated with syndromic cases caused by germline variants of the same genes. Higher-grade meningiomas more often have chromosomal losses involving several chromosomes, *TERT*-promoter mutations, and homozygous *CDKN2A* gene deletions. The latter two alterations are now considered diagnostic of a CNS WHO grade 3 meningioma.

MORPHOLOGY

Meningiomas are usually rubbery, rounded, or bosselated dural masses that compress the underlying brain but are easily separated from it (Fig. 28.53A). They may also grow *en plaque*, in which the tumor spreads in a sheet-like fashion along the surface of the dura; this form is commonly associated with hyperostotic changes in the adjacent bone due to invasion, although this by itself does not signify malignancy. The lesions range from firm to finely gritty (sand-like), the latter typically due to numerous psammomatous calcifications (“psammos” is Greek for sand).

Most meningiomas have a relatively low risk of recurrence or aggressive growth, and so are considered CNS WHO grade 1. Many histologic patterns are observed, with most showing no prognostic significance. The most common include **meningotheial**, containing clusters of epithelioid cells with fuzzy or indiscernible cell membranes; **fibroblastic**, with intersecting fascicles of spindled cells and abundant collagen deposition; **transitional**, with mixed meningotheial and fibroblastic features, often including many whorls (concentric wrapping of tumor cells around one another); and **psammomatous**, with

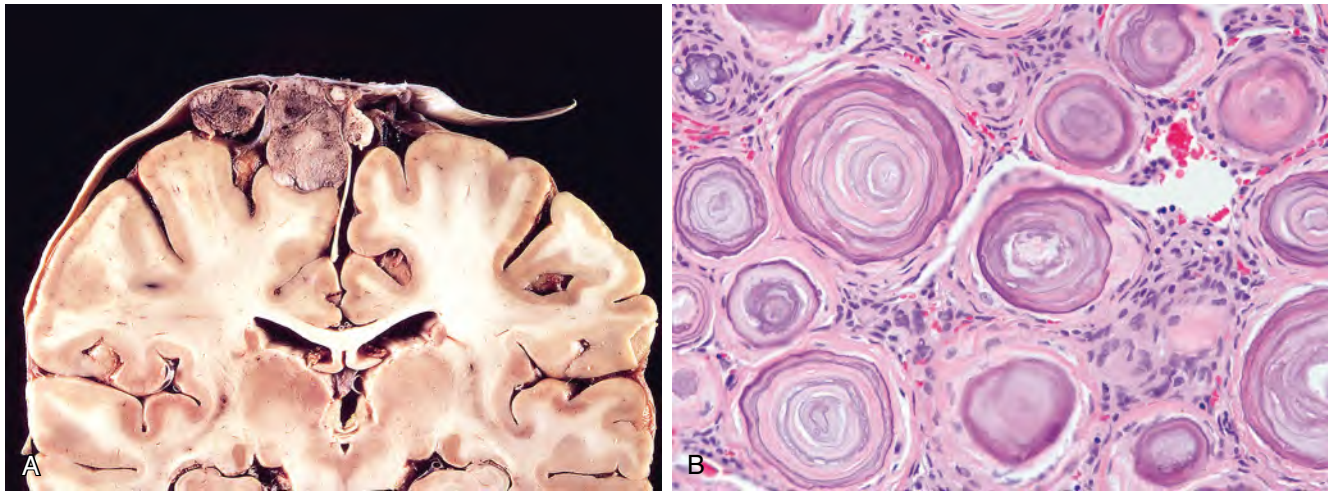


Fig. 28.53 Meningioma. (A) Parasagittal multilobular meningioma attached to the dura with compression of underlying brain. (B) Whorled pattern of cell growth and numerous psammoma bodies (calcifications with concentric rings).

a predominance of psammoma bodies, the latter representing concentric rings of calcification deposited over preexisting whorls (Fig. 28.53B).

Atypical meningiomas (CNS WHO grade 2) represent about one-quarter of meningiomas and are associated with higher recurrence rates and more aggressive local growth, which may require radiation therapy in addition to surgery. They are distinguished from lower grade meningiomas by showing an increased mitotic index, brain invasion, or other aggressive features. Certain histologic patterns (**clear cell** and **chordoid**) are considered CNS WHO grade 2 by definition because of their more aggressive behavior.

Anaplastic (malignant) meningioma (CNS WHO grade 3) is a highly aggressive malignancy that often resembles a carcinoma or sarcoma, but retains some evidence of meningotheial origin. They account for 1% to 3% of all meningiomas and may arise *de novo* or from malignant progression of lower grade meningiomas. Mitotic rates are often markedly elevated.

Clinical Features

Meningiomas are usually slow-growing tumors. Patients present either with vague nonlocalizing symptoms or with focal findings referable to compression of the underlying brain. Common sites of involvement include the parasagittal aspect of the brain convexity, dura over the lateral convexity, wing of the sphenoid, olfactory groove, sella turcica, and foramen magnum. Meningiomas are uncommon in children and show a moderate (2:1) female predominance in adults, although the ratio is 10:1 for spinal meningiomas (which are also commonly psammomatous). Lesions are usually solitary, but when present at multiple sites, especially in association with vestibular schwannoma and/or spinal ependymoma, the possibility of NF2 should be considered. Meningiomas often express a variety of hormone (e.g., progesterone) receptors and may grow more rapidly during pregnancy, only to regress after delivery.

Metastatic Tumors

Metastatic lesions, mostly carcinomas, account for approximately one-quarter to one-half of intracranial tumors in hospitalized patients. The five most common primary sites are lung, breast, skin (melanoma), kidney, and gastrointestinal tract, accounting for about 80% of all metastases combined. Some rare tumors (e.g., choriocarcinoma) have a high likelihood of metastasizing to the brain, whereas other more common tumors (e.g., prostatic adenocarcinoma) almost never do so. The meninges are also a frequent site of involvement by metastatic disease, where they may

either form discrete masses or a more disseminated subarachnoid, meningitis-like pattern known as *meningeal carcinomatosis*; the latter is often highly challenging to treat and is associated with both cranial and spinal neuropathies from nerve root compression. Metastatic tumors present clinically as mass lesions and may occasionally be the first manifestation of systemic cancer. In general, localized treatment of solitary brain metastases improves the quality and duration of the patient's life.

MORPHOLOGY

Intraparenchymal metastases form sharply demarcated masses, often at the junction of gray and white matter, usually surrounded by a zone of edema. The boundary between tumor and brain parenchyma is usually well-defined microscopically, although melanoma does not always follow this rule and may be more infiltrative. The tumors typically resemble their parent tumor type but can be very poorly differentiated in some cases; for the roughly 10% that present as a metastasis of unknown primary, immunohistochemistry is often helpful for further subtyping. Meningeal carcinomatosis, with tumor nodules studding the surface of the brain, spinal cord, and intradural nerve roots, is most often associated with carcinoma of the lung and breast.

Paraneoplastic Syndromes

In addition to the direct and localized effects produced by CNS metastases, peripheral neoplasms sometimes produce *paraneoplastic syndromes* that involve the peripheral and/or central nervous systems and can precede the clinical recognition of the primary tumor. A variety of paraneoplastic syndromes has been described; a shared underlying mechanism appears to be the development of an immune response against tumor antigens that cross-reacts with antigens in the central or peripheral nervous system. The spectrum of known circulating antibodies and target antigens continues to expand; illustrative CNS examples are:

- *Subacute cerebellar degeneration* is associated with destruction of Purkinje cells, gliosis, and a chronic inflammatory cell infiltrate. One group of affected patients has a circulating PCA-1 (Purkinje cell cytoplasmic antibody type 1) antibody that recognizes cerebellar Purkinje cells; this antibody occurs predominantly in patients with ovarian, uterine, or breast carcinoma.
- *Limbic encephalitis* is characterized by subacute dementia and shows perivascular inflammatory cuffs, microglial nodules, some neuronal loss, and gliosis, all of which are most evident in the anterior and

medial portions of the temporal lobe; the microscopic picture resembles that of a viral encephalitis. A comparable process involving the brainstem can be seen in isolation or together with limbic system involvement. Some affected patients have a circulating antibody that recognizes neuronal nuclei in the central and peripheral nervous systems; it is most often associated with small cell carcinoma of the lung. Another group of patients has a circulating antibody that recognizes the NMDA receptor and cross-reacts with hippocampal neurons. Originally identified in patients with ovarian teratomas, the same clinical syndrome is now also recognized in a small proportion of patients with sporadic encephalitis. A third group of patients has a circulating VGKC-complex antibody that recognizes the voltage-gated potassium channel; this antibody may also be associated with peripheral neuropathy.

- *Eye movement disorders*, most commonly opsoclonus, may occur by themselves or in association with other evidence of cerebellar and brainstem dysfunction. In children, this is most often associated with neuroblastoma and is often accompanied by myoclonus.

The peripheral nervous system can also be affected by the following paraneoplastic syndromes:

- *Subacute sensory neuropathy* may be found in association with limbic encephalitis or in isolation. It is marked by loss of sensory neurons and lymphocytic inflammation in the dorsal root ganglia.
- *Lambert-Eaton myasthenic syndrome* is caused by antibodies against the voltage-gated calcium channel in the presynaptic elements of the neuromuscular junction. This syndrome can also be seen in the absence of malignancy.

For some paraneoplastic syndromes, there is evidence that immunotherapy (removal of circulating antibodies and immunosuppression) and tumor removal lead to clinical improvement. In general, clinical syndromes associated with plasma membrane-reactive antibodies (e.g., antibodies that recognize VGKC or NMDA receptor) respond to immunotherapy better than those associated with intracellular antigens.

Familial Tumor Syndromes

A number of inherited syndromes are associated with an increased risk of neoplasms (Chapter 7). Here we discuss several in which tumors of the nervous system are prominent features.

Neurofibromatosis and Schwannomatosis

The two forms of *neurofibromatosis*, NF1 and NF2, are biologically distinct familial autosomal dominant syndromes characterized by tumors of the peripheral and central nervous systems. In contrast, *schwannomatosis* involves the peripheral nervous system only and may be either familial or sporadic. NF1 has a frequency of 1 in 3000 individuals and is characterized by neurofibromas of peripheral nerves, gliomas of the optic nerve, pigmented nodules of the iris (Lisch nodules), and cutaneous hyperpigmented macules (café au lait spots). NF2 is less common (with a frequency of 1 in 40,000 to 50,000 persons) and is characterized by the occurrence of bilateral vestibular (cranial nerve VIII) schwannomas, multiple meningiomas, and ependymomas of the cervical spinal cord. Non-NF2 schwannomatosis, roughly as common as NF2, is the most recently recognized syndrome; it is defined by multiple nonvestibular schwannomas, either throughout the body or limited to one region. These disorders are discussed in greater detail in Chapter 27.

Tuberous Sclerosis Complex

Tuberous sclerosis complex (TSC) is an autosomal dominant syndrome that occurs at a frequency of approximately 1 in 6000 births.

It is characterized by the development of hamartomas and benign neoplasms involving the brain and other tissues; the most frequent clinical manifestations are seizures, autism, and intellectual disability. Hamartomas within the CNS take the form of cortical tubers and subependymal nodules; *subependymal giant cell astrocytomas* (SEGA) are benign neoplasms that appear to develop from the hamartomatous nodules in the same locations. Cortical tubers are often epileptogenic, and surgical resection can be beneficial when medical management of the seizures fails. Elsewhere in the body, renal angiomyolipomas, retinal glial hamartomas, pulmonary lymphangioleiomyomatosis, and cardiac rhabdomyomas develop during childhood and adolescence. Cysts may be found at various sites, including the liver, kidneys, and pancreas. Cutaneous lesions include angiofibromas, localized leathery thickenings (shagreen patches), hypopigmented areas (ash-leaf patches), and subungual fibromas.

One tuberous sclerosis gene (*TSC1*) is found on chromosome 9q34 and encodes a protein known as hamartin; the more commonly mutated gene (*TSC2*) is on 16p13.3 and encodes tuberlin. These two proteins form a complex that inhibits the kinase mTOR, a key regulator of protein synthesis and other aspects of anabolic metabolism. Mutations in *TSC1* or *TSC2* disrupt this control and lead to increased and unregulated mTOR activity. Of note, mTOR controls cell size, and the tumor cells associated with TSC, particularly the subependymal giant cell astrocytomas, are remarkable for having voluminous cytoplasm. Treatment is usually symptomatic, including anticonvulsant therapy for control of seizures; however, treatment with mTOR inhibitors has sometimes resulted in dramatic clinical improvement.

MORPHOLOGY

Cortical hamartomas of TSC are firm areas that, in contrast to the softer adjacent cortex, have been likened to potatoes (hence the appellation “**tubers**”). The tubers are composed of haphazardly arranged neurons that lack the normal laminar organization. In addition, some dysplastic cells have an appearance intermediate between glia and neurons (large vesicular nuclei with nucleoli, resembling neurons, and abundant eosinophilic cytoplasm resembling gemistocytic astrocytes); they often express intermediate filaments of both neuronal (neurofilament) and glial (GFAP) types. Similar hamartomatous features are present in the subependymal nodules, where the large astrocyte-like cells cluster beneath the ventricular surface. These *small* wax-like masses that bulge into the ventricular system gave rise to the term **candle gutterings**. Subependymal giant cell astrocytoma is histologically similar, but larger; it often presents with obstructive hydrocephalus.

Von Hippel-Lindau Disease

Individuals with this autosomal dominant disease develop hemangioblastomas of the CNS; cysts that involve the pancreas, liver, and kidneys; renal cell carcinomas; and pheochromocytomas. Hemangioblastomas are most common in the cerebellum and retina but may also occur in other CNS locations. The disease frequency is 1 in 30,000 to 40,000 individuals.

The gene associated with von Hippel-Lindau disease, called *VHL*, is a tumor-suppressor gene located on chromosome 3p25.3; it encodes a protein (VHL) that is a component of a ubiquitin ligase complex that down-regulates transcription factors of the hypoxia-induced factor (HIF) family. HIFs regulate the expression of VEGF, erythropoietin, and other growth factors. VEGF leads to excessive vessel growth, contributing to the development of hemangioblastomas. Erythropoietin overexpression leads to polycythemia in about 10% of hemangioblastoma patients. HIFs also regulate the expression of genes that control

cellular metabolism and cell growth, activities that likely contribute to tumor formation. Therapy is directed at the symptomatic neoplasms, including resection of the cerebellar hemangioblastomas and laser therapy for retinal hemangioblastomas.

MORPHOLOGY

Hemangioblastomas are highly vascular neoplasms that form a mural nodule associated with a large fluid-filled cyst. The lesion consists of numerous capillary-size or somewhat larger thin-walled vessels with intervening neoplastic cells that have vacuolated, lipid-rich cytoplasm. The neoplastic stromal cells express erythropoietin, VEGF, and inhibin (a diagnostically useful marker), and are believed to be derived from an early mesenchymal progenitor cell capable of differentiating into both endothelial and hemopoietic cells.

KEY CONCEPTS

Tumors

- Tumors of the CNS may arise from the brain coverings (meningiomas), the brain parenchyma (gliomas, neuronal tumors, choroid plexus tumors), and other resident CNS cell populations (primary CNS lymphoma); they can also originate elsewhere in the body (metastases).
- Even low-grade or benign tumors can have poor clinical outcomes, depending on where they occur in the brain.
- Distinct types of tumors affect specific brain regions (e.g., cerebellum for medulloblastoma) and specific age populations (medulloblastoma and pilocytic astrocytoma in pediatric age groups; glioblastoma and lymphoma in older patients).
- Glial tumors are broadly classified into astrocytomas, oligodendrogliomas, and ependymomas. Increasing tumor malignancy is associated with more cytologic anaplasia, increased cell density, necrosis, and mitotic activity. Diagnostic, prognostic, and/or predictive genetic alterations have been recognized in many of these tumors.
- Metastatic spread of brain tumors to other regions of the body is rare, but the brain is a common site of metastatic disease from systemic malignancies. Carcinomas are most common.

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